

Machine learning for visualization (ml4vis)

CS424: Visualization & Visual Analytics
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evl

electronic
visualization
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Machine learning in many domains



Banks are using tabular data to identify fraud or calculate loan risk.



Machine translation has improved remarkably over the past few years.



Healthcare providers look to interpretable models to guide decisions.

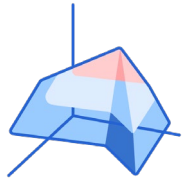
Understanding the gap

- Model understanding has yet to catch up to model performance improvements.
- Visualization is key to improve model understanding to a wide range of audiences.
- There are many rich domains to apply VisML techniques, including industry and research.

Where does visualization fit in?



Many tasks in machine learning inevitably involve some form of visualization.



In other cases, visualization is a byproduct of the process.



We have also seen an increase in the number of interactive visualization systems, designed with machine learning as a critical component.

Visual Analytics in Deep Learning | Interrogative Survey Overview

§4 WHY

Why would one want to use visualization in deep learning?

Interpretability & Explainability
Debugging & Improving Models
Comparing & Selecting Models
Teaching Deep Learning Concepts

§6 WHAT

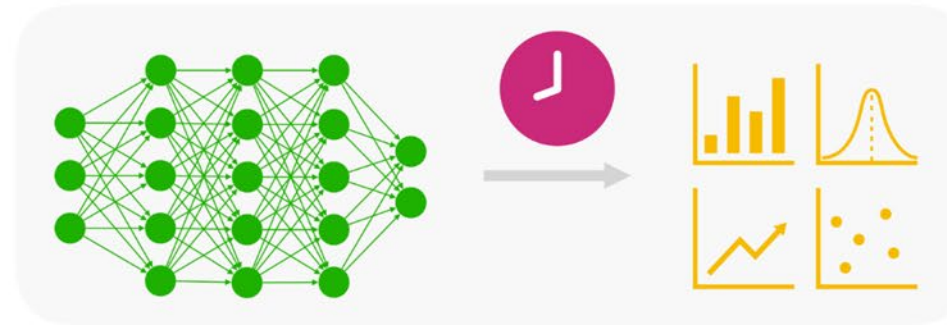
What data, features, and relationships in deep learning can be visualized?

Computational Graph & Network Architecture
Learned Model Parameters
Individual Computational Units
Neurons In High-dimensional Space
Aggregated Information

§8 WHEN

When in the deep learning process is visualization used?

During Training
After Training



§5 WHO

Who would use and benefit from visualizing deep learning?

Model Developers & Builders
Model Users
Non-experts

§7 HOW

How can we visualize deep learning data, features, and relationships?

Node-link Diagrams for Network Architecture
Dimensionality Reduction & Scatter Plots
Line Charts for Temporal Metrics
Instance-based Analysis & Exploration
Interactive Experimentation
Algorithms for Attribution & Feature Visualization

§9 WHERE

Where has deep learning visualization been used?

Application Domains & Models
A Vibrant Research Community

[Hohman et al., 2018]

Vis4ML and ML4Vis



Visualization for machine learning

- Understand model behavior and failure cases
- Build trust: uncertainty and explainability
- What to visualize?
 - Embeddings
 - Loss / metric curves
 - Confusion matrix
 - Feature importance
 - ...

TensorBoard SCALARS GRAPHS DISTRIBUTIONS HISTOGRAMS TIME SERIES

Histogram mode

OVERLAY **OFFSET**

Offset time axis

STEP RELATIVE WALL

Runs

Write a regex to filter runs

☒ train ☐ test

TOGGLE ALL RUNS

logs/fit

Q Filter tags (regular expressions supported)

dense_2

dense_2/bias_0 train

dense_2/kernel_0 train

dense_3

dense_3/bias_0 train

dense_3/kernel_0 train

The figure displays the TensorBoard interface for visualizing histograms of model weights. The top navigation bar includes 'SCALARS', 'GRAPHS', 'DISTRIBUTIONS', 'HISTOGRAMS' (selected), and 'TIME SERIES'. The left sidebar contains controls for histogram mode (Overlay/Offset), time axis (Step/Relative/Wall), and run filtering (Train/Test). The main area shows four histograms for layers dense_2 and dense_3, specifically for bias_0 and kernel_0. Each histogram displays multiple overlapping distributions for different runs, with the 'train' run selected. The x-axis represents the weight values, and the y-axis represents the frequency. The distributions for dense_2 are centered around 0, while those for dense_3 are centered around 0.1. The histograms for dense_2/kernel_0 are centered around 0, and those for dense_3/kernel_0 are centered around 0.2.



Visualization for machine learning

NEO: Generalizing Confusion Matrix Visualization to Hierarchical and Multi-Output Labels

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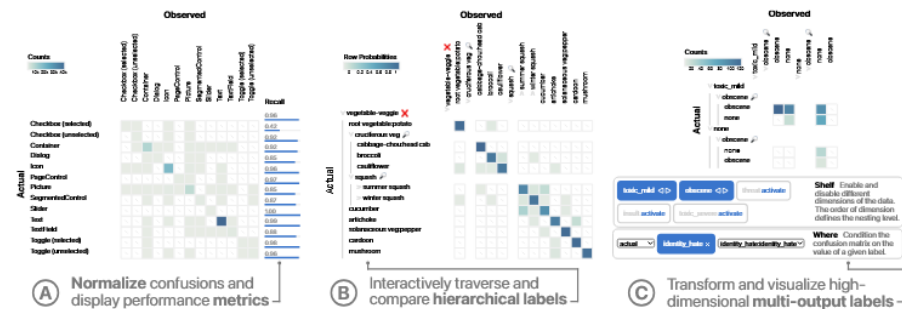


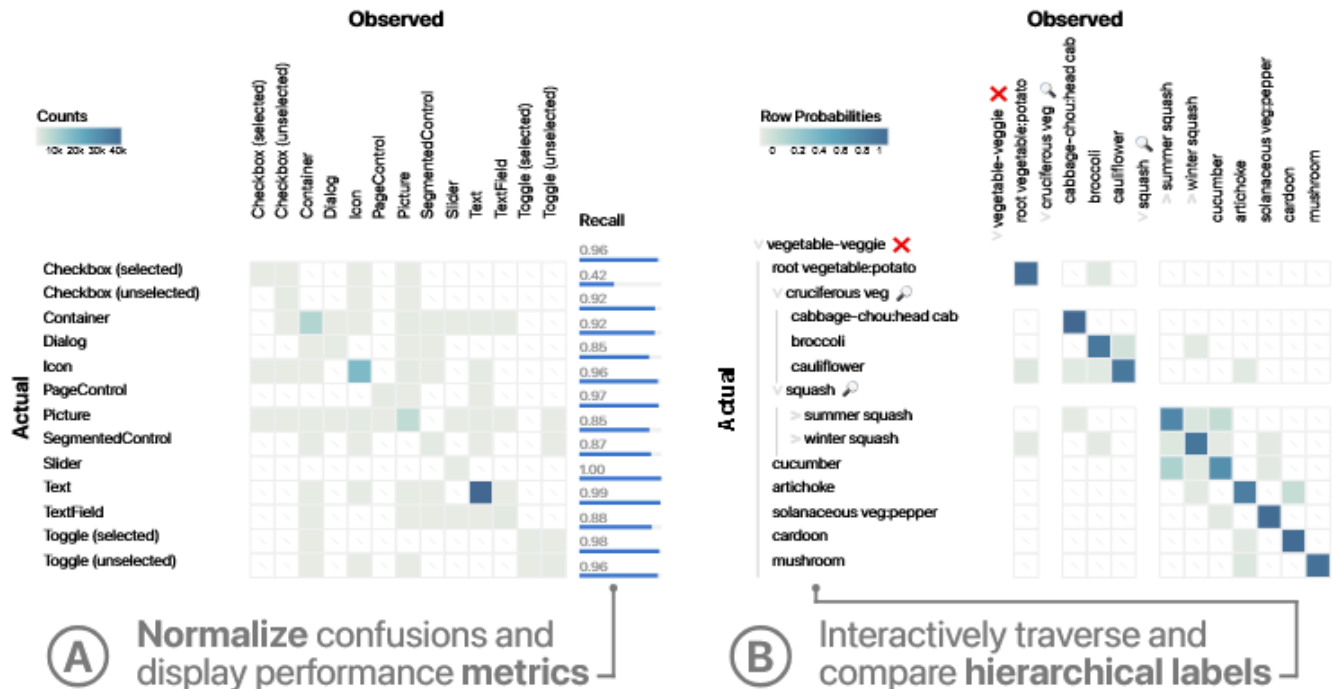
Figure 1: NEO generalizes conventional confusion matrices and enables machine learning practitioners to find hidden confusions, visualize per class metrics, traverse hierarchical labels on tiered axes, and transform high-dimensional, multi-output labels for model evaluation. (A) This confusion matrix for an object detection model computes and shows user-specified performance metrics, such as recall, per class alongside the matrix visualization. (B) This sub-hierarchy of a confusion matrix for a 1,000-class image classifier compares the confusions of different vegetables, such as *cucumber* and *mushroom*, against the sub-hierarchies of *cruciferous vegetables* and *squash*. (C) This confusion matrix for a naive multi-output online toxicity detector conditions and filters by *identity hate* confusions, nests *obscene* confusions under *mild toxic* confusions, and finds that the model misses to identify many *obscene* comments.

ABSTRACT

The confusion matrix, a ubiquitous visualization for helping people evaluate machine learning models, is a tabular layout that compares predicted class labels against actual class labels over all data instances. We conduct formative research with machine learning practitioners at Apple and find that conventional confusion matrices do not support more complex data-structures found in modern-day applications, such as hierarchical and multi-output labels. To express such variations of confusion matrices, we design an algebra that transforms confusion matrices as probability distributions. Based

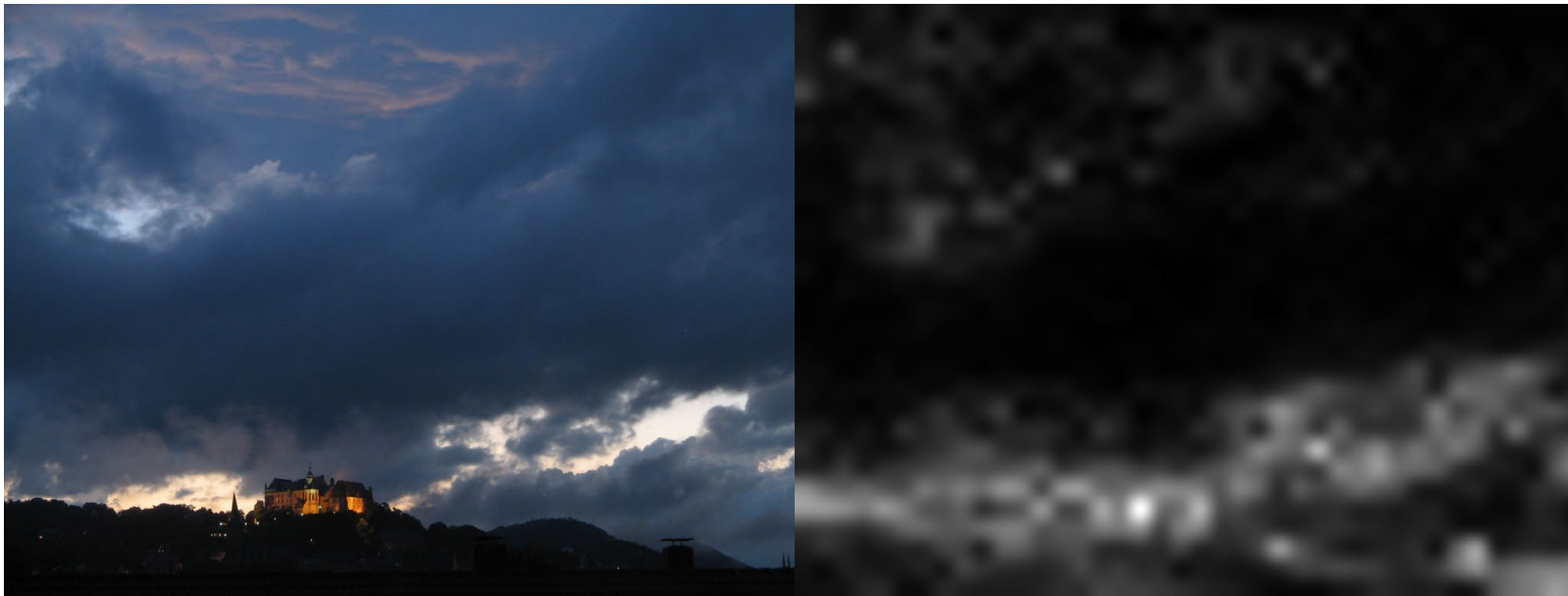
*Work done at Apple.

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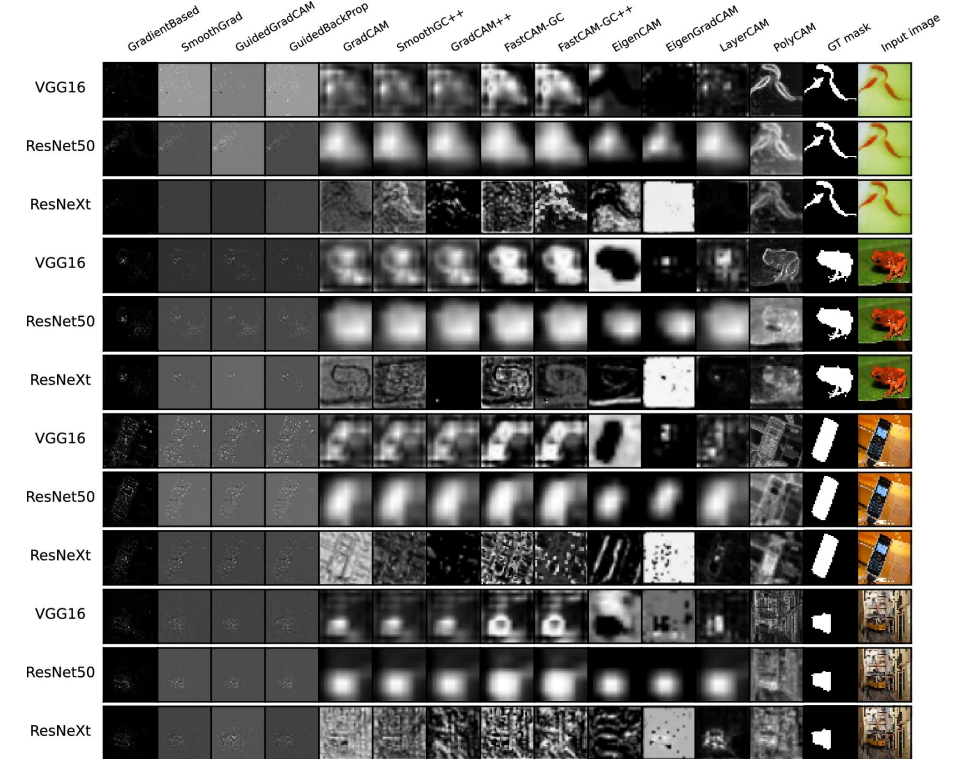
Visualization for machine learning

- Saliency maps (a.k.a. sensitivity maps)
- Idea: sensitivity of class to each pixel



Visualization for machine learning

- Saliency maps (a.k.a. sensitivity maps)
- Idea: sensitivity of class to each pixel
- Heatmaps highlight input regions most influential for a model's prediction.
- You have a model $f(x)$ and a target class score $s_c(x)$
 - Which parts of (x) would most change $s_c(x)$ if I nudge them a bit?
 - Compute $\nabla_x s_c(x)$ as a per-feature sensitivity
 - For images: take the per-pixel gradient magnitude $|\nabla_x s_c(x)|$



Visualization for machine learning

Visualizing and Understanding Convolutional Networks

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Abstract

Large Convolutional Network models have recently demonstrated impressive classification performance on the ImageNet benchmark (Krizhevsky et al., 2012). However there is no clear understanding of why they perform so well, or how they might be improved. In this paper we address both issues. We introduce a novel visualization technique that gives insight into the function of intermediate feature layers and the operation of the classifier. Used in a diagnostic role, these visualizations allow us to find model architectures that outperform Krizhevsky *et al.* on the ImageNet classification benchmark. We also perform an ablation study to discover the performance contribution from different model layers. We show our ImageNet model generalizes well to other datasets: when the softmax classifier is retrained, it convincingly beats the current state-of-the-art results on Caltech-101 and Caltech-256 datasets.

1. Introduction

Since their introduction by (LeCun et al., 1989) in the early 1990's, Convolutional Networks (convnets) have demonstrated excellent performance at tasks such as hand-written digit classification and face detection. In the last year, several papers have shown that they can also deliver outstanding performance on more challenging visual classification tasks. (Ciresan et al., 2012) demonstrate state-of-the-art performance on NORB and CIFAR-10 datasets. Most notably, (Krizhevsky et al., 2012) show record beating performance on the ImageNet 2012 classification benchmark, with their convnet model achieving an error rate of 16.4%, compared to the 2nd place result of 26.1%. Several factors are responsible for this renewed inter-

est in convnet models: (i) the availability of much larger training sets, with millions of labeled examples; (ii) powerful GPU implementations, making the training of very large models practical and (iii) better model regularization strategies, such as Dropout (Hinton et al., 2012).

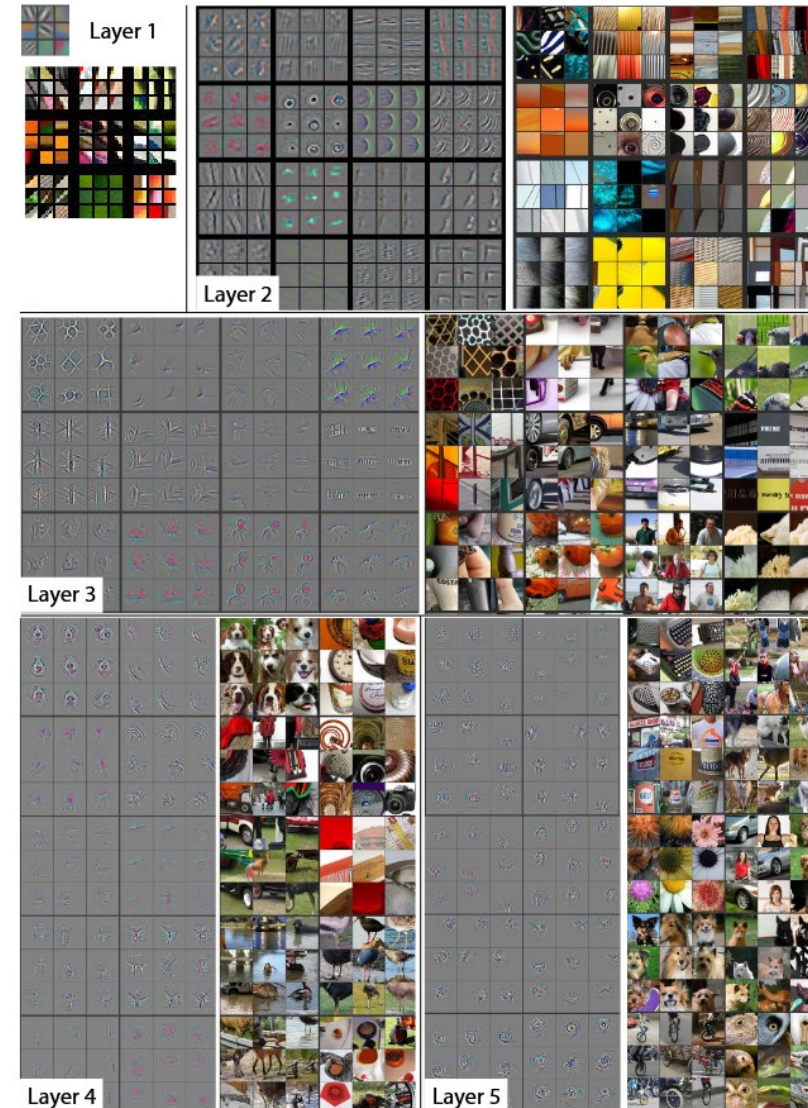
Despite this encouraging progress, there is still little insight into the internal operation and behavior of these complex models, or how they achieve such good performance. From a scientific standpoint, this is deeply unsatisfactory. Without clear understanding of how and why they work, the development of better models is reduced to trial-and-error. In this paper we introduce a visualization technique that reveals the input stimuli that excite individual feature maps at any layer in the model. It also allows us to observe the evolution of features during training and to diagnose potential problems with the model. The visualization technique we propose uses a multi-layered Deconvolutional Network (deconvnet), as proposed by (Zeiler et al., 2011), to project the feature activations back to the input pixel space. We also perform a sensitivity analysis of the classifier output by occluding portions of the input image, revealing which parts of the scene are important for classification.

Using these tools, we start with the architecture of (Krizhevsky et al., 2012) and explore different architectures, discovering ones that outperform their results on ImageNet. We then explore the generalization ability of the model to other datasets, just retraining the softmax classifier on top. As such, this is a form of supervised pre-training, which contrasts with the unsupervised pre-training methods popularized by (Hinton et al., 2006) and others (Bengio et al., 2007; Vincent et al., 2008). The generalization ability of convnet features is also explored in concurrent work by (Donahue et al., 2013).



Visualization for machine learning

- Activation maps (or feature maps)
 - The tensors a hidden layer outputs for a given input.
 - *See what patterns fire where.*
 - Different from saliency maps.
 - Saliency maps give information on the most important parts of an image.
 - Class-discriminative saliency maps help distinguish between classes.



Machine learning for visualization



Machine learning for visualization

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A Survey on ML4VIS: Applying Machine Learning Advances to Data Visualization

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Abstract—Inspired by the great success of machine learning (ML), researchers have applied ML techniques to visualizations to achieve a better design, development, and evaluation of visualizations. This branch of studies, known as ML4VIS, is gaining increasing research attention in recent years. To successfully adapt ML techniques for visualizations, a structured understanding of the integration of ML4VIS is needed. In this article, we systematically survey 88 ML4VIS studies, aiming to answer two motivating questions: “*what visualization processes can be assisted by ML?*” and “*how ML techniques can be used to solve visualization problems?*” This survey reveals seven main processes where the employment of ML techniques can benefit visualizations: *Data Processing4VIS*, *Data-VIS Mapping*, *Insight Communication*, *Style Imitation*, *VIS Interaction*, *VIS Reading*, and *User Profiling*. The seven processes are related to existing visualization theoretical models in an ML4VIS pipeline, aiming to illuminate the role of ML-assisted visualization in general visualizations. Meanwhile, the seven processes are mapped into main learning tasks in ML to align the capabilities of ML with the needs in visualization. Current practices and future opportunities of ML4VIS are discussed in the context of the ML4VIS pipeline and the ML-VIS mapping. While more studies are still needed in the area of ML4VIS, we hope this article can provide a stepping-stone for future exploration. A web-based interactive browser of this survey is available at <https://ml4vis.github.io>.

Index Terms—ML4VIS, machine learning, data visualization, survey

1 INTRODUCTION

DATA visualization (VIS), which maps data into visual presentations (e.g., position, color), is a powerful approach to uncover hidden insights and communicate compelling stories in data. However, achieving an effective visualization is typically challenging, requiring a large amount of human effort and having a high reliance on expertise, such as graphic design, user experience design, and data analysis.

Machine Learning (ML), on the other hand, provides a practical opportunity to relieve the reliance on experts in visualization. By automatically learning knowledge from data, ML enables task completion without explicit instructions from humans. The application of ML techniques can benefit a variety of visualization-related problems. In this paper, we define visualization-related problems as problems that are related to the process of creating, interacting with, and evaluating visualizations. We refer to studies that *apply ML techniques to solve visualization-related problems using the knowledge extracted from data as ML4VIS* in this paper. ML4VIS studies can be traced back to 1986 [89] and have been reignited by the recent advances in ML. A series of

ML4VIS studies are emerging, covering a wide range of visualization problems (e.g., graph layout [50], visualization assessment [54]) and ML techniques (e.g., variational auto-encoders [49], Mask-RCNN [60]).

While ML4VIS generates a stream of new opportunities, it also poses a series of challenges. First, it is not always clear, among various problems related to visualization, which one can be significantly improved by existing ML techniques and which one still requires a high level of human intervention. Roughly applying ML techniques to unsuitable visualization problems may only impose the drawbacks of ML (e.g., uncertainty, inexplainability) and threaten the validity of a visualization without bringing any benefit. Second, given a visualization-related task, selecting a proper ML technique and employing necessary adaptation are crucial yet challenging. There are a plethora of ML techniques, most of which are exclusively suitable for a certain type of problems. For example, to automatically interpret infographics, ML techniques developed for natural-image-understanding are more helpful and relevant than ML techniques developed for speech recognition. Therefore, the success of ML4VIS hinges on a better understanding of both the visualization and the ML, as well as the integration between the two.

Researchers have contributed a series of frameworks and surveys about the integration between visualization and ML, but they mainly investigate how visualization can assist ML in data analysis [90], [91], [92], [93], [94], [95] rather than how ML can be used to solve visualization-related problems. For example, in a state-of-the-art report, Endert *et al.* [92] depicted a closer integration between visualization and ML techniques through interactive ML. However, the goal of this integration is still to facilitate data analysis in different application domains (e.g., text analytics, multimedia analytics). Some recent surveys have reviewed

ML4Vis: “*Studies that apply **ML techniques** to solve visualization-related problems using the knowledge extracted from data*”

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Machine learning for visualization

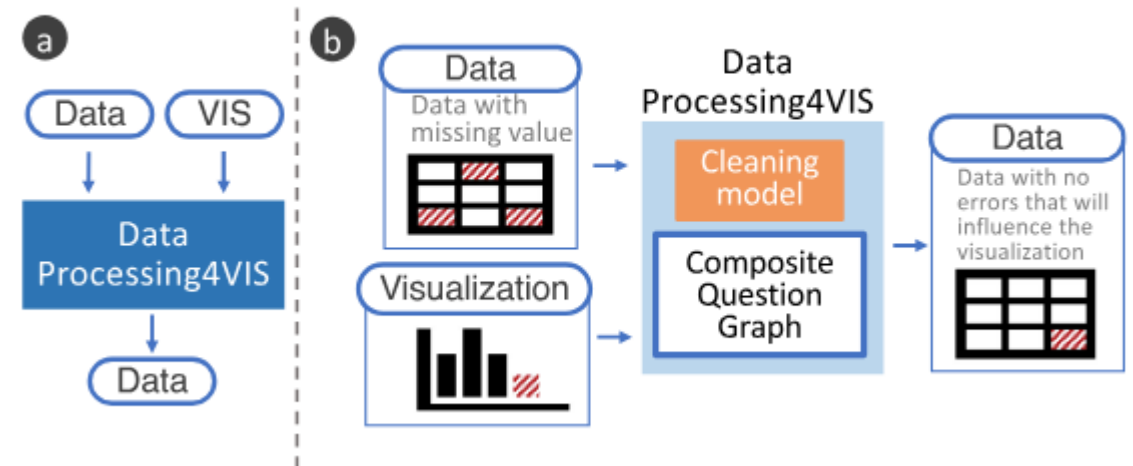
	Process	Description	Problems Solved by ML	Representative Examples
DATA	Data Processing4VIS	raw data is transformed into a format that better suits the following visualization processes	improve the efficiency of data processing	ML cleans data for a specified visualization by asking questions to users [82]
			enhance perception of generated visualizations	ML transforms high dimension data to 2D and maximizes the human-perceived separation among classes [29]
VIS	Data-VIS Mapping	data fields are mapped into visual channels	generate suitable visualizations	ML generates Vega-Lite visualization specifications for a given dataset [41]
			improve the efficiency of visualization creation	ML synthesizes intermediate density map images between given density maps without storing and visualizing data of all time steps [48]
	Insight Communication	insights are embedded in visualizations to be effectively communicated	interpret insights	ML recognizes entities from the user-provided text insights [59]
			generate suitable visualizations	ML predicts suitable visualization specifications for communicating given insights [31]
USER	Style Imitation	styles are extracted from the given examples and applied to the created visualization	imitate the color selection	ML generates color ramps by imitating designers' practices in choosing colors [62]
			imitate the layouts	ML generates graph drawings with similar layout styles of the given examples [50]
	VIS Interaction	users interact with a visualization and transformed it into a new stage through user actions	refine the result of the current interaction	ML refines the selected points of a lasso selection in 3D point clouds [46]
			understand natural interaction	ML interprets the given text descriptions and extracts the corresponding visual patterns [25]
	User Profiling	user actions with visualizations are logged and analyzed to better understand users	predict user behavior	ML predicts a user's next click in a scatter plot [66]
			predict user characteristics	ML predicts a user's task performance and personality based on mouse interaction [6]
	VIS Reading	users read visualizations and obtain useful information	extract content	ML extracts an extensible visualization template from a timeline infographics [60]
			interpret content	ML answers questions about a given bar chart visualization [38]
			estimate human perception	ML predicts the distribution of users' attention on an infographic [18]

Machine learning for visualization

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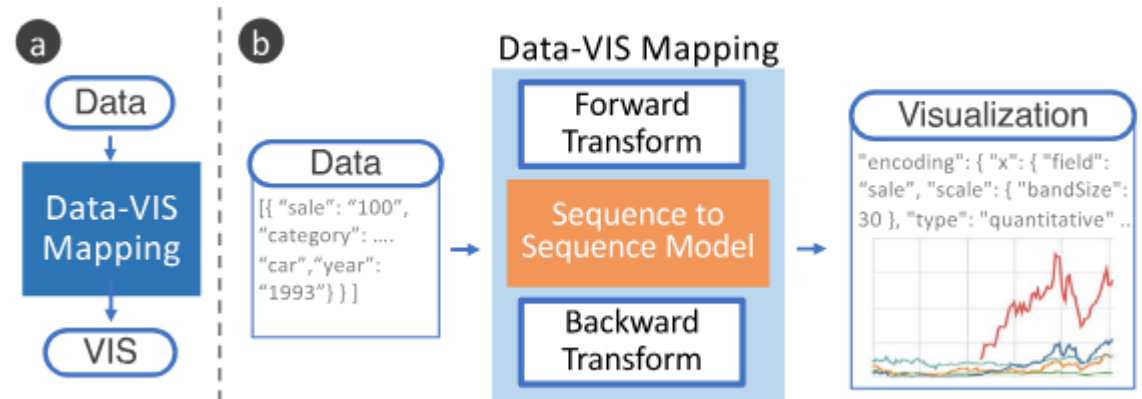
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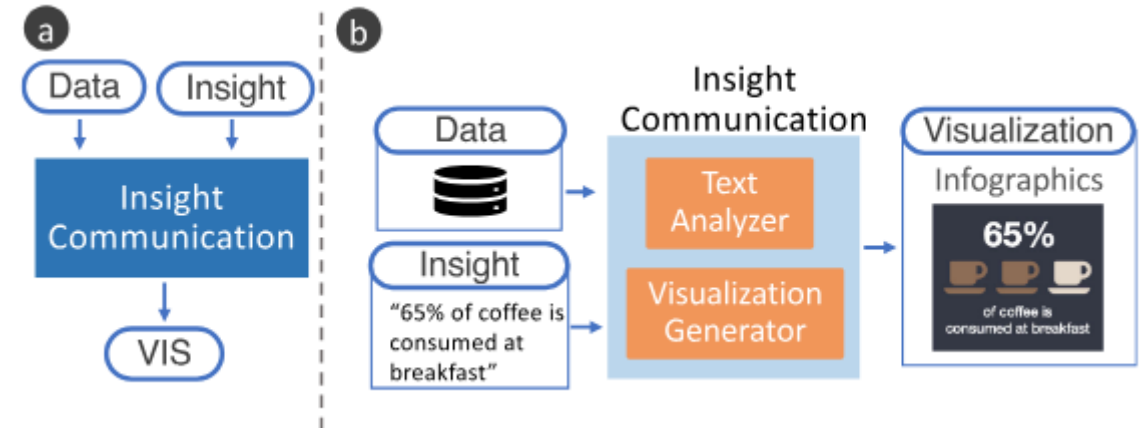
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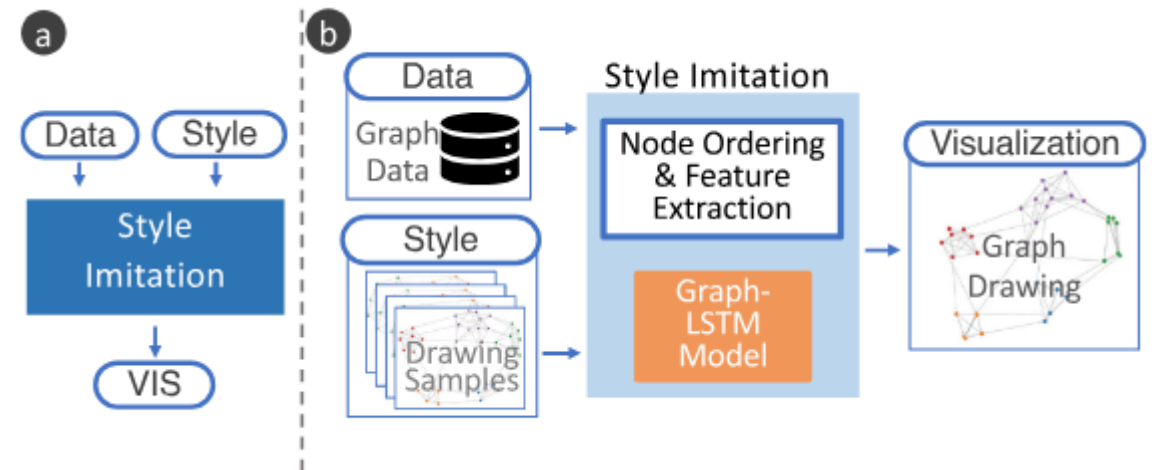
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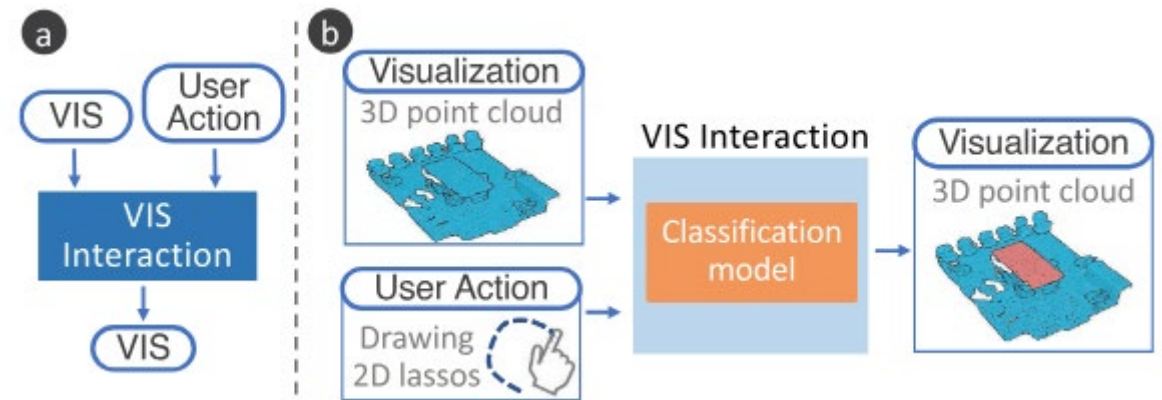
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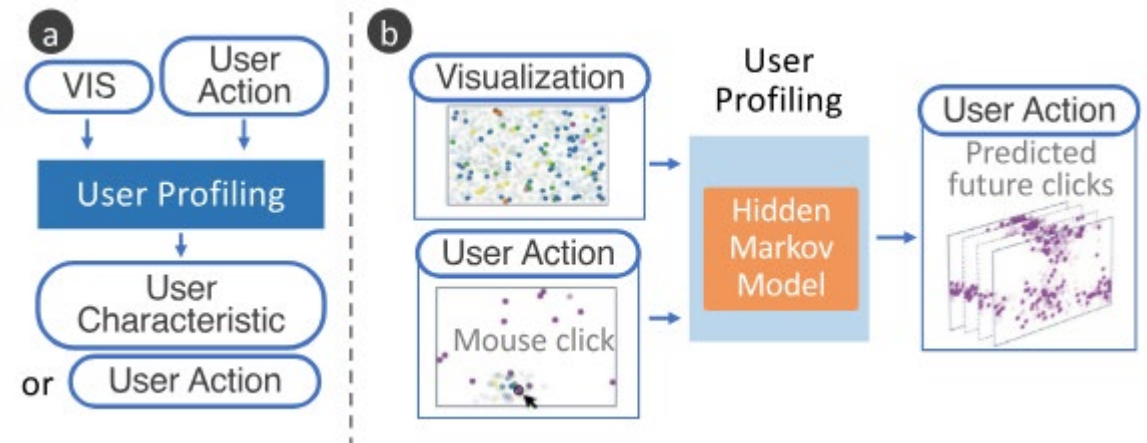
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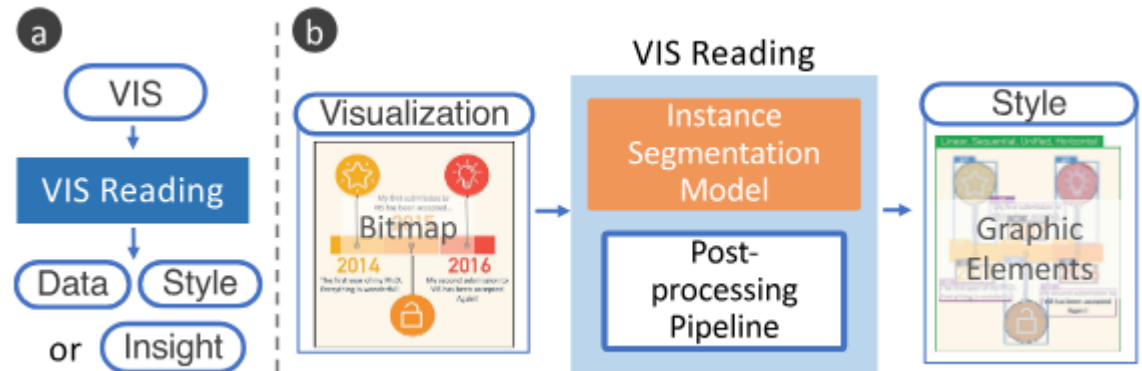
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Machine learning for visualization

LassoNet: Deep Lasso-Selection of 3D Point Clouds

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Lingyun Yu, Chi-Wing Fu, *Member, IEEE*, and Huamin Qu, *Member, IEEE*

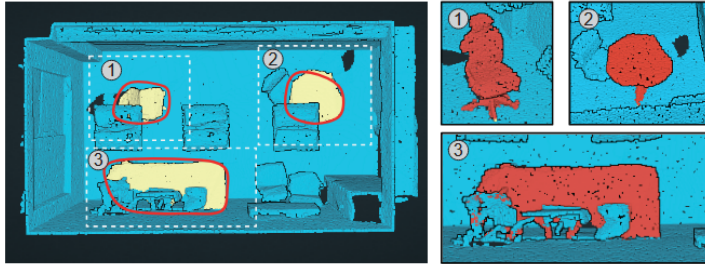


Fig. 1. LassoNet enables effective lasso-selection of 3D point clouds based on a latent mapping from viewpoint and lasso to target point clouds. LassoNet is particularly efficient for selecting multiple regions (insets 1, 2, 3) in a complex scene (left), since no viewpoint changing is required to select occluded points. Notice here the insets are viewed from different viewpoints.

Abstract—Selection is a fundamental task in exploratory analysis and visualization of 3D point clouds. Prior researches on selection methods were developed mainly based on heuristics such as local point density, thus limiting their applicability in general data. Specific challenges root in the great variabilities implied by point clouds (e.g., dense vs. sparse), viewpoint (e.g., occluded vs. non-occluded), and lasso (e.g., small vs. large). In this work, we introduce LassoNet, a new deep neural network for lasso selection of 3D point clouds, attempting to learn a latent mapping from viewpoint and lasso to point cloud regions. To achieve this, we couple user-target points with viewpoint and lasso information through 3D coordinate transform and naive selection, and improve the method scalability via an intention filtering and farthest point sampling. A hierarchical network is trained using a dataset with over 30K lasso-selection records on two different point cloud data. We conduct a formal user study to compare LassoNet with two state-of-the-art lasso-selection methods. The evaluations confirm that our approach improves the selection effectiveness and efficiency across different combinations of 3D point clouds, viewpoints, and lasso selections. *Project Website:* <https://lassonet.github.io>

Index Terms—Point Clouds, Lasso Selection, Deep Learning

1 INTRODUCTION

Vast amounts of 3D point clouds have been collected from various sources, such as LiDAR scanning and particle simulation. Exploratory analysis and visualization of point clouds show benefits in many applications, including astronomy, autonomous navigation, and scene reconstruction. Selection is a fundamental task in exploratory analysis of point clouds. However, designing effective selection for 3D point clouds is challenging, especially when the visualization is projected onto a planar 2D surface [19]. The challenge comes from several perspectives: 1) *data*: a point cloud often consists of a set of unlabeled points, i.e., no information of what label each point holds; 2) *visual-*

ization: projecting points in 3D space to 2D surface can easily cause occlusion and visual cluttering; 3) *interaction*: input devices such as mouses and touchscreens operate in a limited 2D space.

Compared to other selection means of picking and brushing, lasso-selection is considered more appropriate for 3D point clouds [43]. The interactive process derives an appropriate subset of points (P_s) from a point cloud dataset (P), based on a lasso (L) input drawn on a 2D surface and the current viewpoint (V). This work aims to develop an advanced lasso-selection method for exploring 3D point clouds. We consider that the requirements for such a lasso-selection method include:

- **Efficient**: The method should enable users to complete point selection as soon as possible, for which a necessary condition is that the computation should be finished in a short time.
- **Effective**: The derived subset of points P_s is expected to match with the target points of a user's intention (P_t). Here, we employ Jaccard distance (d_J) to measure difference between P_s and P_t .
- **Robust**: The method should be robust to variability implied by data (e.g., dense vs sparse, viewpoint (e.g., occluded vs non-occluded), and lasso drawing (e.g., small vs big)).

Recently, a number of lasso-selection methods for point clouds have been proposed [30, 42, 43]. The methods can be categorized as *structure-aware* that depend on characteristics of point clouds, e.g.,



How to select objects in such a dense and unstructured environment?

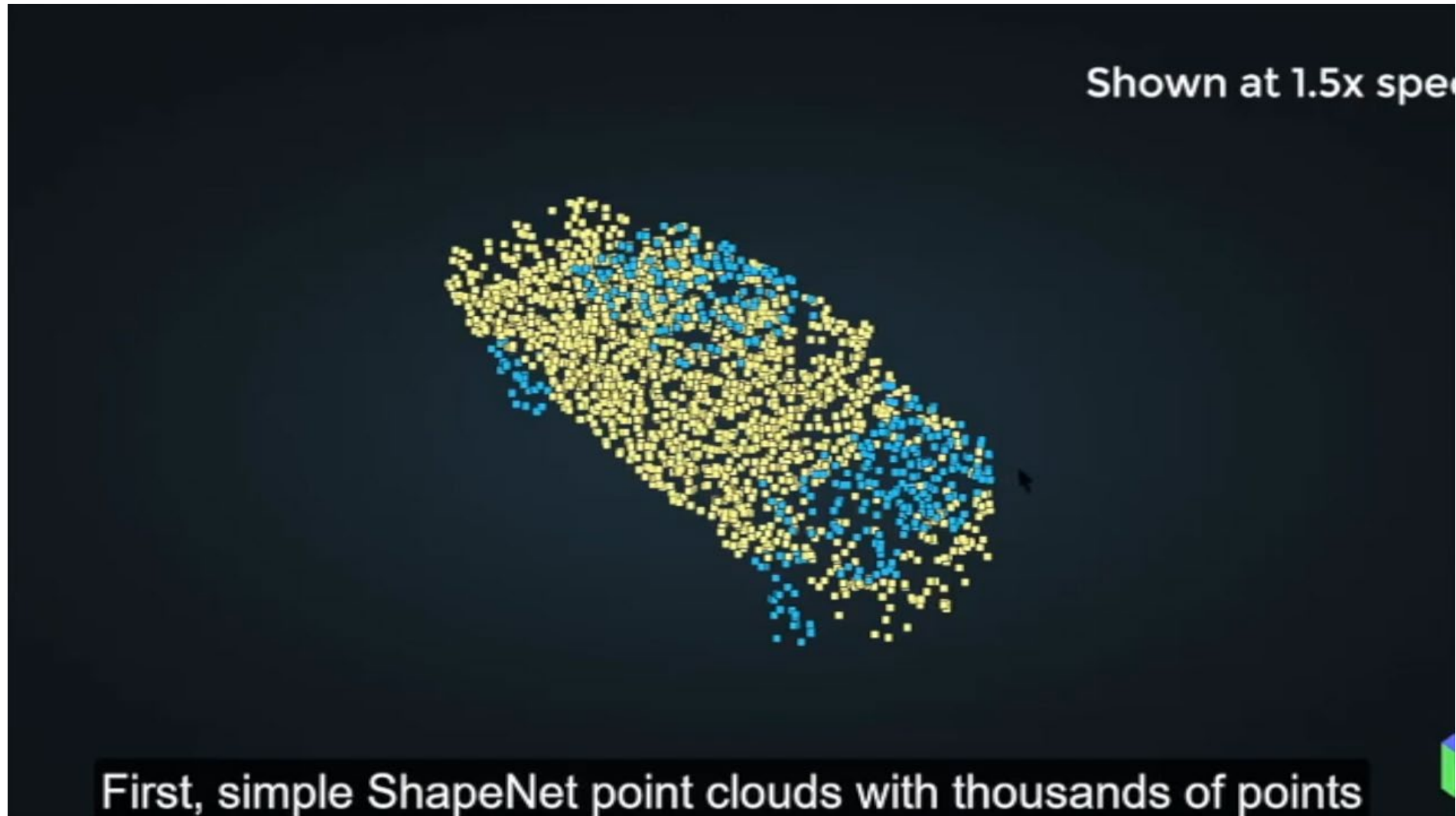
In this work, we regard lasso-selection as a mapping $f(P, V, L)$ that retrieves a subset of points $P_s \subseteq P$ based on the current viewpoint V and lasso drawing L . To be *effective*, the mapping function f should minimize difference between P_s and target points P_t , i.e.,

$$\arg \min_f \{d_J(P_s, P_t) | f(P, V, L) \rightarrow P_s\} \quad (1)$$

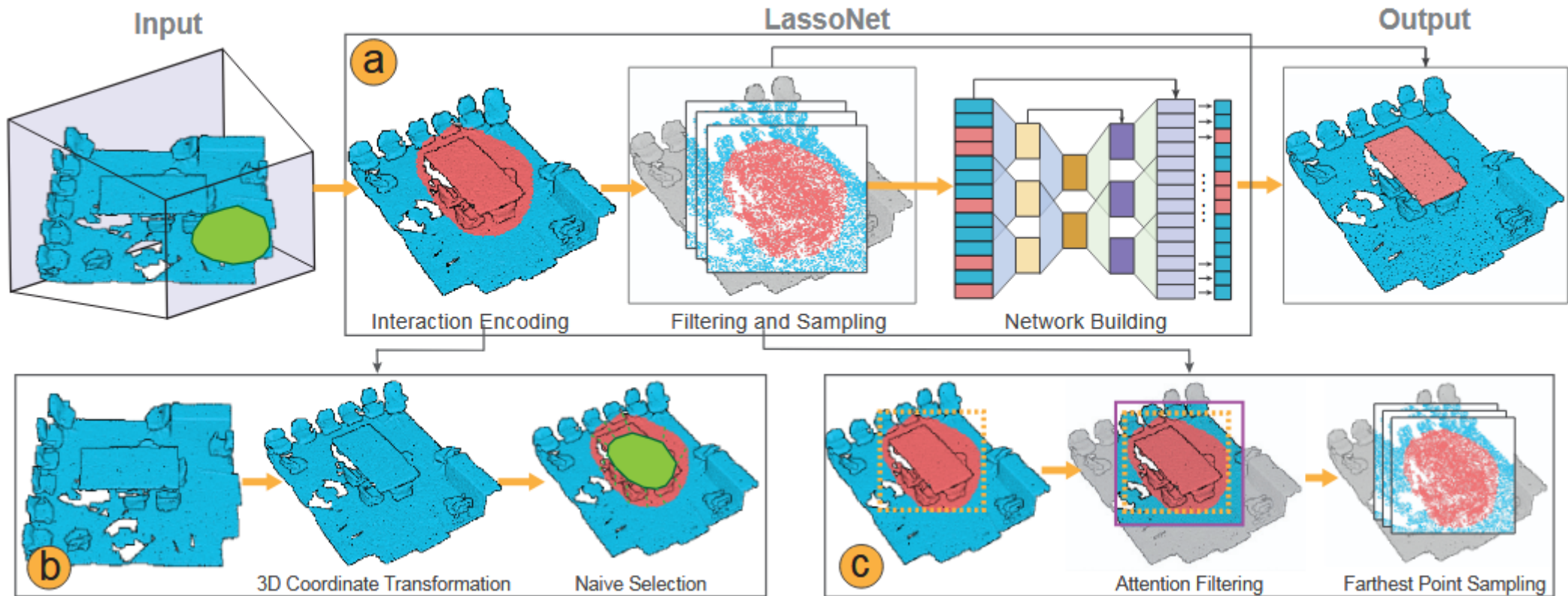
Manuscript received 31 Mar. 2019; accepted 1 Aug. 2019.
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Digital Object Identifier no. 10.1109/TVCG.2019.2934332

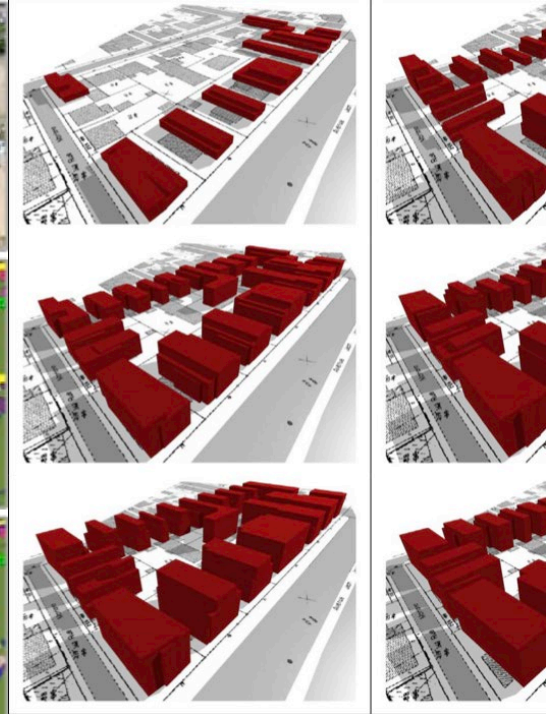
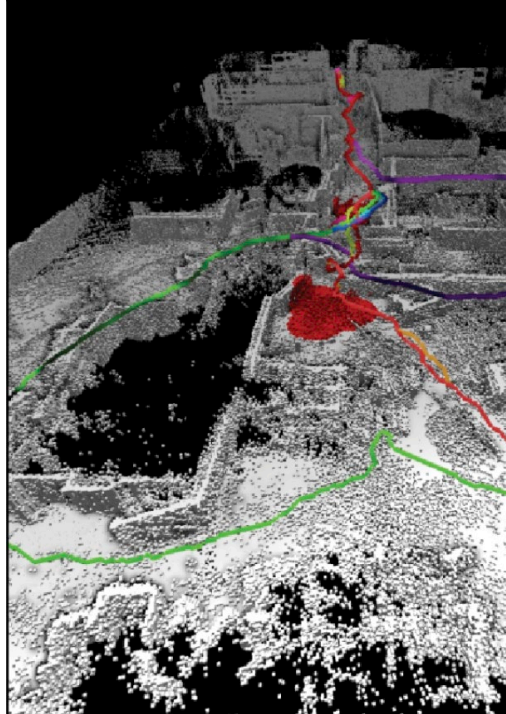
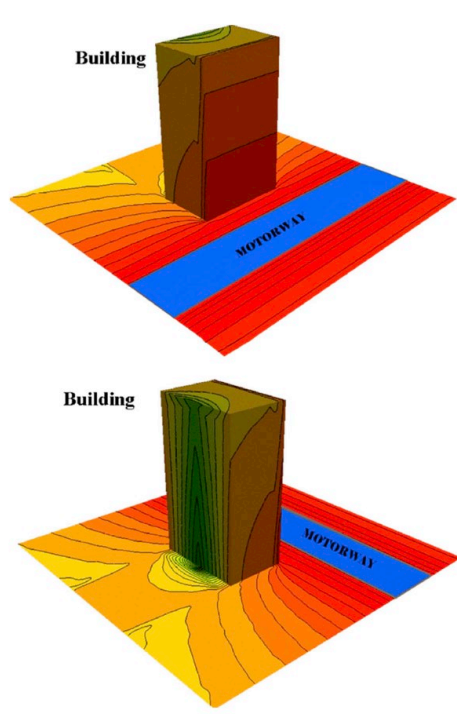
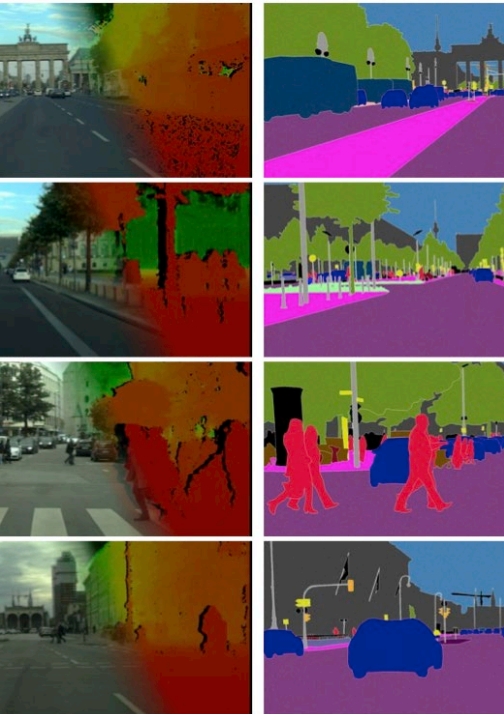
• Chen Zhu-Tian and Huamin Qu are with the Hong Kong University of Science and Technology. E-mail: zhuatian.chen@connect.ust.hk and huamin@cse.ust.hk.
• Wei Zeng and Zhiguang Yang are with Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences. Wei Zeng is the corresponding author. E-mail: wei.zeng_cq.yang@sia.ac.cn.
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Machine learning for visualization

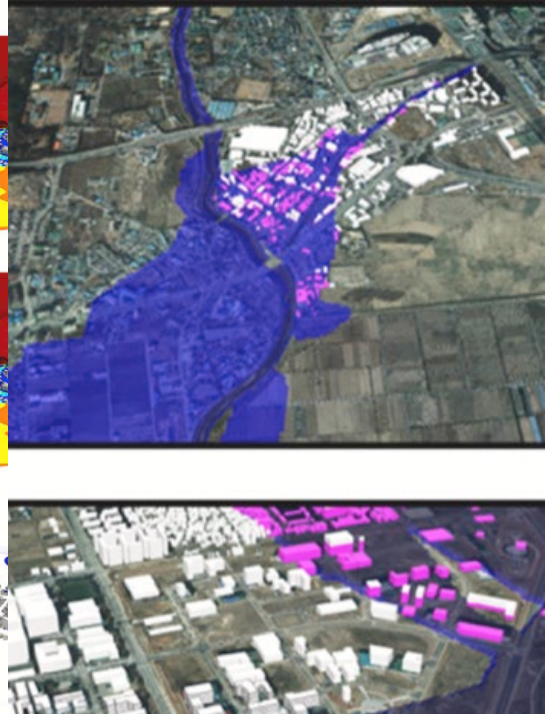
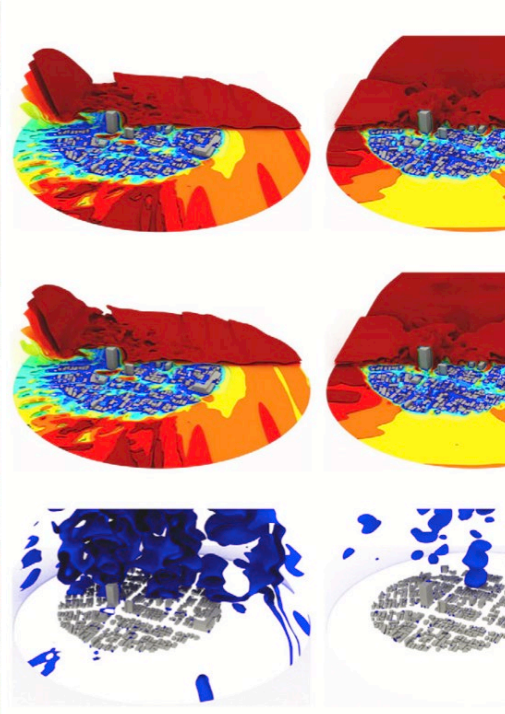
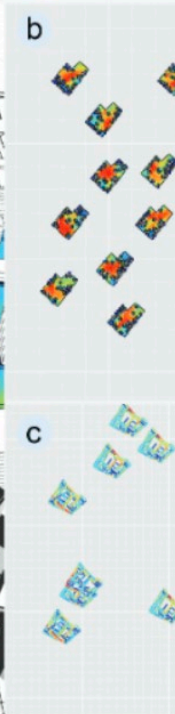
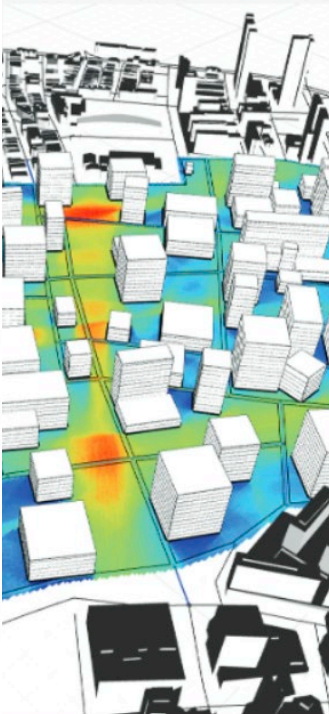
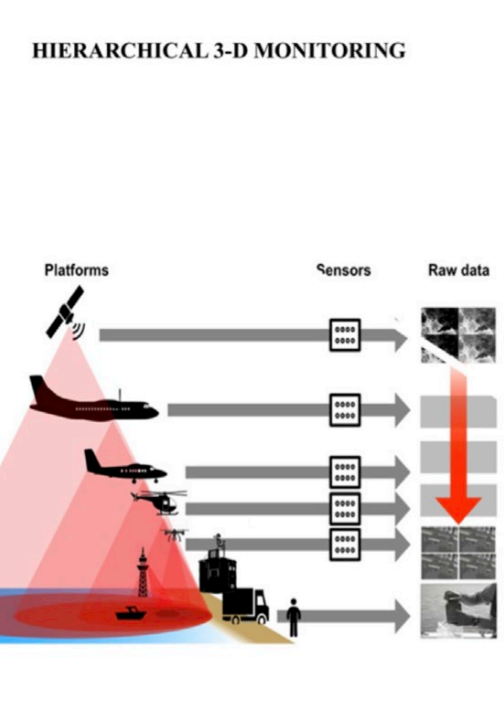


Machine learning for visualization

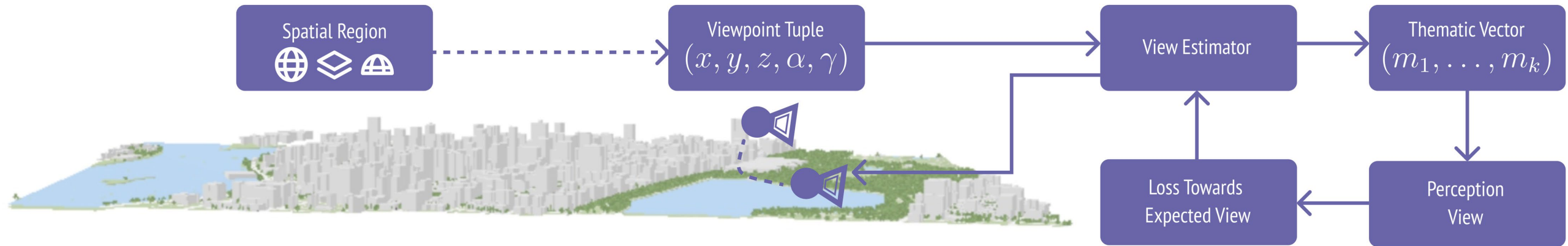




HIERARCHICAL 3-D MONITORING



Neural fields for 3D data exploration



Challenges

- ❖ Physical and thematic data modelling.

ML-based solutions:

- ❖ Neural modeling.

Challenges

- ❖ Physical and thematic data modelling.

ML-based solutions:

- ❖ Neural modeling.

Challenges

- ❖ Physical and thematic data modelling.
- ❖ Computational complexity.

ML-based solutions:

- ❖ Neural modeling.
- ❖ Direct queries.

Challenges

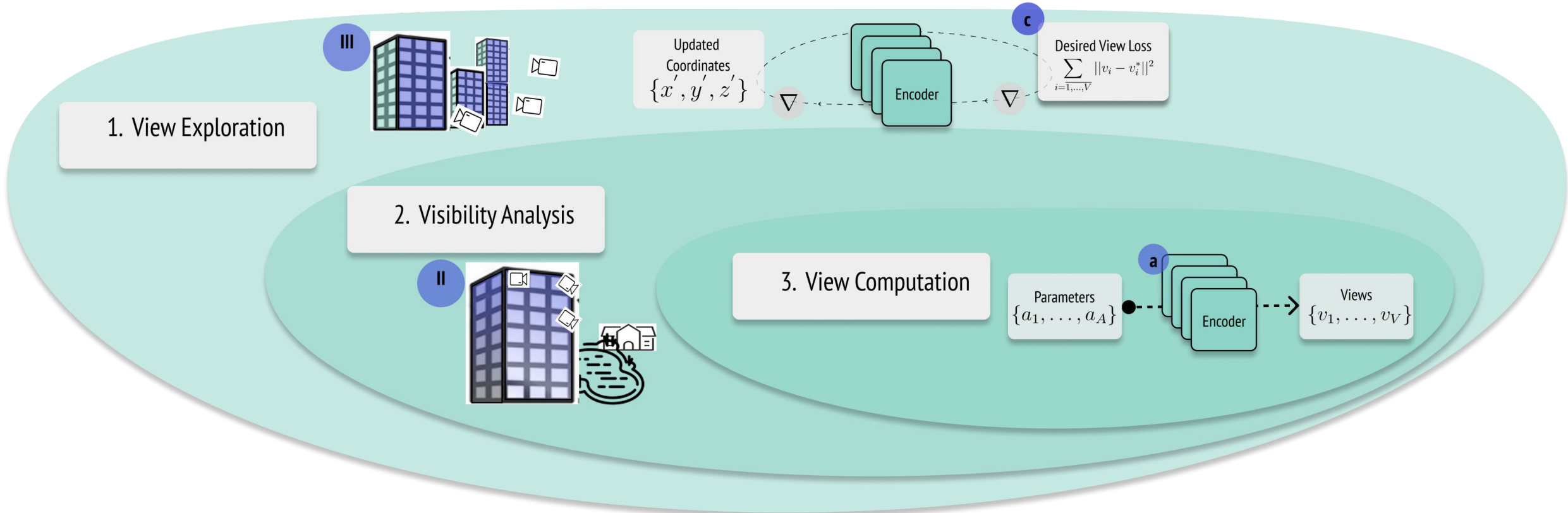
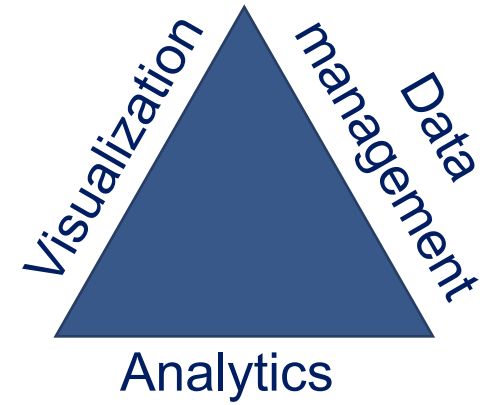
- ❖ Physical and thematic data modelling.
- ❖ Computational complexity.
- ❖ Viewpoint selection.

ML-based solutions:

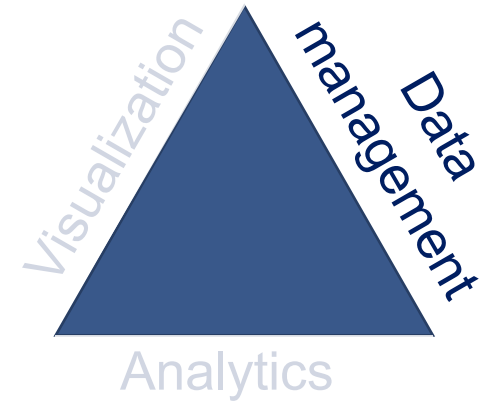
- ❖ Neural modeling.
- ❖ Direct queries.
- ❖ Inverse queries.



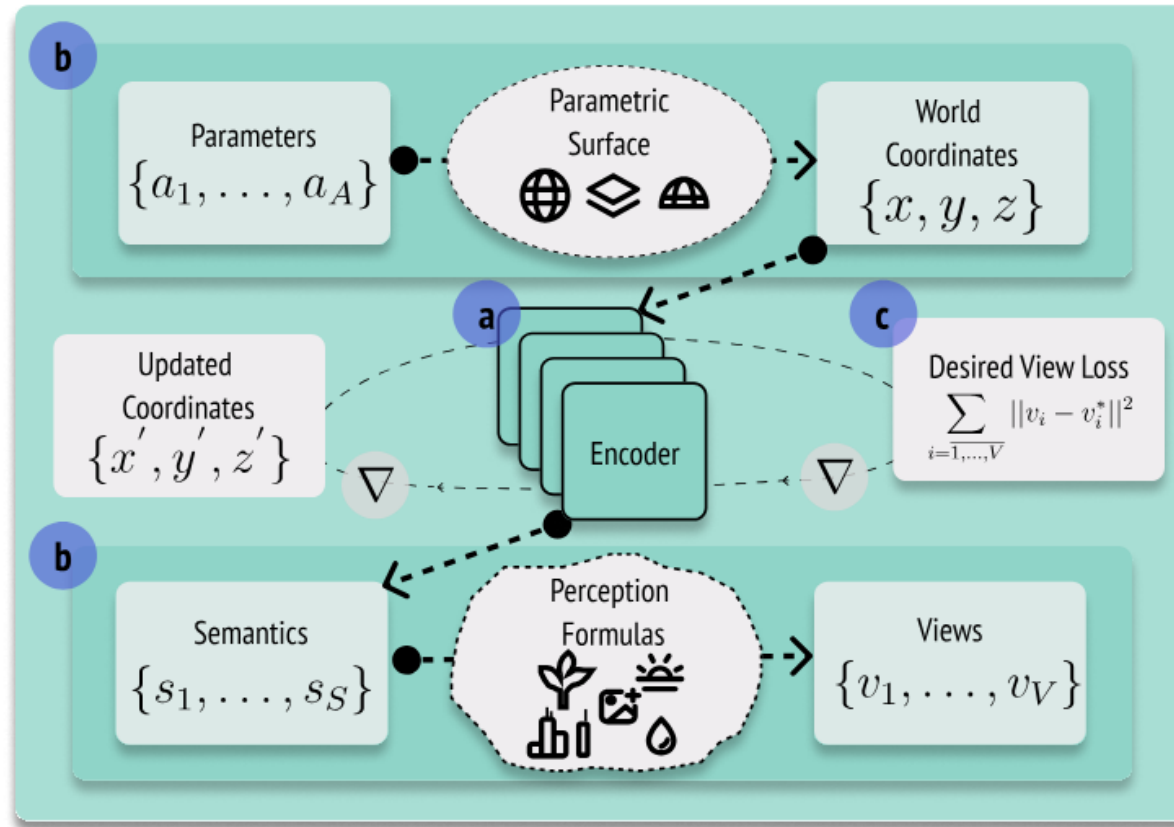
Visual analytics pipeline



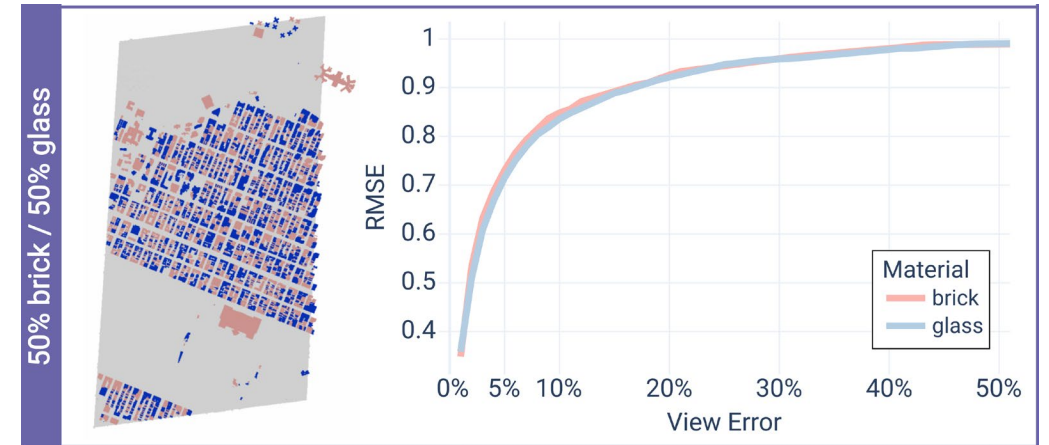
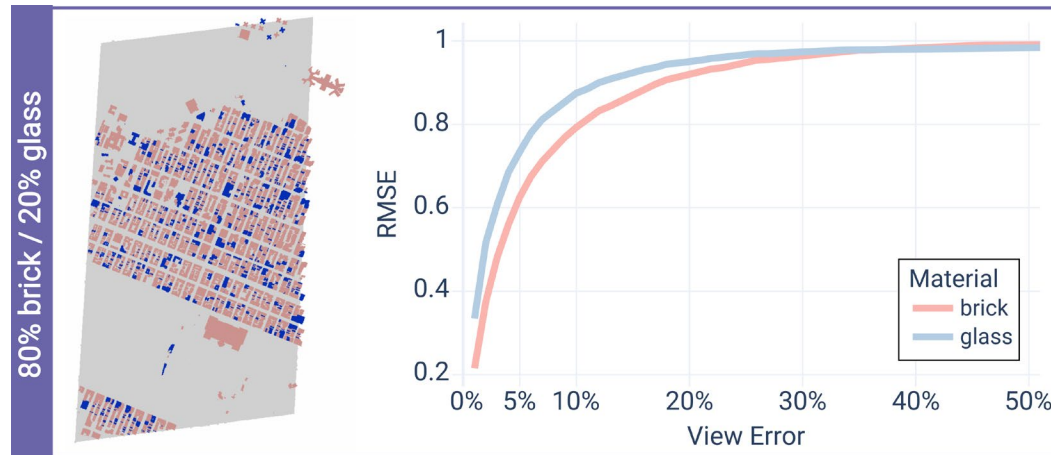
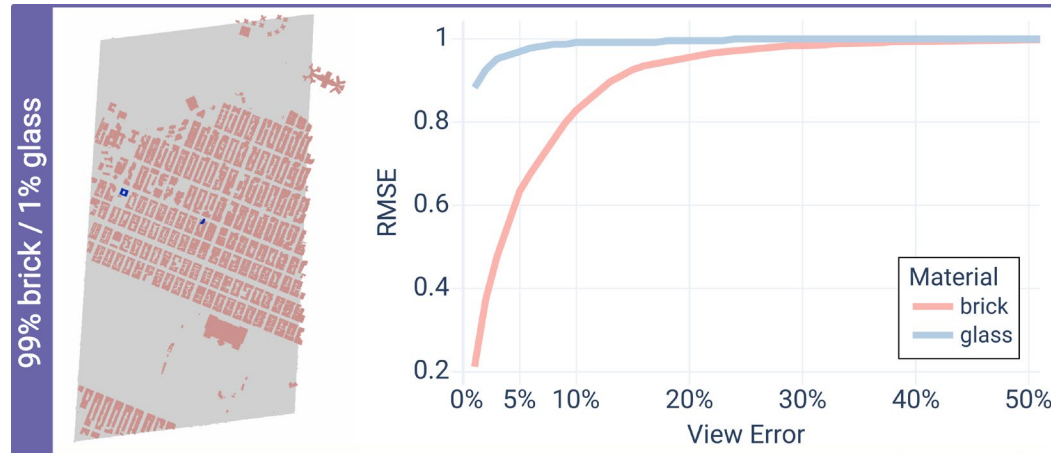
Visual analytics pipeline



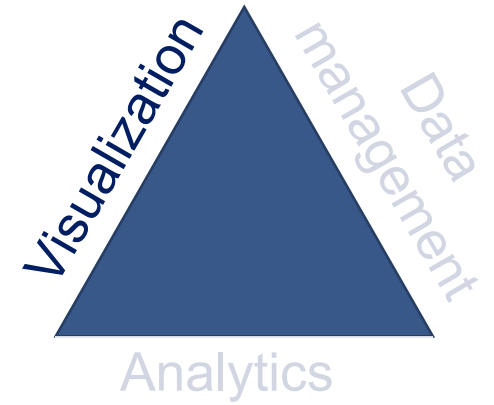
I. View Optimization



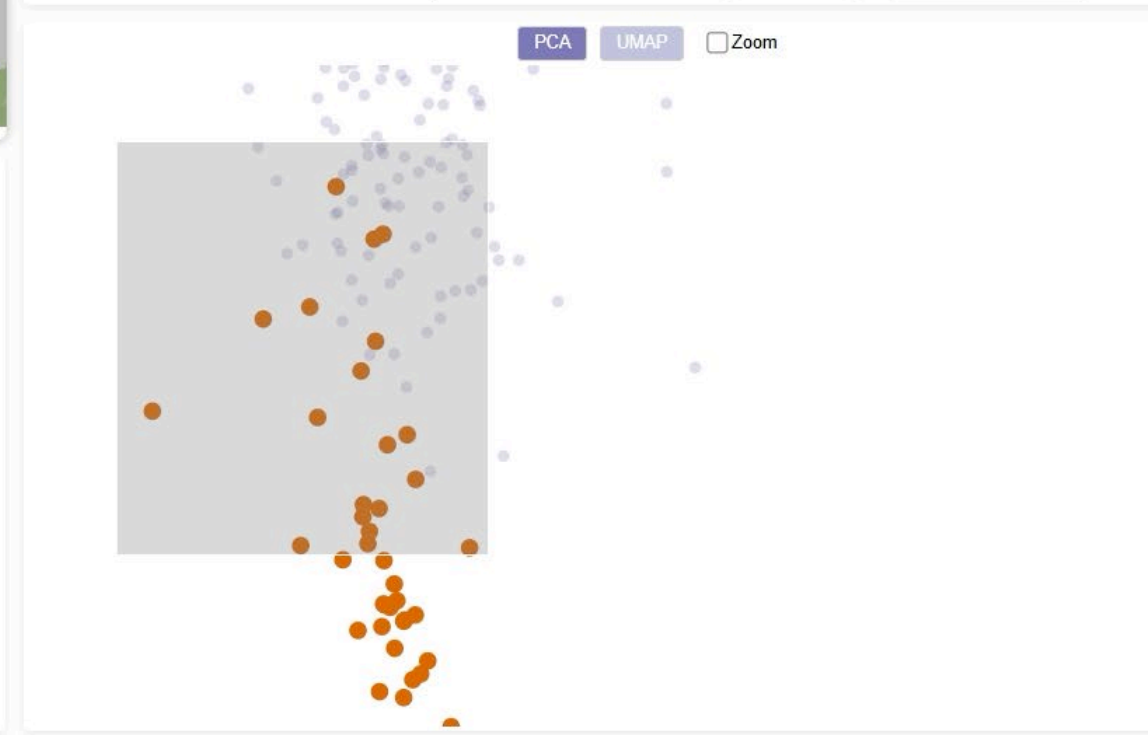
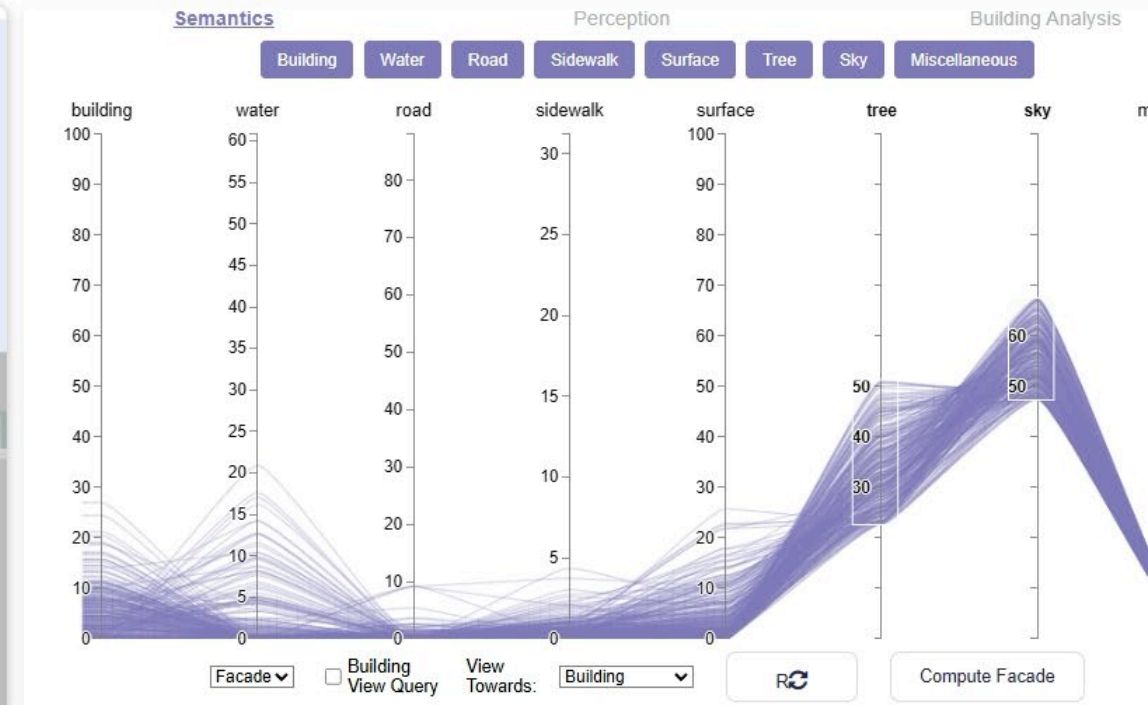
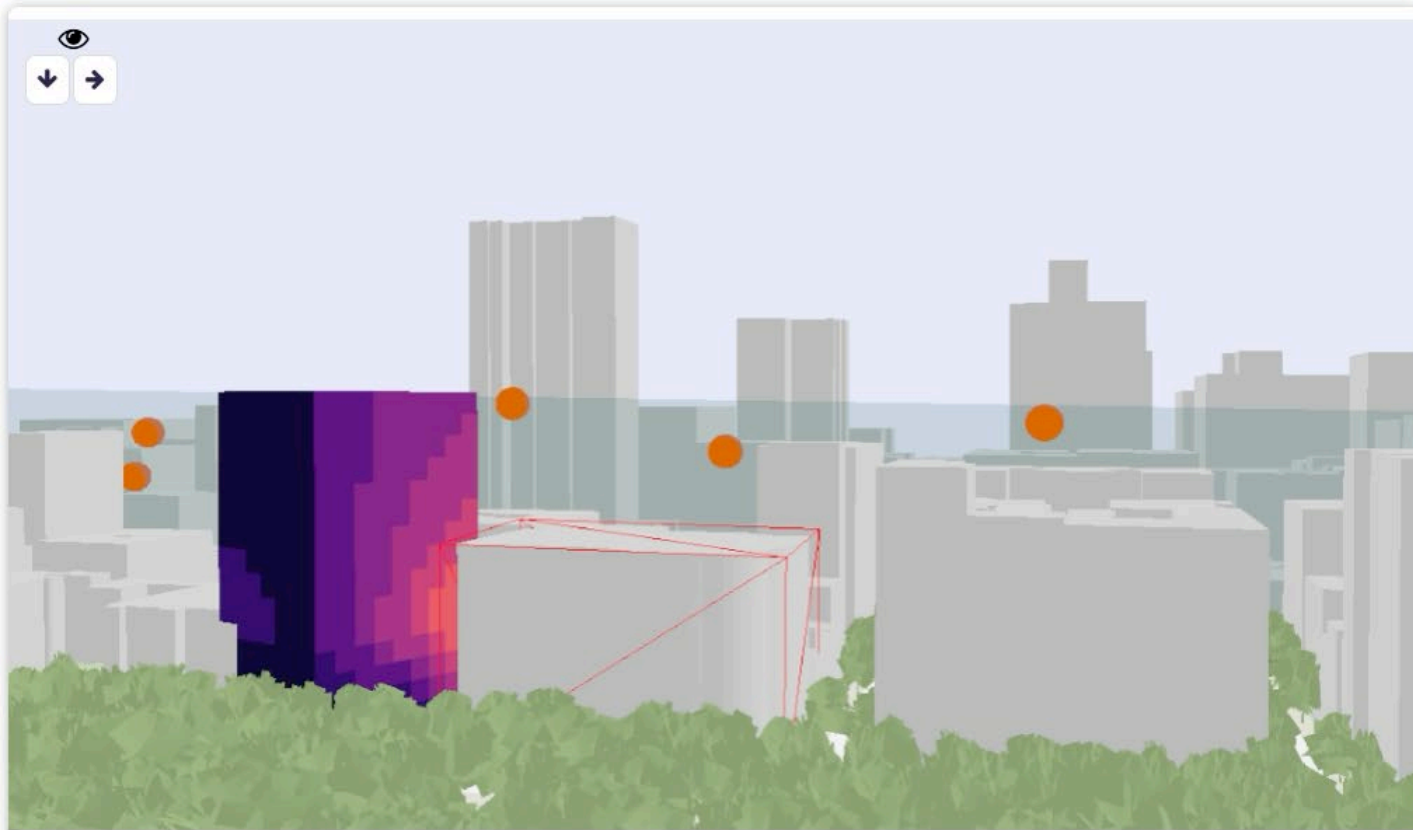
Visual analytics pipeline

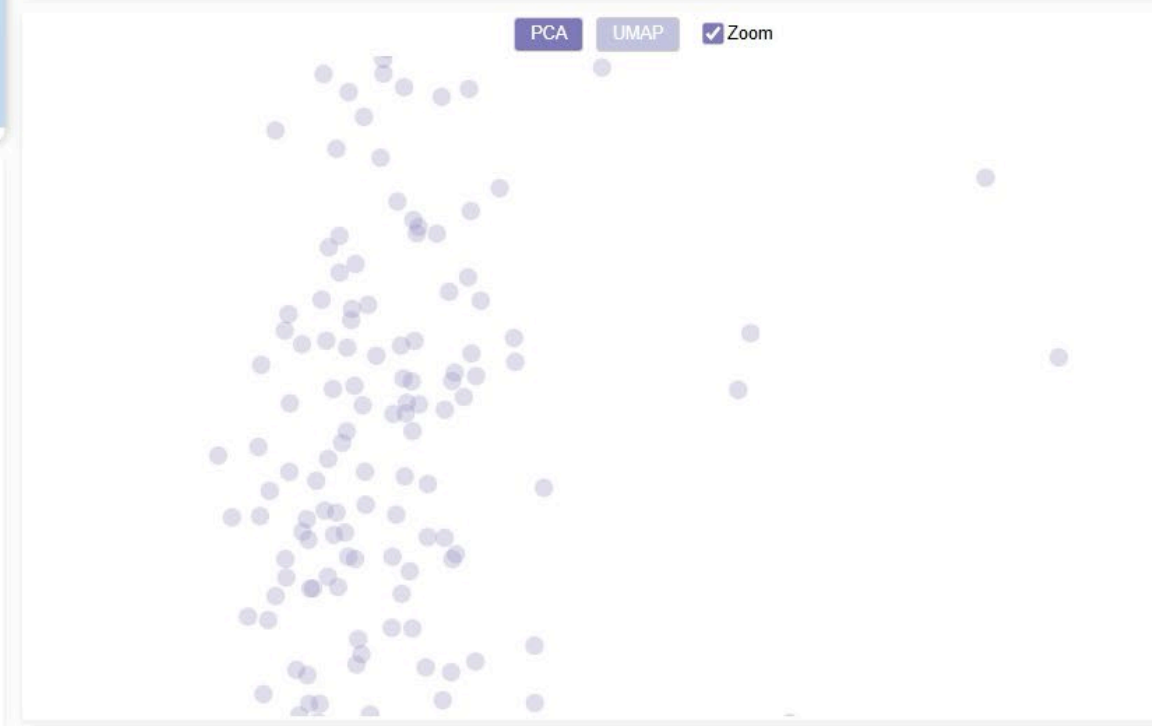
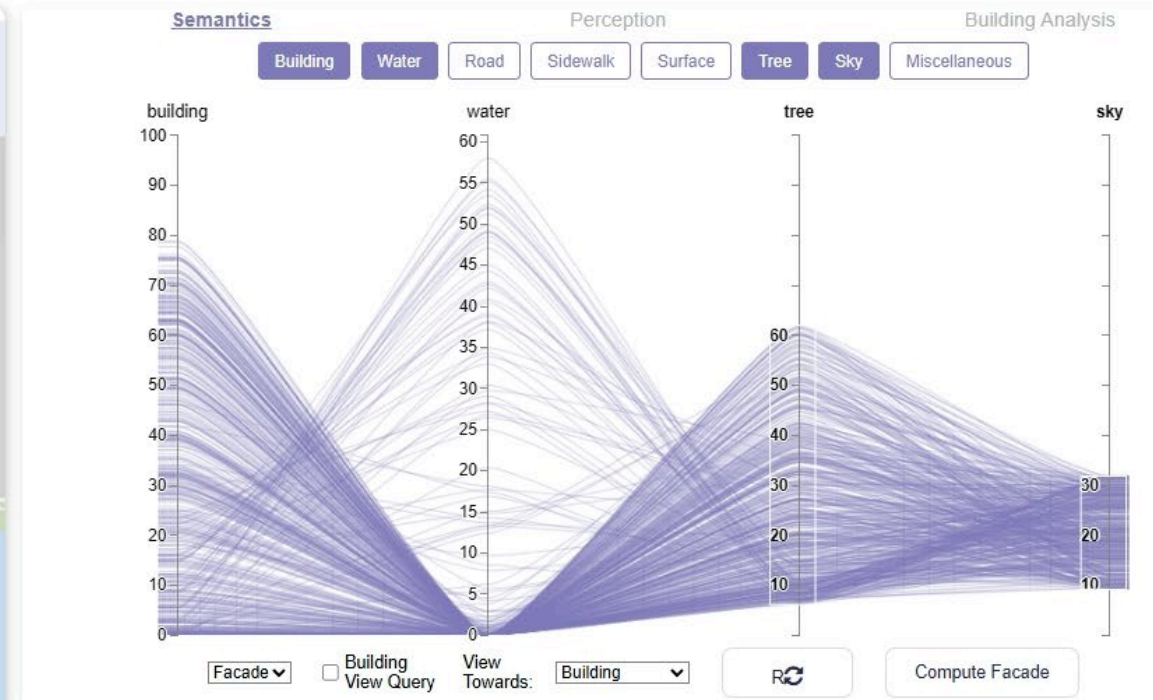
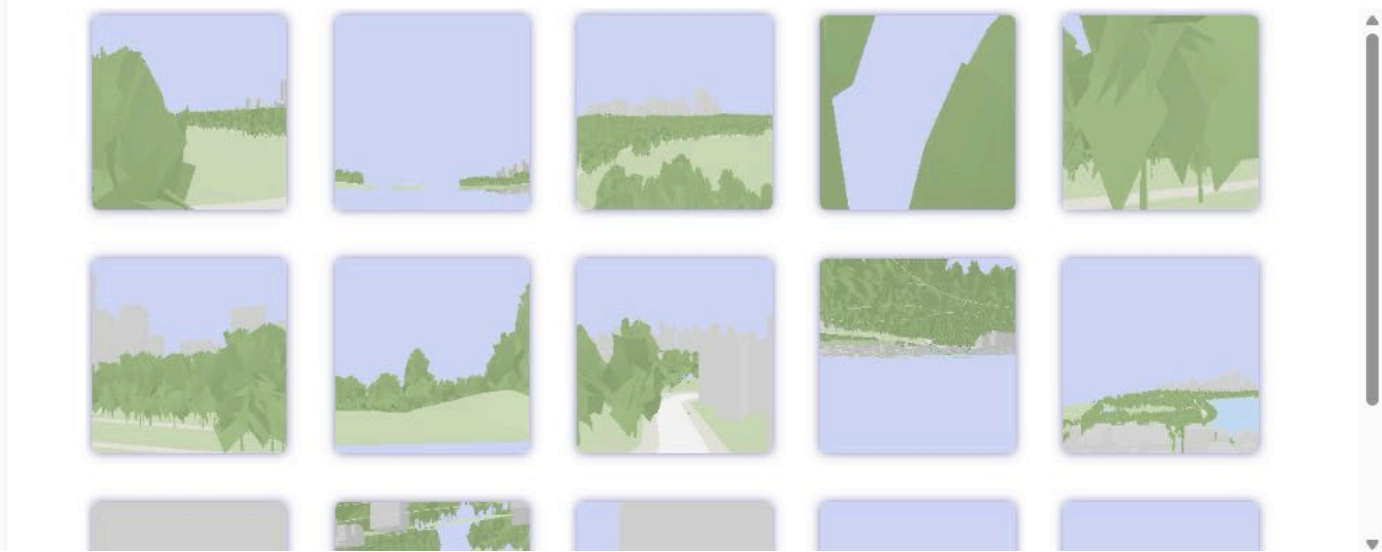


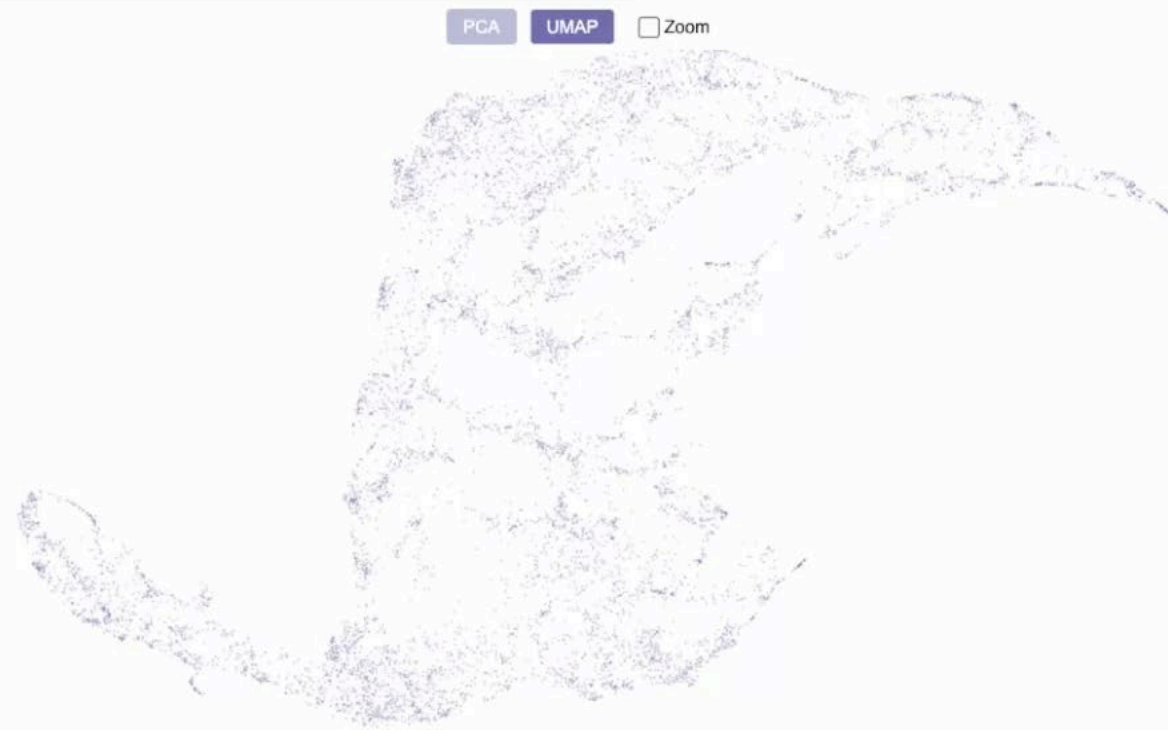
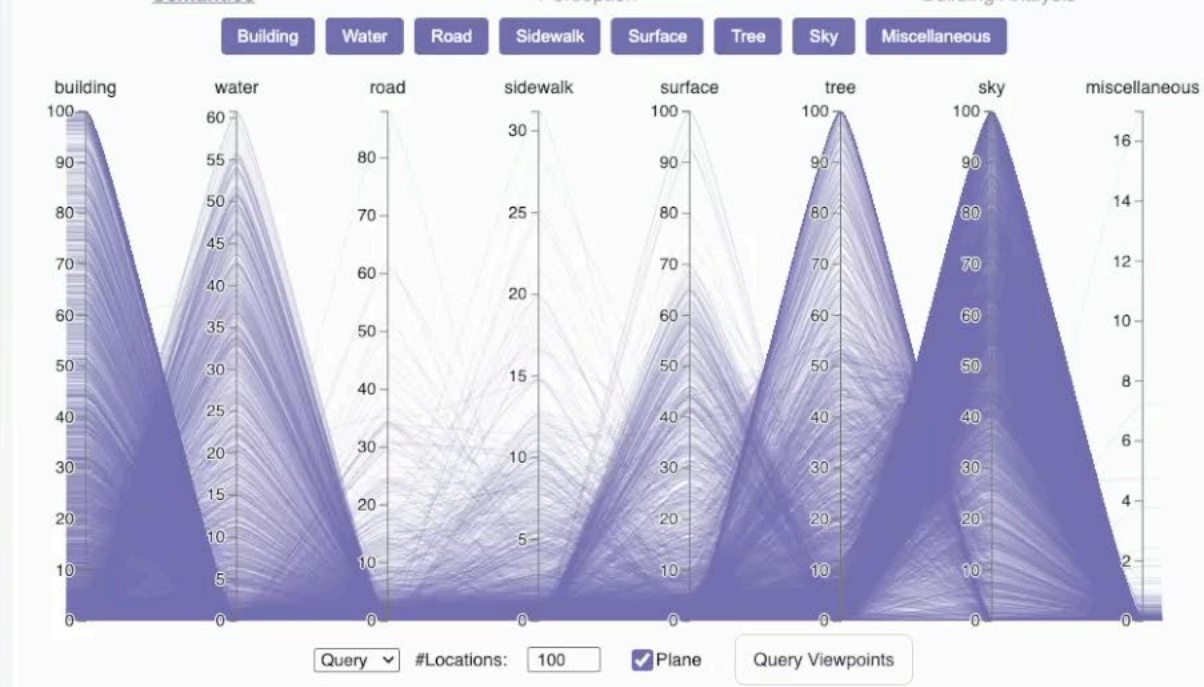
Visual analytics pipeline



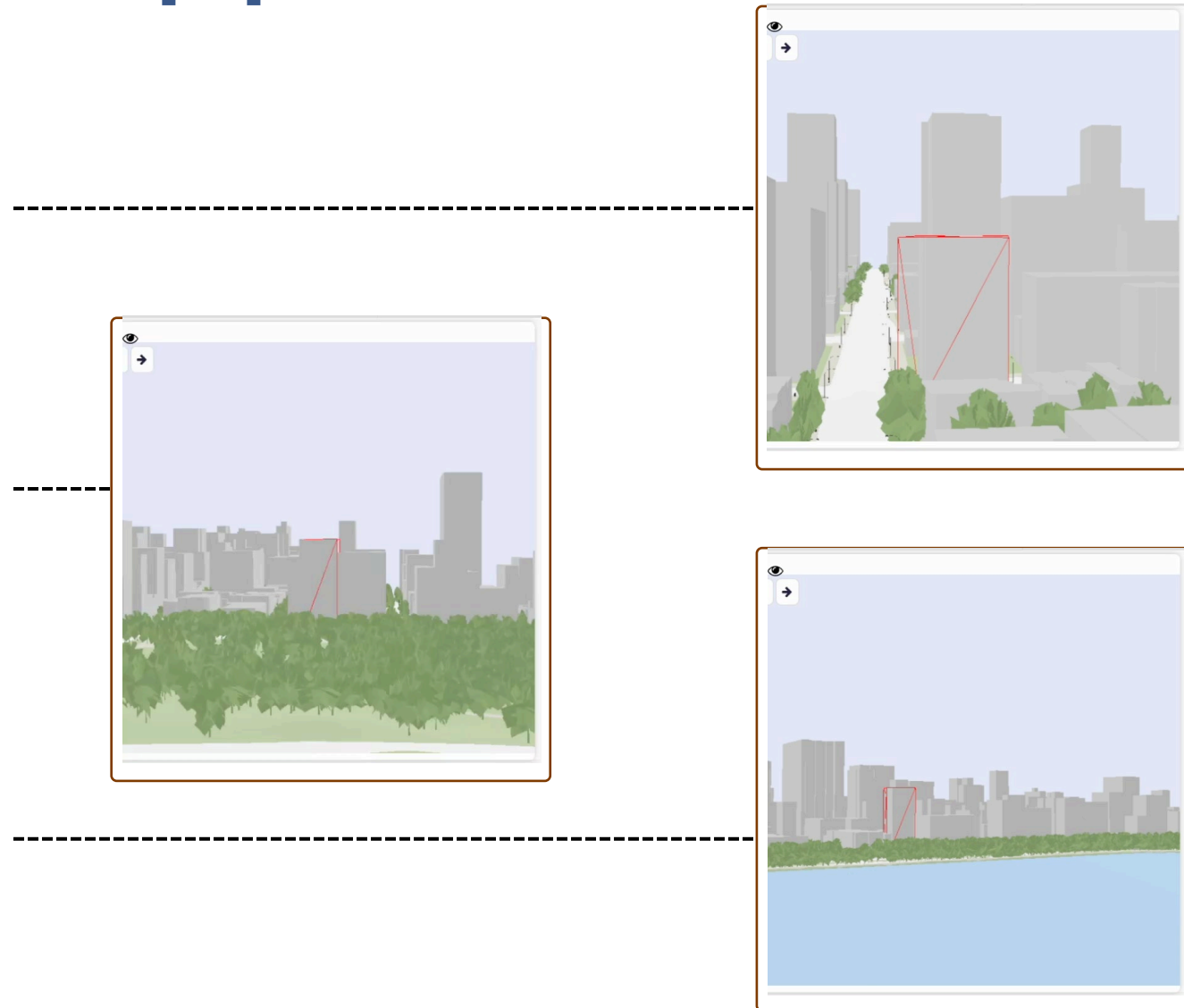
- Data Visualization







Visual analytics pipeline



Machine learning for visualization

NeuralCubes: Deep Representations for Visual Data Exploration

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Matthew Berger Dept. of Computer Science Vanderbilt University Nashville, TN, USA matthew.berger@vanderbilt.edu	Joshua A. Levine Dept. of Computer Science University of Arizona Tucson, AZ, USA josh@email.arizona.edu	Remco Chang Dept. of Computer Science Tufts University Medford, MA, USA remco@cs.tufts.edu	Carlos Scheidegger Dept. of Computer Science University of Arizona Tucson, AZ, USA cscheid@email.arizona.edu

Abstract—Visual exploration of large multi-dimensional datasets has seen tremendous progress in recent years, allowing users to express rich data queries that produce informative visual summaries, all in real time. Techniques based on data cubes are some of the most promising approaches. However, these techniques usually require a large memory footprint for large datasets. To tackle this problem, we present NeuralCubes: neural networks that predict results for aggregate queries, similar to data cubes. NeuralCubes learns a function that takes as input a given query, for instance, a geographic region and temporal interval, and outputs the result of the query. The learned function serves as a real-time, low-memory approximator for aggregation queries. Our models are small enough to be sent to the client side (e.g. the web browser for a web-based application) for evaluation, enabling data exploration of large datasets without database/network connection. We demonstrate the effectiveness of NeuralCubes through extensive experiments on a variety of datasets and discuss how NeuralCubes opens up opportunities for new types of visualization and interaction.

I. INTRODUCTION

Interactive visual exploration is becoming increasingly essential for making sense of large multi-dimensional datasets. It is common for datasets to have billions of data items that contain a variety of attributes of geographic, temporal, and categorical nature. Due to this size and complexity, issuing data queries in real-time is often not feasible as it will result in an unreasonable amount of latency. Instead, efficient data structures are often designed for real-time exploration, in lieu of making raw database queries. These data structures are pre-computed and optimized around queries frequently used by the visualization, such as performing data summaries [33], [45], ranking [38], and applying multivariate statistics [57].

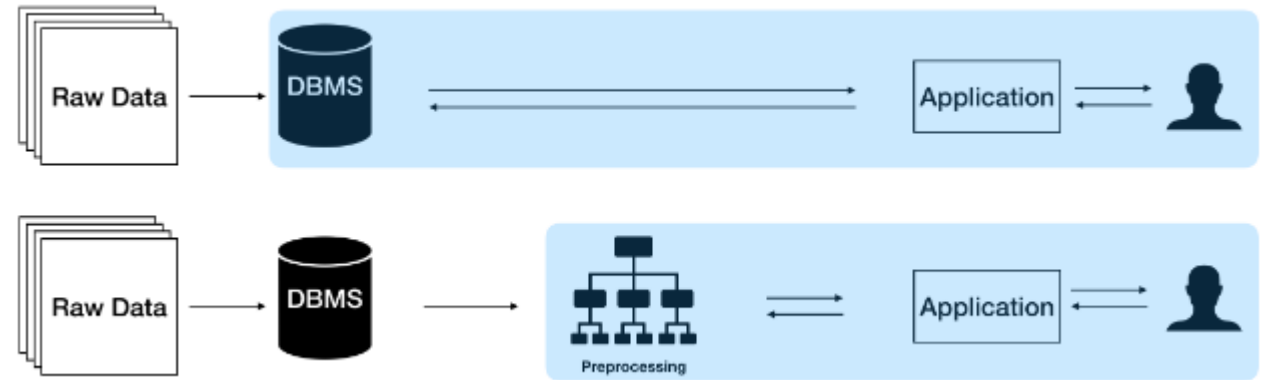
However, while these data structures are effective, they can still be prohibitively large as data size increases. Worse, when data complexity increases (i.e. in terms of the number of dimensions in the data), the sizes of many of these data structures grow exponentially. As a result, these data structures



Fig. 1. NeuralCubes frames approximate query processing over a database as a neural network, one that takes a query as input, e.g. selection of a temporal interval, and maps it to an aggregation result as output, e.g. a count of the number of records, enabling real-time visual exploration for large-scale data. In the figures above, the blue lines represent visualizations generated with data from the database and the orange lines are generated using NeuralCubes. Note that the two are nearly indistinguishable, but NeuralCubes uses only a fraction of the memory footprint as the database.

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Machine learning for visualization

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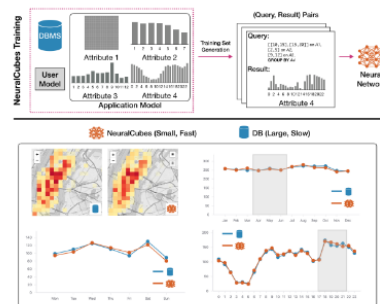
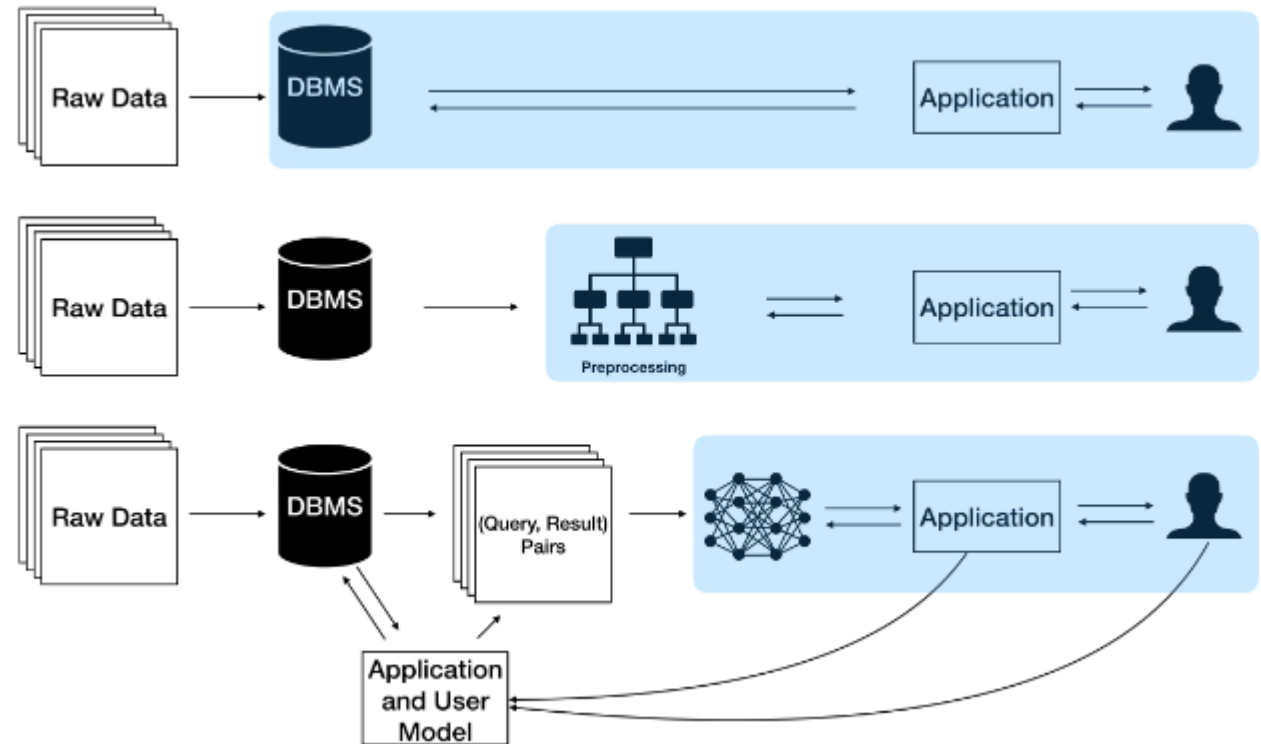


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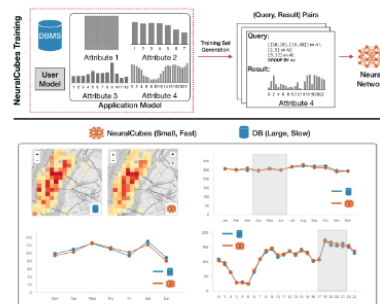
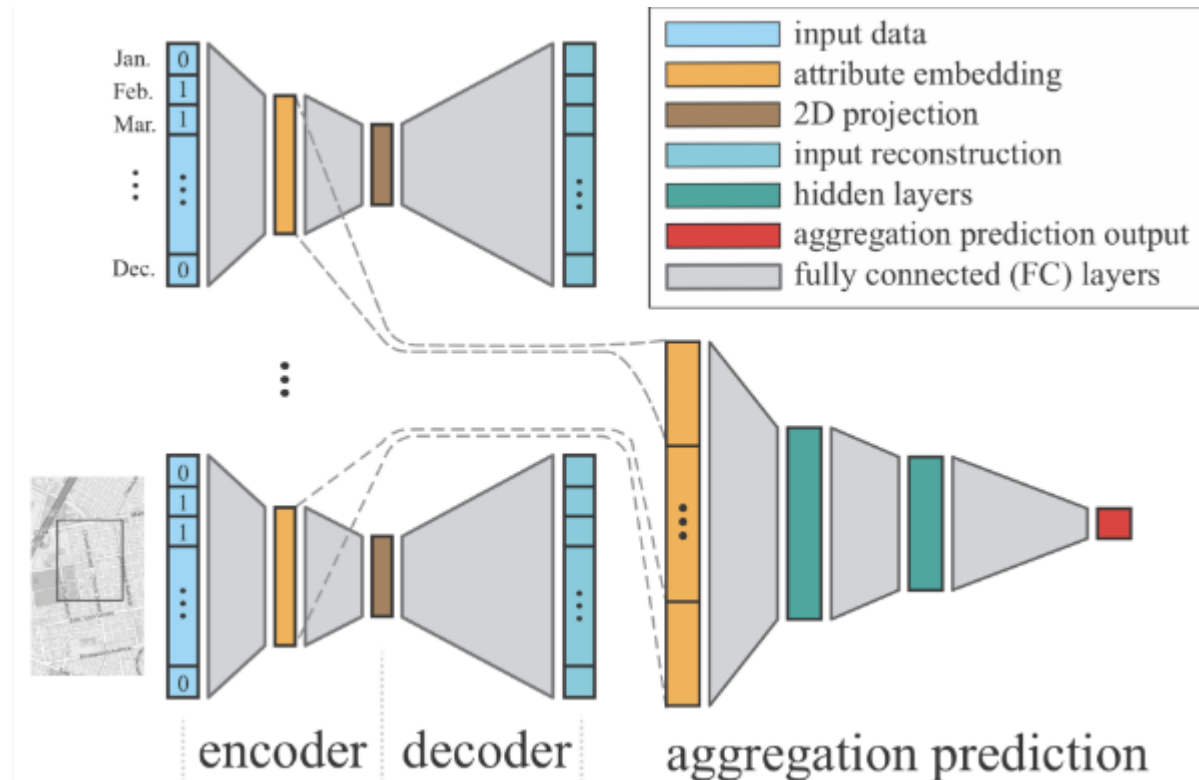


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Dimensionality reduction

- Input data may have thousands or millions of dimensions
 - E.g., text, images, videos, ...
- **Dimensionality reduction** aims to represent data with fewer dimensions.
 - Easier learning: fewer parameters.
 - Discover “intrinsic dimensionality” of data.
 - High dimensional data that is truly lower dimensional.
 - Noise reduction.
 - Visualization: show high-dimensional data in a visual space (2D or 3D).
 - How are the points spread?
 - Are there well-defined groups of similar instances?



Technique	Taxonomy												
	Data Types					Linearity	Supervision	Multi-level	Locality	Steerability	Stability	OOC	Comp.
	Di	Or	Ca	Ne	Ct								
PCA [65]			✓			✓					•	✓	$O(p^3)$
LDA [50]			✓			✓	✓					✓	$O(np^2+p^3)$
Classical MDS [155]	✓		✓	•		✓	•			•	•		$O(n^3)$
Kruskal [79]	✓	✓	✓	•			•			•		•	$O(in^2)$
NLM [132]	✓		✓	•			•			•		•	$O(in^2)$
MCA [17]			•		✓								$O(n^3)$
Smacof [42]	✓	✓	✓	•			•					•	$O(in^2)$
SOM [126]			✓									✓	$O(l^2np+l^2)$
FastMap [48]	✓		✓	•			•					•	$O(n)$
Chalmers [32]	✓		✓	•			•	•		•	◦	✓	$O(in)$
GTM [22]			✓									•	$O((lnp)^3+m^3)$
Pekalska [120]	✓		✓	•		✓	•						$O(r^3+rn)$
CCA [43]	✓		✓	•			•					✓	$O(l^2)$
LLE [129]	✓		✓	•			•		✓			•	$O(n^3)$
Isomap [153]	✓		✓	✓			•					•	$O(n^3)$
Lapl. Eigenmaps [15]	✓		✓	✓			•		✓			•	$O(n^3)$
Force-Directed [152]	✓		✓				•	•				✓	$O(in^2)$
LTSA [180]			✓			•			✓				$O(n^3)$
MVU [169]	✓		✓	•			•		✓			•	$O(n^3)$
LSP [117]	✓		✓	✓			•	•	✓	✓	◦		$O(n^3)$
SNE [64]	✓		✓	•			•	•		◦		•	$O(in^2)$
PLMP [118]			✓			•	•	•	✓	✓	◦		$O(n^3)$
LAMP [73]			✓				•	•	✓	✓	✓	✓	$O(pn)$
RBF-MP [2]	✓		✓	•			•			✓	◦	•	$O(r^3+n)$
LoCH [47]	✓		✓	•			•		✓	✓		✓	$O(n\sqrt{(n)})$
ClassiMap [90]	✓		✓	•			✓					•	$O(in^2)$
Kelp [13]	✓		✓	•		•	•		✓	✓	◦	•	$O(r^3)$

Technique	Taxonomy												
	Data Types					Linearity	Supervision	Multi-level	Locality	Steerability	Stability	OOC	Comp.
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LDA [50]						✓	✓					✓	$O(np^2+p^3)$
Classical MDS [155]						✓	•				•	•	
Kruskal [79]	✓	✓	✓	•			•			•		•	$O(in^2)$
NLM [132]	✓		✓	•			•			•		•	$O(in^2)$
MCA [17]			•		✓								$O(n^3)$
Smacof [42]	✓	✓	✓	•			•					•	$O(in^2)$
SOM [126]			✓									✓	$O(l^2np+l^2)$
FastMap [48]	✓		✓	•			•					•	$O(n)$
Chalmers [32]	✓		✓	•			•	•		•	◦	✓	$O(in)$
GTM [22]			✓									•	$O(lnp^3+m^3)$
Pekalska [120]	✓		✓	•		✓	•						$O(r^3+rn)$
CCA [43]	✓		✓	•			•					✓	$O(l^2)$
LLE [129]	✓		✓	•			•		✓			•	$O(n^3)$
Isomap [153]	✓		✓	✓			•					•	$O(n^3)$
Lapl. Eigenmaps [15]	✓		✓	✓			•		✓			•	$O(n^3)$
Force-Directed [152]	✓		✓				•	•				✓	$O(in^2)$
LTSA [180]			✓			•			✓				$O(n^3)$
MVU [169]	✓		✓	•			•		✓			•	$O(n^3)$
LSP [117]	✓		✓	✓			•	•	✓	✓	◦		$O(n^3)$
SNE [64]	✓		✓	•			•	•		◦		•	$O(in^2)$
PLMP [118]			✓			•	•	•	✓	✓	◦		$O(n^3)$
LAMP [73]			✓				•	•	✓	✓	✓	✓	$O(pn)$
RBF-MP [2]	✓		✓	•			•			✓	◦	•	$O(r^3+n)$
LoCH [47]	✓		✓	•			•		✓	✓		✓	$O(n\sqrt{n})$
ClassiMap [90]	✓		✓	•			✓					•	$O(in^2)$
Kelp [13]	✓		✓	•		•	•		✓	✓	◦	•	$O(r^3)$

Technique	Taxonomy												
	Data Types					Linearity	Supervision	Multi-level	Locality	Steerability	Stability	OOC	Comp.
	Di	Or	Ca	Ne	Ct								
PCA [65]	Core Techniques					✓					•	✓	$O(p^3)$
LDA [50]						✓	✓					✓	$O(np^2+p^3)$
Classical MDS [155]						✓	•			•	•		$O(n^3)$
Kruskal [79]	✓	✓	✓	•			•			•		•	$O(in^2)$
NLM [132]	✓		✓	•			•			•		•	$O(in^2)$
MCA [17]			•		✓								$O(n^3)$
Smacof [42]	✓	✓	✓	•			•					•	$O(in^2)$
SOM [126]			✓									✓	$O(l^2np+l^2)$
FastMap [48]	✓		✓	•			•					•	$O(n)$
Chalmers [32]	✓		✓	•		Core Method	Variants					Many Variants	$O(in)$
GTM [22]			✓			PCA [65]	see [181] for variants; iPCA [71], PCP [182]						$O((lnp)^3+m^3)$
Pekalska [120]	✓		✓	•		LDA [50]	see [61] for variants						$O(r^3+rn)$
CCA [43]	✓		✓	•		Classical MDS [155]	L-MDS [144], Pivot-MDS [24], CFMDS [113]						✓ $O(l^2)$
LLE [129]	✓		✓	•		MCA [17]	see [154] for variants						• $O(n^3)$
Isomap [153]	✓		✓	✓		SOM [126]	see [150] for variants						• $O(n^3)$
Lapl. Eigenmaps [15]	✓		✓	✓		Chalmers [32]	Glimmer [70]						• $O(n^3)$
Force-Directed [152]	✓		✓			LLE [129]	SLLE [178]						✓ $O(in^2)$
LTSA [180]			✓			Isomap [153]	S-Isomap [53]						$O(n^3)$
MVU [169]	✓		✓	•		LTSA [180]	LLTSA [179]						• $O(n^3)$
LSP [117]	✓		✓	✓		MVU [169]	MUHSIC [147]						✓ $O(n^3)$
SNE [64]	✓		✓	•		LSP [117]	PLP [115], E-LSP [33], Hipp [116]						◦ $O(in^2)$
PLMP [118]			✓			SNE [64]	NeRV [162], t-SNE [158], DS t-SNE [77], Q-SNE [69]						✓ $O(n^3)$
LAMP [73]			✓				A-tSNE [122], H-SNE [121], BH-tSNE [157]						✓ $O(pn)$
RBF-MP [2]	✓		✓	•			•			✓	◦		•
LoCH [47]	✓		✓	•			•		✓	✓		✓	$O(n\sqrt{n})$
ClassiMap [90]	✓		✓	•			✓					•	$O(in^2)$
Kelp [13]	✓		✓	•		•	•		✓	✓	◦	•	$O(r^3)$

Dimensionality reduction

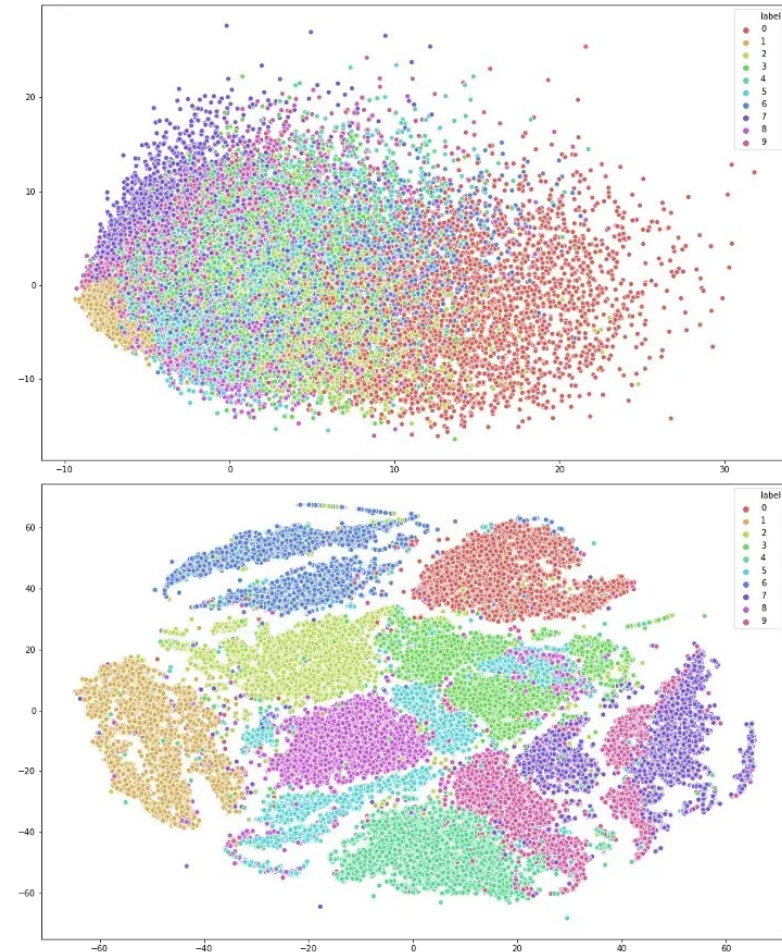
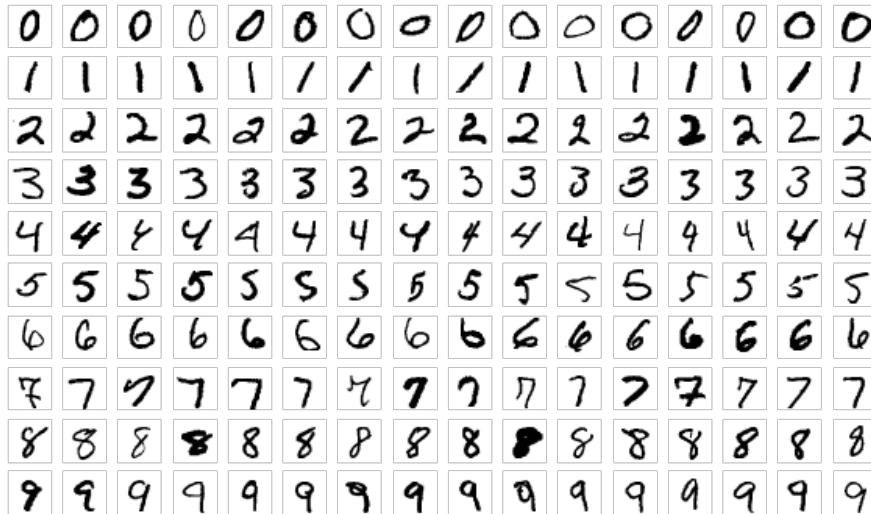
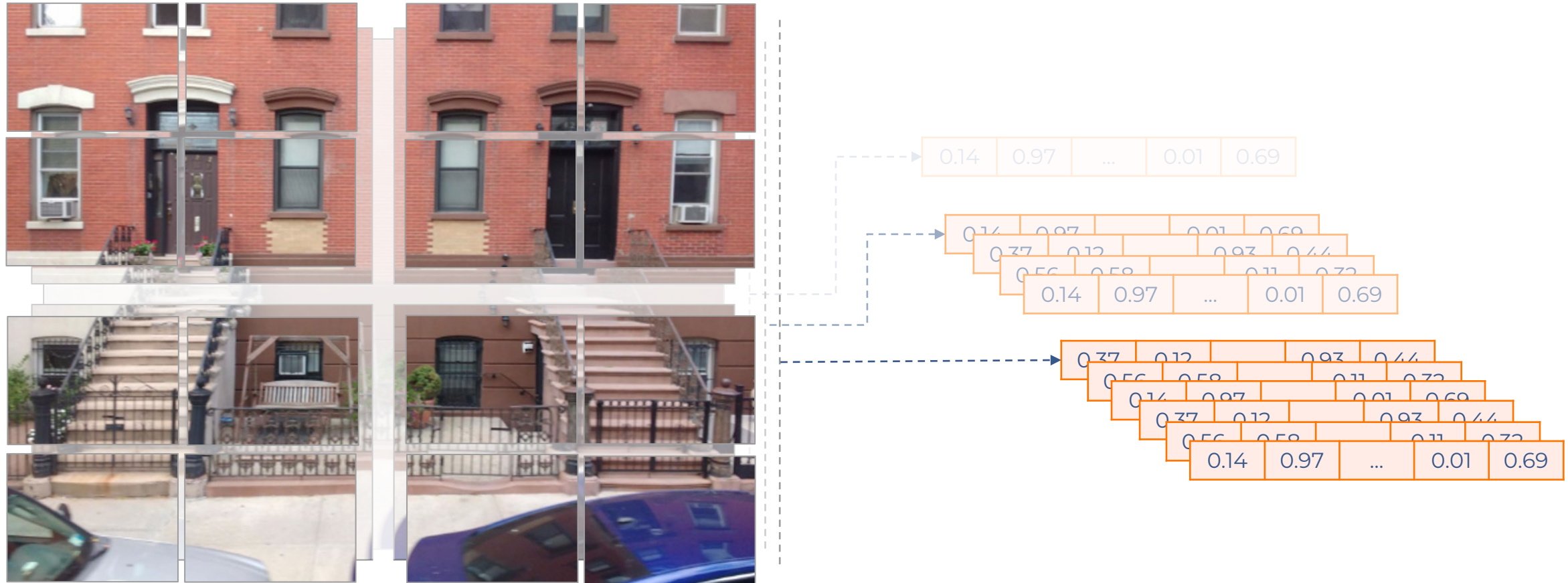
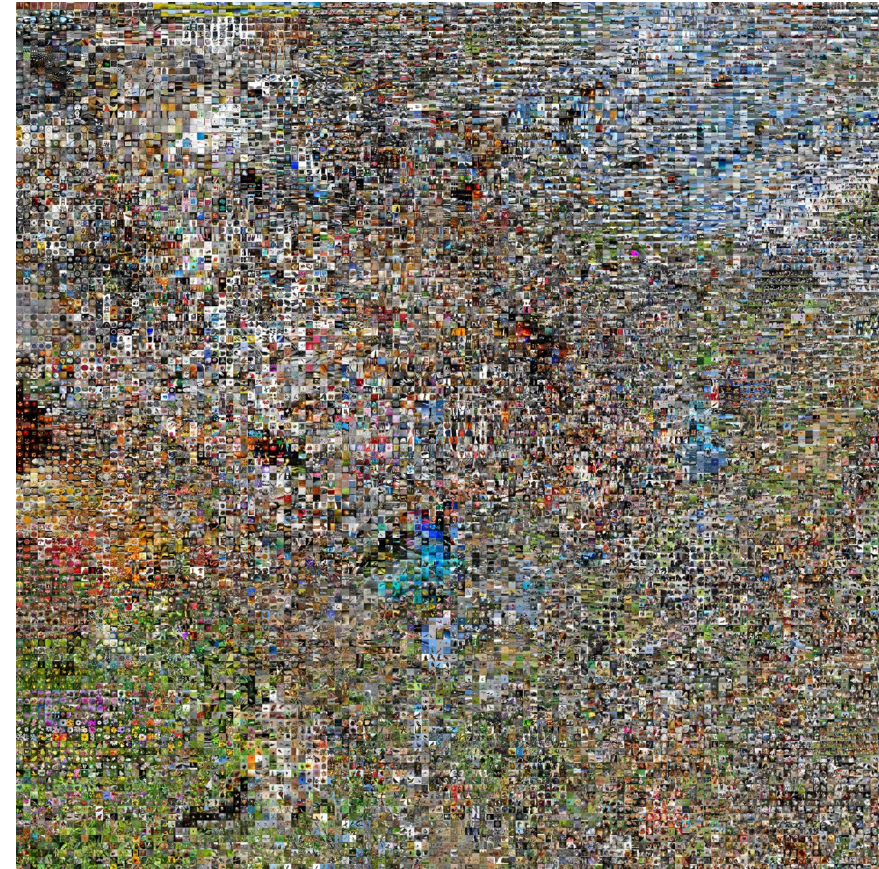


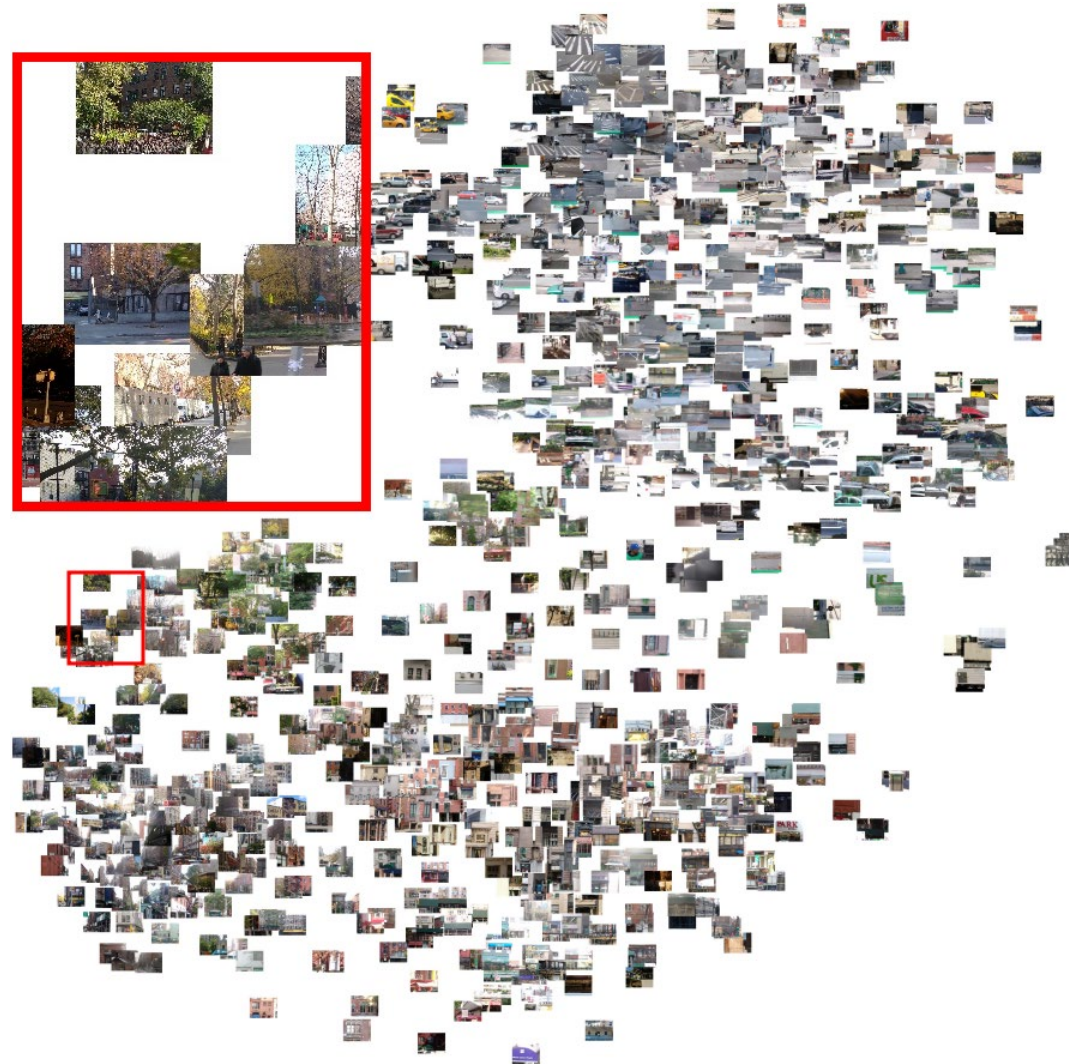
Image embeddings



Dimensionality reduction



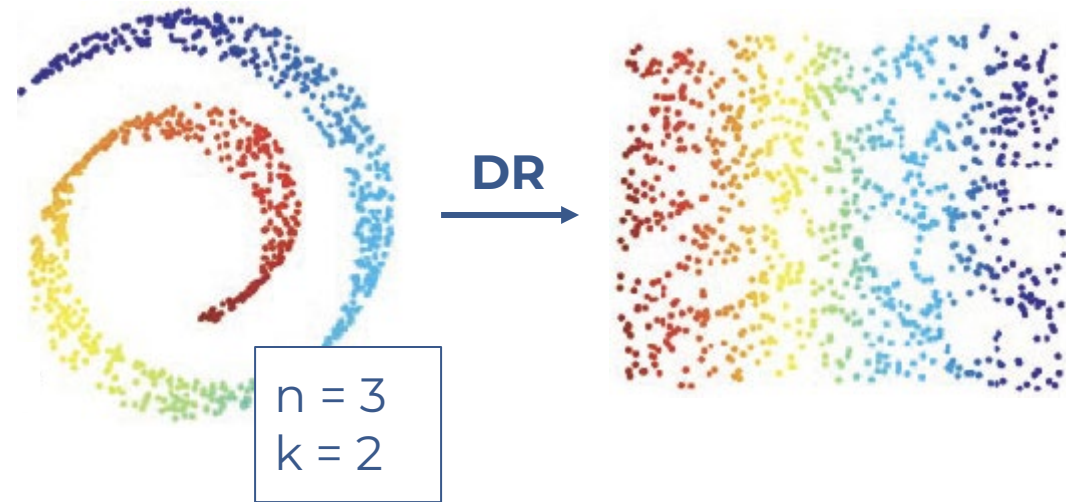
Dimensionality reduction



Dimensionality reduction

- High-dimensional data typically has an intrinsic dimension smaller than the space in which it is embedded.

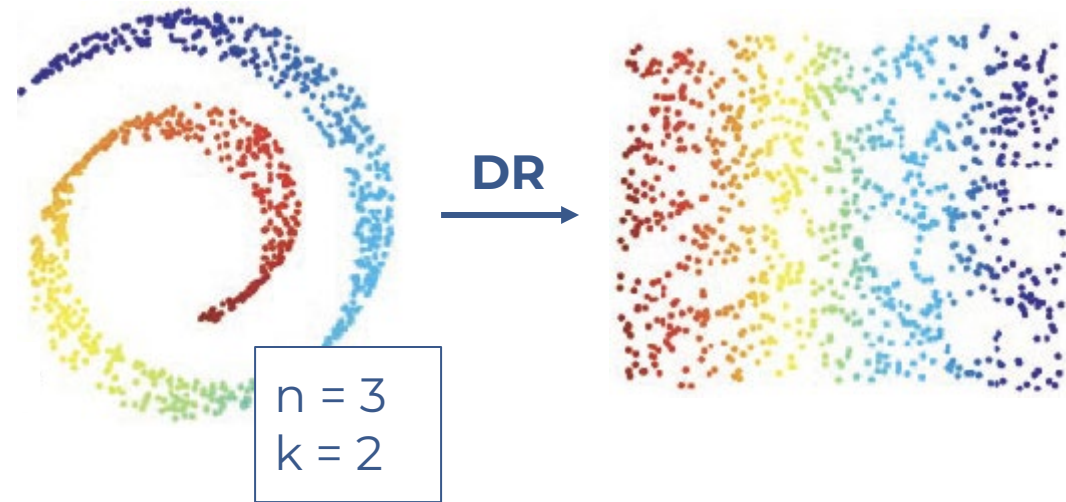
DR comprises a category of unsupervised methods that aim to capture, to a certain extent, the intrinsic dimension of the data.



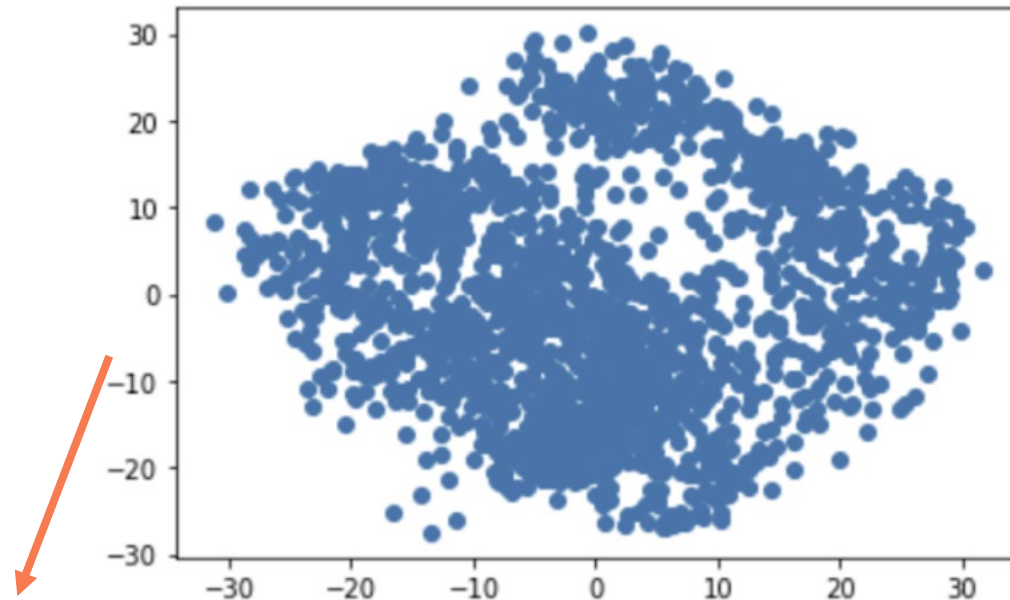
Dimensionality reduction

- High-dimensional data typically has an intrinsic dimension smaller than the space in which it is embedded.

The goal is to perform the dimensionality reduction preserving some similarity information among data instances.



Dimensionality reduction



It is not possible to interpret the meaning of the axes.

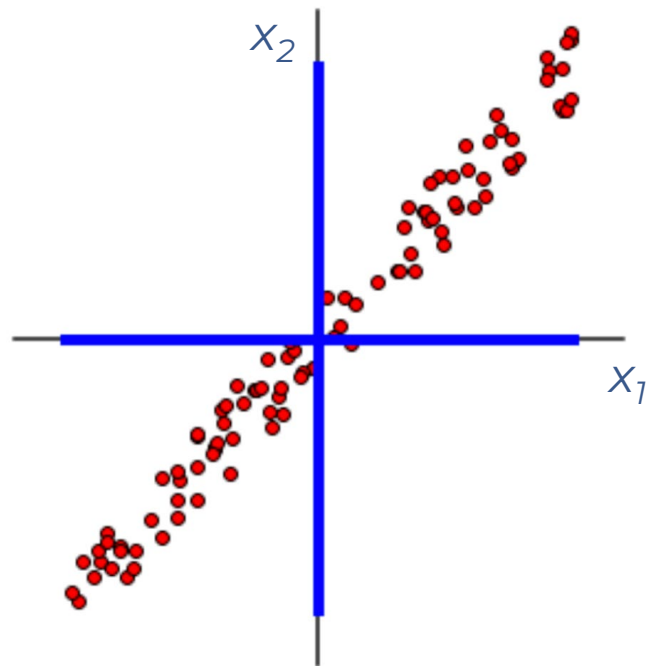
Dimensionality reduction techniques embed data in a latent space where the point coordinates (axes) do not have a semantic meaning.

$$\Omega \rightarrow \mathbb{R}^p$$

- Dimensionality reduction is a mapping between two spaces.
- For visualization purposes, usually $p = 2$ or $p = 3$.
- Principal Component Analysis (PCA).
- Multidimensional Scaling (MDS).
- t-distributed Stochastic Neighbor Embedding (t-SNE).
- Uniform Manifold Approximation and Projection (UMAP).
- Local Affine Multidimensional Projection (LAMP).

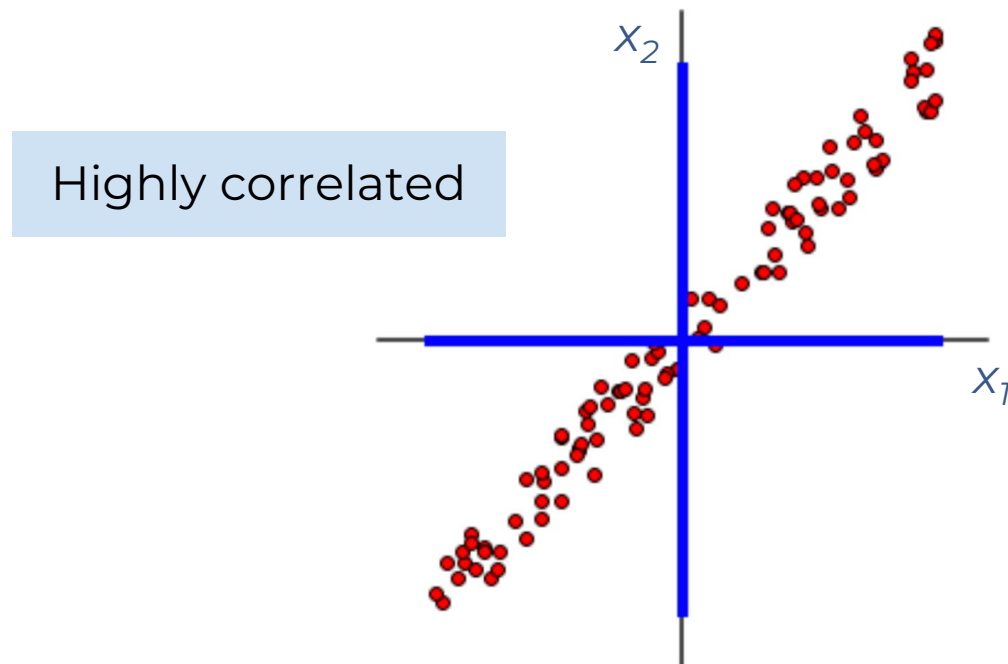
Principal Component Analysis (PCA)

- Given data matrix $\mathbf{X}$, PCA finds a new basis to represent the data so that the coordinates of the points in the new basis are uncorrelated.



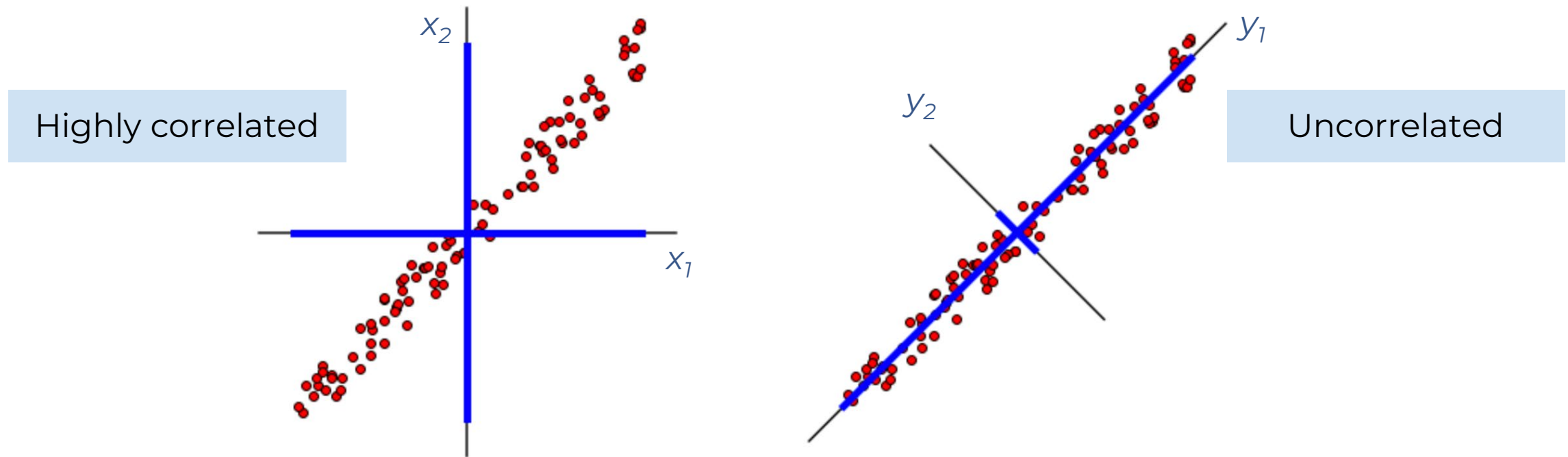
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Principal Component Analysis (PCA)

- Given data matrix $\mathbf{X}$, PCA finds a new basis to represent the data so that the coordinates of the points in the new basis are uncorrelated.



Principal Component Analysis (PCA)

- Dimensionality reduction method that transforms data from a high-dimensional space into a new lower dimensional space.
- Steps:
 1. Standardization of the values in the dataset to zero mean and unit variance.
 2. Create the covariance matrix.
 3. Calculate the Eigen values and Eigen vectors from the covariance matrix.
 4. Extract the principal components.
 5. Project the data points in the direction of the principal components.

Principal Component Analysis (PCA)

Given a dataset of n points $a_1, \dots, a_n \in \mathbb{R}^d$:

- $d = 1$
 - Mean

$$\mu = \frac{1}{n} \sum_{i=1}^n a_i \in \mathbb{R}$$

- Variance

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (a_i - \mu)^2 \in \mathbb{R}$$

Principal Component Analysis (PCA)

Given a dataset of n points $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{R}^d$:

- $d = 1$
 - Mean
- $d \geq 2$
 - Mean

$$\mu = \frac{1}{n} \sum_{i=1}^n a_i \in \mathbb{R}$$

- Variance

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (a_i - \mu)^2 \in \mathbb{R}$$

$$\boldsymbol{\mu} = \frac{1}{n} \sum_{i=1}^n \mathbf{a}_i \in \mathbb{R}^d$$

- Covariance matrix

$$\mathbf{S} = \frac{1}{n} \sum_{i=1}^n (\mathbf{a}_i - \boldsymbol{\mu})(\mathbf{a}_i - \boldsymbol{\mu})^T \in \mathbb{R}^{d \times d}$$

$$= \frac{1}{n} \sum_{i=1}^n \mathbf{a}_i \mathbf{a}_i^T \quad \text{if } \boldsymbol{\mu} = \mathbf{0}$$

Principal Component Analysis (PCA)

1. **Standardization of the values in the dataset to zero mean and unit variance.**

$$x_i = a_i - \mu$$
$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}. \quad \mathbf{Z} \in \mathbb{R}^{n \times d}$$

2. Create the covariance matrix.
3. Calculate the Eigen values and Eigen vectors from the covariance matrix.
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Principal Component Analysis (PCA)

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4. Extract the principal components.
5. Project the data points in the direction of the principal components.

$$\mathbf{S} = \frac{1}{n} \mathbf{Z}^T \mathbf{Z} \in \mathbb{R}^{d \times d}.$$

Principal Component Analysis (PCA)

1. Standardization of the values in the dataset to zero mean and unit variance.
2. Create the covariance matrix.
- 3. Calculate the Eigen values and Eigen vectors from the covariance matrix.**
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5. Project the data points in the direction of the principal components.

$$S = V\Lambda V^T$$

where:

$V = [v_1, \dots, v_p]$ has **orthonormal eigenvectors**

$\Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$ has **eigenvalues** sorted $\lambda_1 \geq \dots \geq \lambda_p \geq 0$.

Principal Component Analysis (PCA)

1. Standardization of the values in the dataset to zero mean and unit variance.
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- 4. Extract the principal components.**
5. Project the data points in the direction of the principal components.

The principal component directions are the eigenvectors:

$$PC_j \equiv v_j, j = 1, \dots, p.$$

Principal Component Analysis (PCA)

1. Standardization of the values in the dataset to zero mean and unit variance.
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5. **Project the data points in the direction of the principal components.**

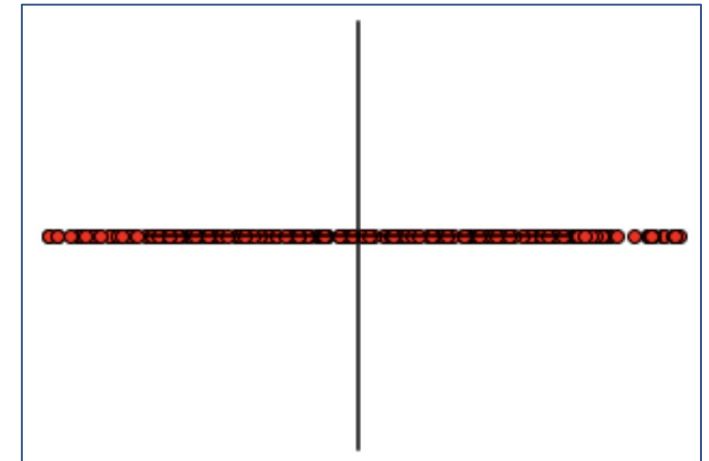
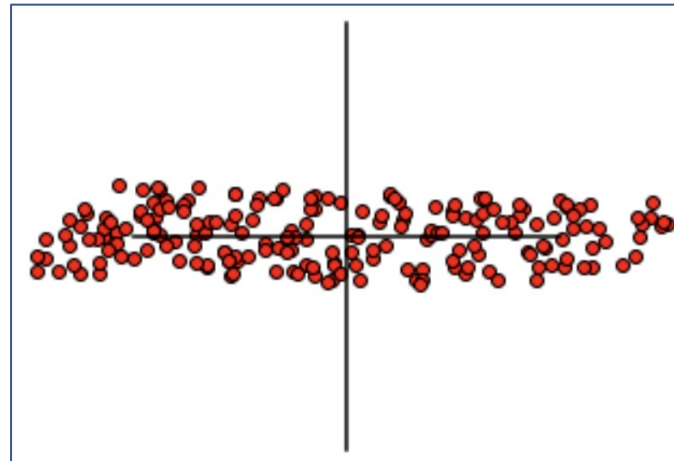
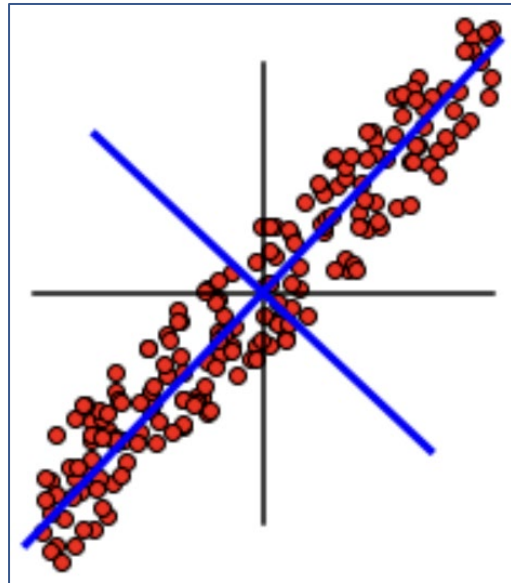
k -dimensional projection (using the top k PCs) is

$$T_k = ZV_k,$$

where $V_k = [v_1, \dots, v_k] \in \mathbb{R}^{d \times k}$.

Principal Component Analysis (PCA)

- Typically, some principal components have **very small eigenvalues** (i.e., near-zero variance). Coordinates (scores) along those directions are then dominated by noise and can often be disregarded.



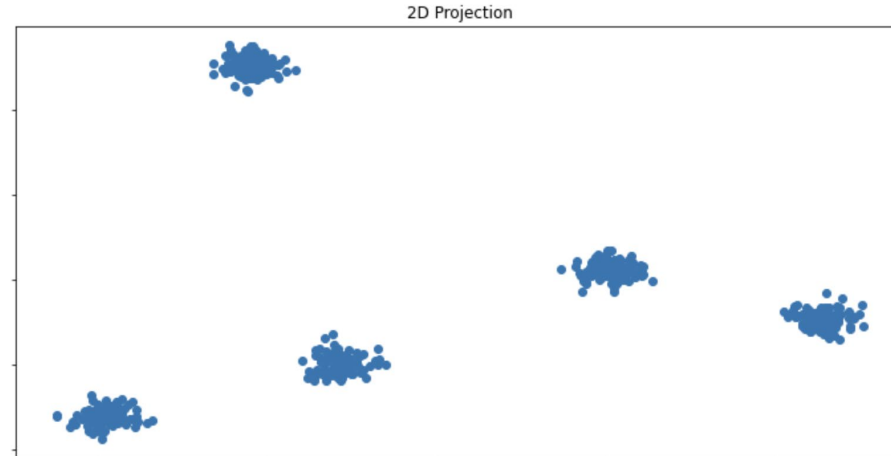
Principal Component Analysis (PCA)

```
from sklearn.decomposition import PCA

data_pca=PCA(svd_solver='full')    # svd_solver='full' computes all the principal directions
data_transformed = data_pca.fit(X) # Computes the PC basis

var = data_transformed.explained_variance_
expl_var = data_transformed.explained_variance_ratio_

Y = np.dot(data_transformed.components_[ :2, :], X.T) # projects original data onto the 2 main PC
```



Principal Component Analysis (PCA)

Eigenfaces [Turk, Pentland '91]

- Input images:



- Principal components:

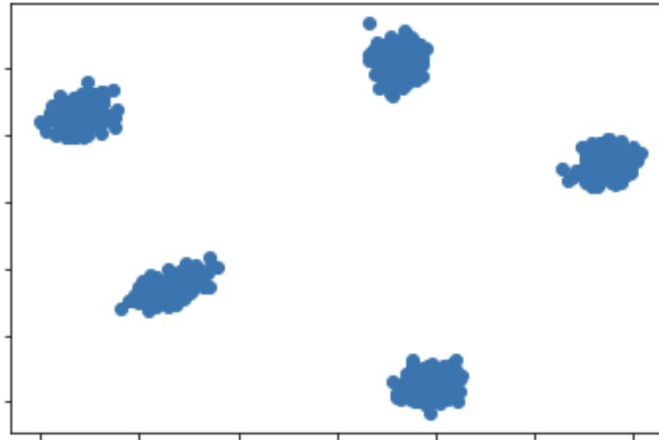


Multidimensional Scaling (MDS)

```
from sklearn.manifold import MDS
from sklearn.metrics.pairwise import euclidean_distances

D = euclidean_distances(X) # pairwise distance matrix

embedding = MDS(n_components=2, dissimilarity='precomputed') # 2D embedding from D
Y = embedding.fit_transform(D) # computes the embedding
```



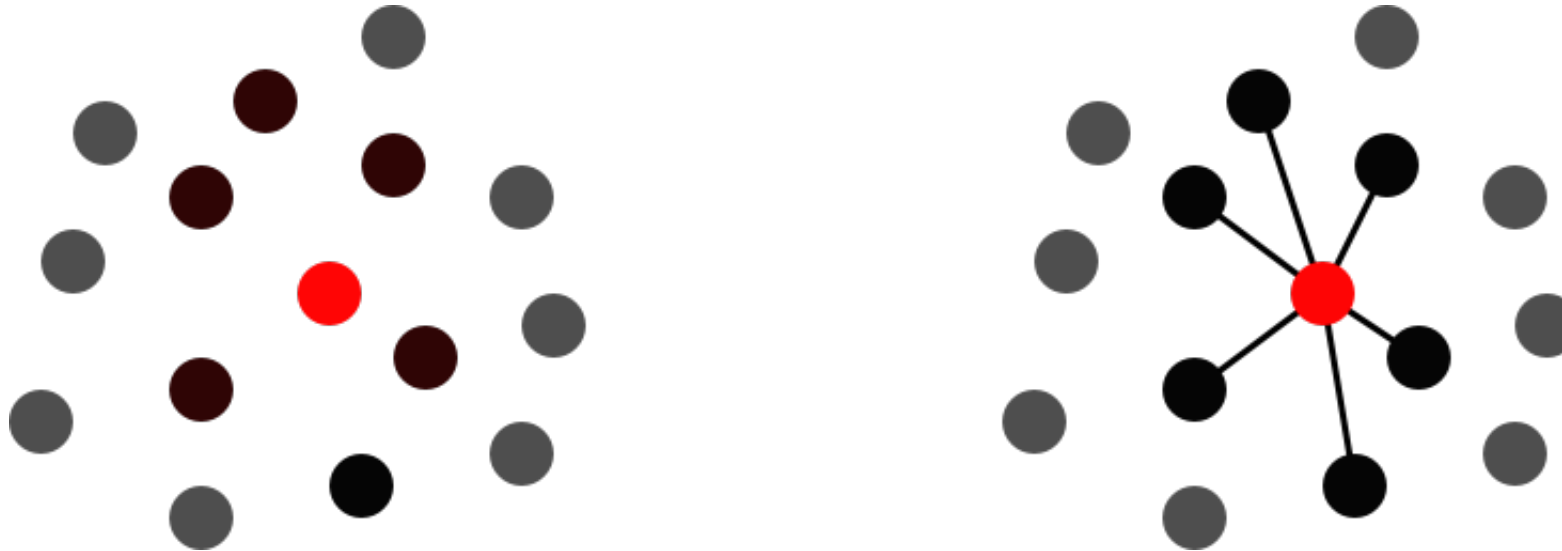
tSNE

- tSNE: tool for dimensionality reduction of high-dimensional data.
- Main idea:
 - Similarity between data points in the high-dimensional space treated as distribution P .
 - Similarity between data points in the low-dimensional space Q .
 - Achieve a representation (embedding) in the low dimension where Q faithfully represents P .
- Steps:
 1. Computation of similarities ($O(n^2)$).
 2. Minimization of cost function (divergence between P and Q).



tSNE

- The core idea of Stochastic Neighbor Embedding (SNE) is to convert the pairwise Euclidean distances between data points into conditional probabilities that represent similarities:



tSNE

- The core idea of Stochastic Neighbor Embedding (SNE) is to convert the pairwise Euclidean distances between data points into conditional probabilities that represent similarities:

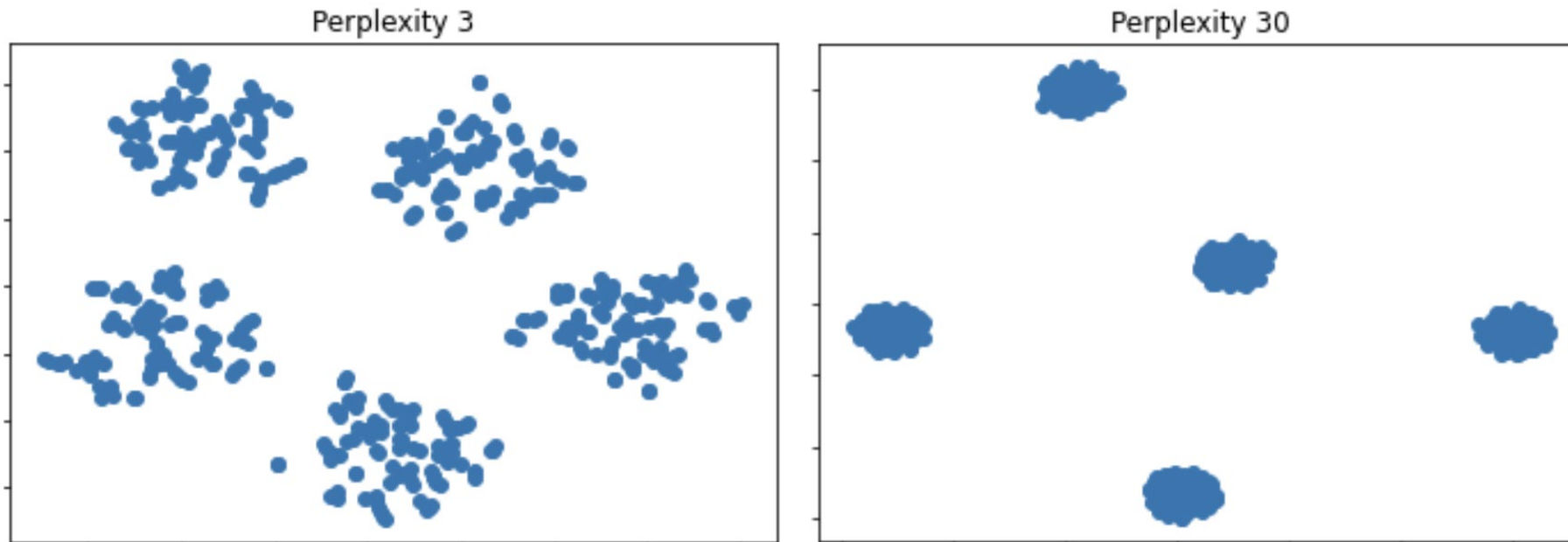
$$p_{j|i} = \frac{\exp(-\|x_i - x_j\|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\|x_i - x_k\|^2 / 2\sigma_i^2)}$$

$$p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n}$$

$$q_{ij} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k \neq l} (1 + \|y_k - y_l\|^2)^{-1}}$$

tSNE

```
from sklearn.manifold import TSNE  
  
t_sne = TSNE(n_components=2, perplexity=3)  
Y = t_sne.fit_transform(X)
```

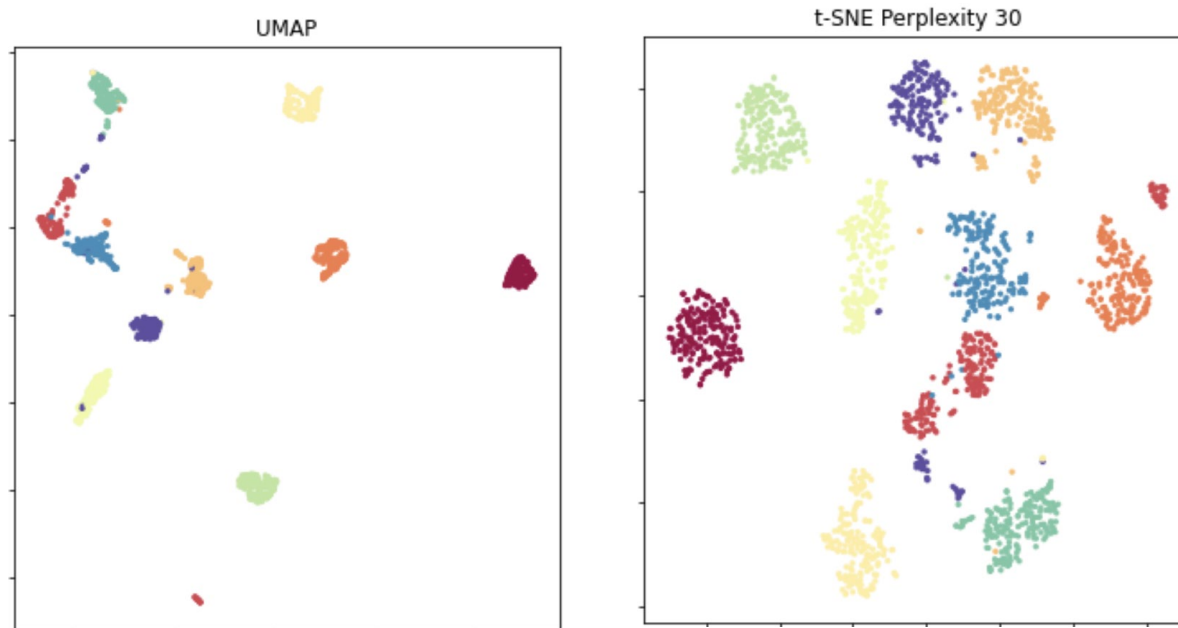


UMAP

```
conda install -c conda-forge umap-learn
```

```
import umap

umap_r = umap.UMAP()
Y = umap_r.fit_transform(X)
```



What about interactivity?

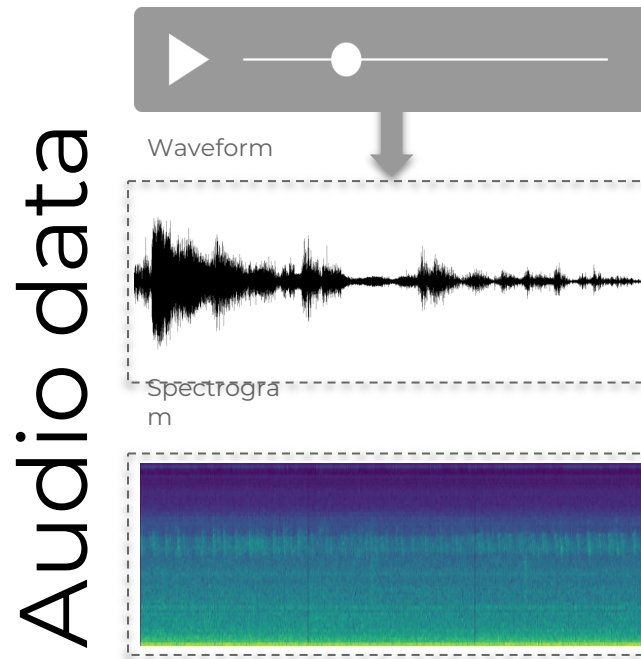
- Dimensionality reduction methods discussed previously are unsupervised, so users can only tune hyperparameters.
- **What if users want to have some control where certain points must be in the visual space while preserving their neighborhood?**

What about interactivity?

- Dimensionality reduction methods discussed previously are unsupervised, so users can only tune hyperparameters.
- **What if users want to have some control where certain points must be in the visual space while preserving their neighborhood?**
- Interactive dimensionality reduction methods are designed to enable users with some control over the mapping.

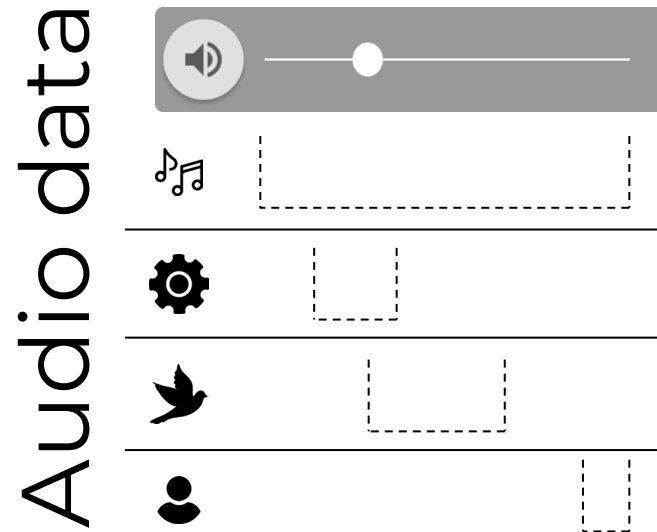
Urban audio data exploration

- Audio recordings are consumed in a **serial** way by us.
- To understand events happening in a 10-second audio snippet, users **must listen through the entire recording**.
- Although the **visualization of specific frequencies or loudness** can help identify interest periods of the recording, it is still difficult to build a **semantic understanding** of the recording.



Urban audio data exploration

- Unlike images, where visual objects are opaque, sound objects are conceptually **transparent**, meaning that multiple objects (sound sources) can have energy at the same frequency.
- At any given instant in time, a sound recording might have a **mixture of background** (birds, dog barks) and **foreground sounds** (party, sirens).

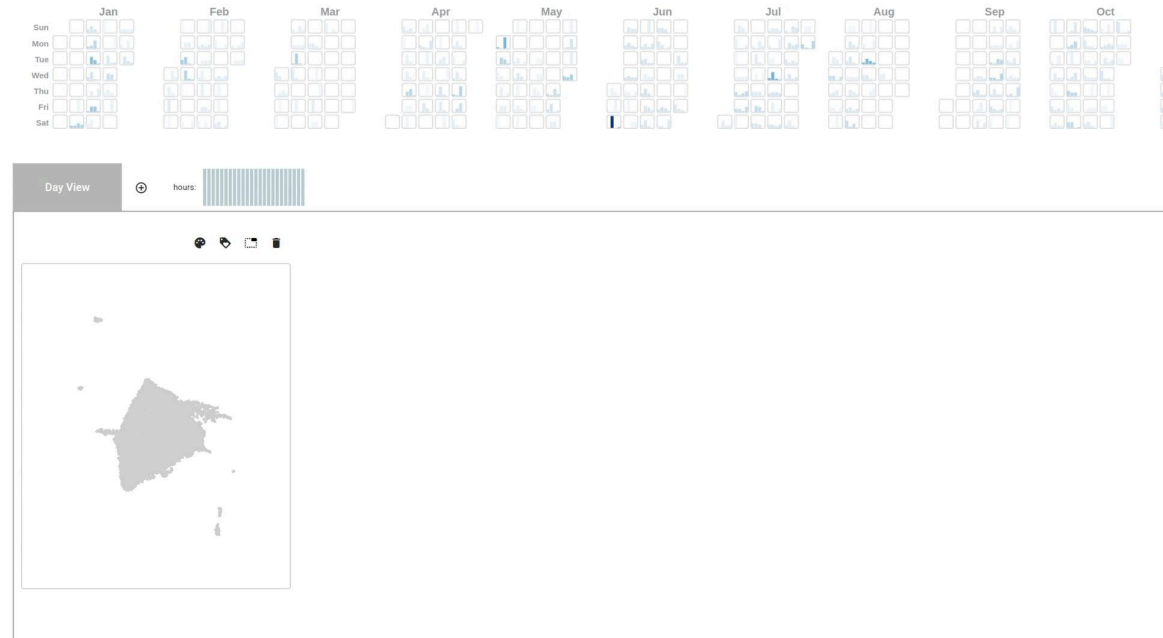


Urban Rhapsody: Creating prototypes

Binary Classification Model

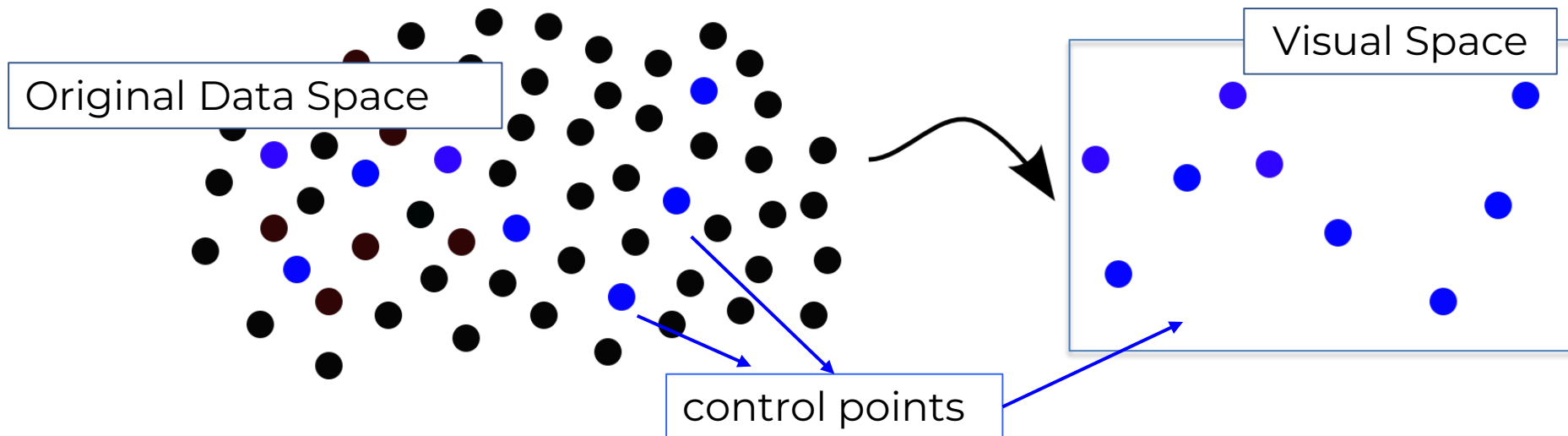


Representatives



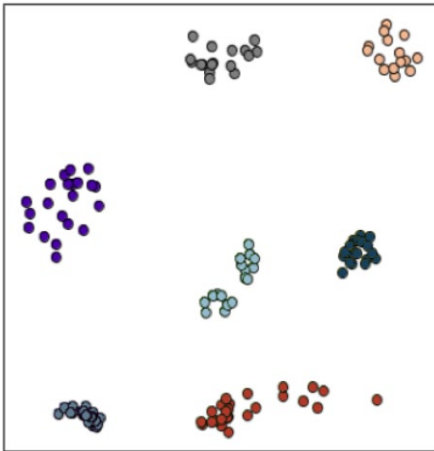
Interactive dimensionality reduction

- There are several interactive dimensionality reduction methods:
 - Least Squares Projection (LSP).
 - Piecewise Laplacian Projection (PLP)
 - Local Affine Multidimensional Projection (LAMP)

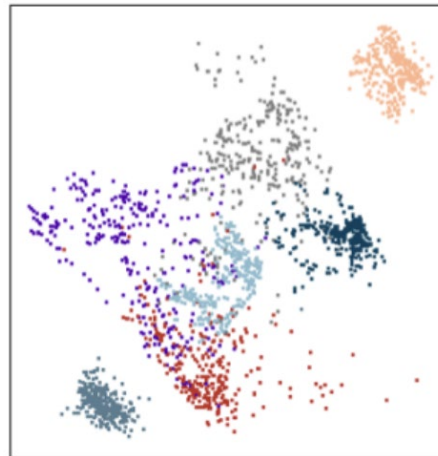


LAMP

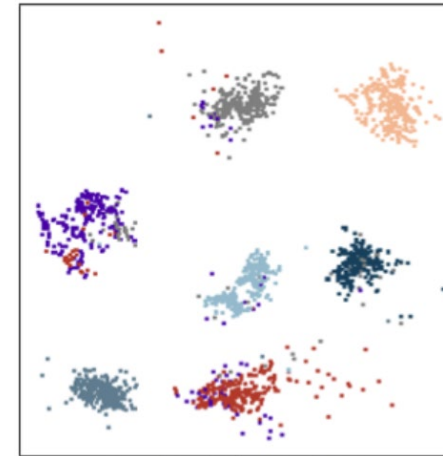
Control
points



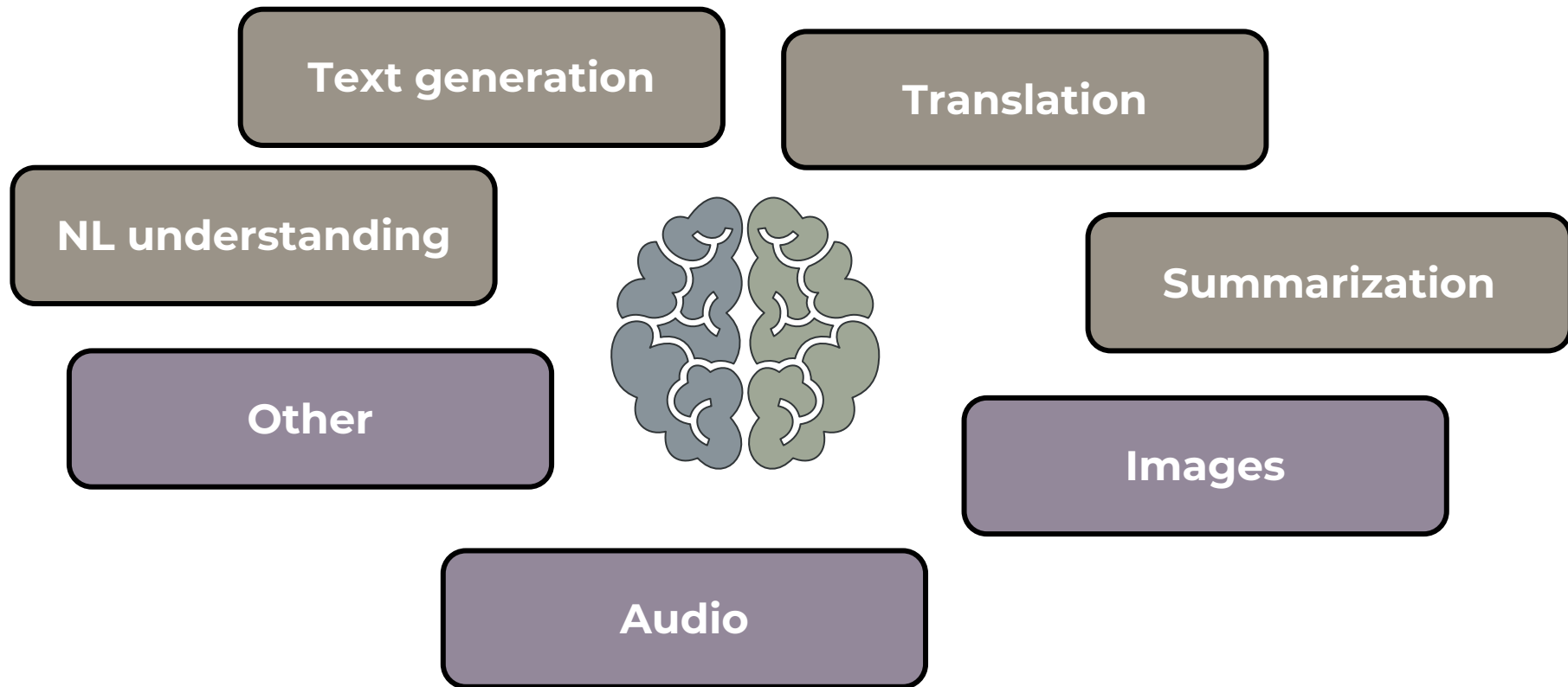
Tuning the
hyperparameter



σ_i



Opportunities with large language models



Urbanite

Urbanite: A Dataflow-Based Framework for Human-AI Interactive Alignment in Urban Visual Analytics

Gustavo Moreira, Leonardo Ferreira, Carolina Veiga, Maryam Hosseini, and Fabio Miranda

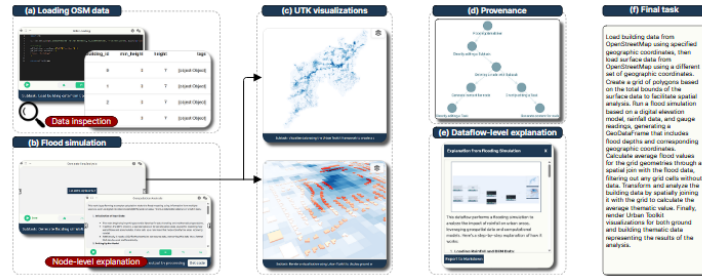


Fig. 1: Using Urbanite to analyze flood simulations. (a) The user loads a building layer and examines outputs through **provenance** and **data inspection**. (b) They create flood simulation nodes and document them using **dataflow** and **node-level explanations**. (c) Results and impacted buildings are visualized using UTK nodes. (d) The user modifies simulation parameters, branching the dataflow with **provenance** and **data inspection**. (e) To enhance documentation, they describe the dataflow using **dataflow** and **node-level explanations**. (f) The final task is generated via **task & subtask definition**, summarizing the dataflow.

Abstract—With the growing availability of urban data and the increasing complexity of societal challenges, visual analytics has become essential for deriving insights into pressing real-world problems. However, analyzing such data is inherently complex and iterative, requiring expertise across multiple domains. The need to manage diverse datasets, distill intricate workflows, and integrate various analytical methods presents a high barrier to entry, especially for researchers and urban experts who lack proficiency in data management, machine learning, and visualization. Advancements in large language models offer a promising solution to lower the barriers to the construction of analytics systems by enabling users to specify intent rather than define precise computational operations. However, this shift from explicit operations to intent-based interaction introduces challenges in ensuring alignment throughout the design and development process. Without proper mechanisms, gaps can emerge between user intent, system behavior, and analytical outcomes. To address these challenges, we propose Urbanite, a framework for human-AI collaboration in urban visual analytics. Urbanite leverages a dataflow-based model that allows users to specify intent at multiple scopes, enabling interactive alignment across the specification, process, and evaluation stages of urban analytics. Based on findings from a survey to uncover challenges, Urbanite incorporates features to facilitate explainability, multi-resolution definition of tasks across dataflows, nodes, and parameters, while supporting the provenance of interactions. We demonstrate Urbanite's effectiveness through usage scenarios created in collaboration with urban experts. Urbanite is available at urbantk.org/urbanite.

Index Terms—Urban analytics, urban data, dataflow, large language models, visualization framework, visualization system.

1 INTRODUCTION

The growing availability of urban data, coupled with increasing societal challenges, has driven the need for systems to support data-driven tasks across a range of domains, including urban planning, architecture, environmental and climate sciences, and public health. Urban visual

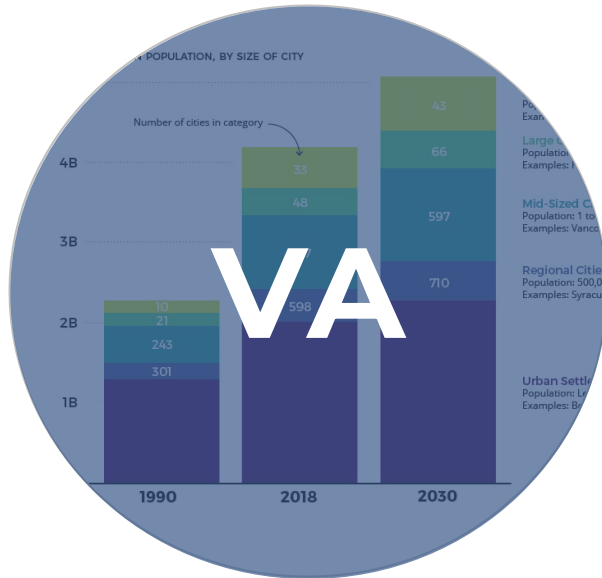
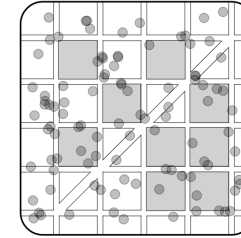
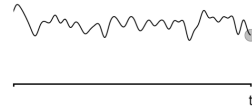
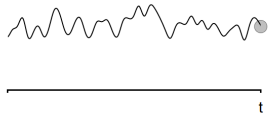
analytics (VA) systems have been repeatedly shown to be a key component in supporting these tasks [12, 15, 16, 34, 68], distilling complex data analytics workflows into accessible visual interfaces. These systems enable urban experts to gain insights [27], detect patterns [19], or evaluate potential urban interventions [31]. Despite their growing importance, urban VA systems are still complex and resource-intensive, requiring expertise in urban science and multiple areas of computer science. Beyond technical challenges, their design is an iterative, collaborative process that must align expectations across diverse stakeholders [1].

Recently, the visualization community has been highlighting tensions and technical hurdles in the process of authoring these systems [1, 21, 58, 59]. These can be broadly categorized into three key interconnected areas. First, urban data is inherently complex, encompassing heterogeneous, multi-scale, spatiotemporal datasets from diverse sources such as sensors, satellite and street-level imagery, and authoritative records. The domain tasks are equally intricate, involving

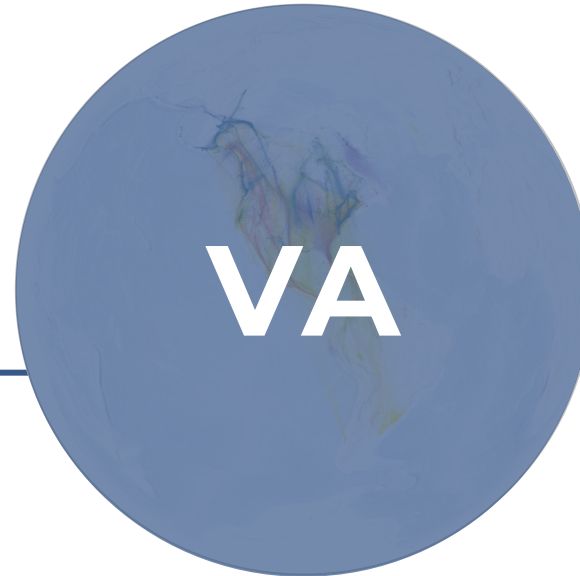
- Gustavo Moreira, Leonardo Ferreira, and Fabio Miranda are with the University of Illinois Chicago. E-mail: {gmorei3, lferr10, fabiom}@uic.edu.
- Carolina Veiga is with the University of Illinois Urbana-Champaign. E-mail: carolyf@illinois.edu.
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A human-in-the-loop dataflow framework that supports human-AI interactive alignment



UN World Urbanization Prospects 2018



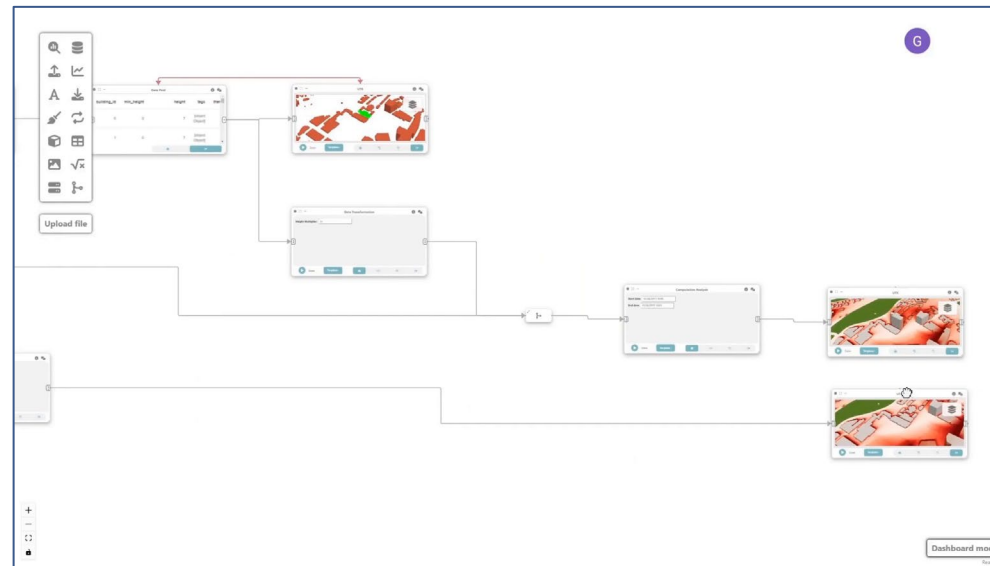
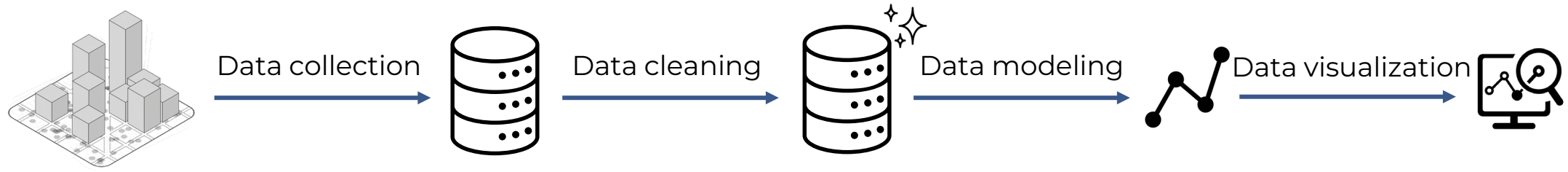
Bird Migration Explorer (Audubon)



Downtown Chicago by Jason Stoker (USGS)



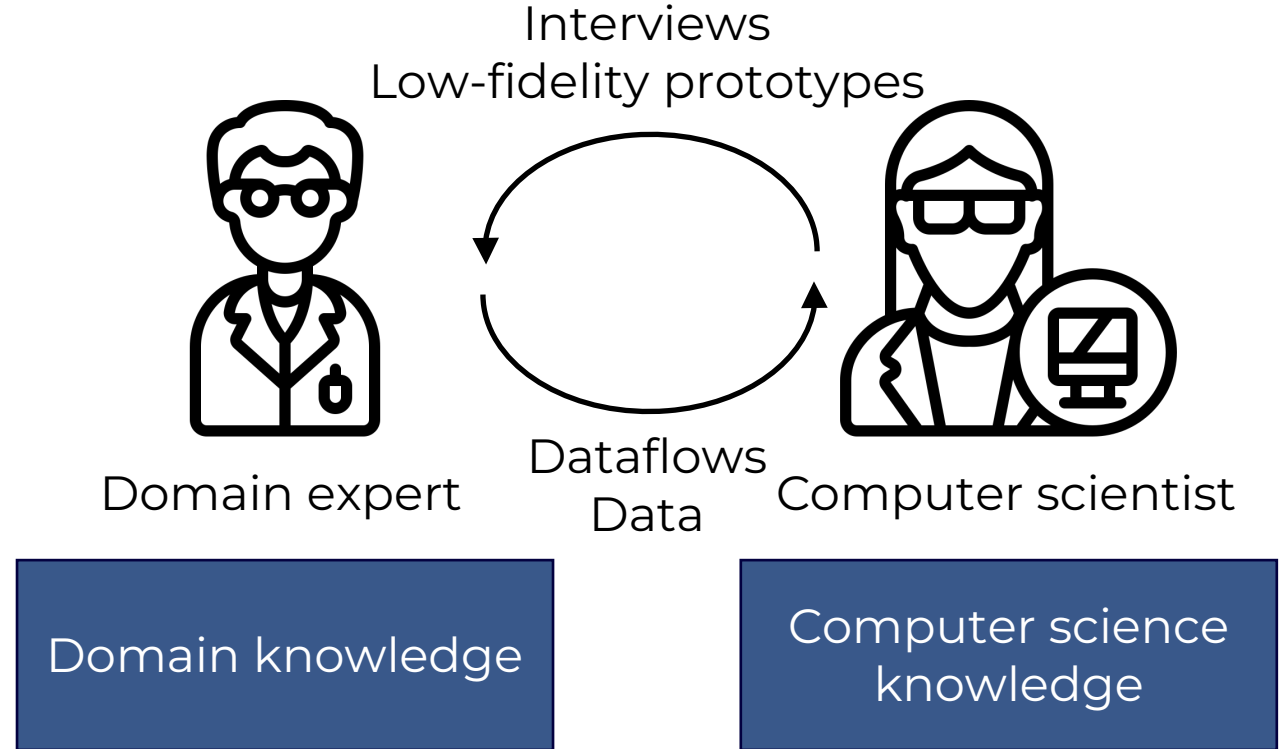
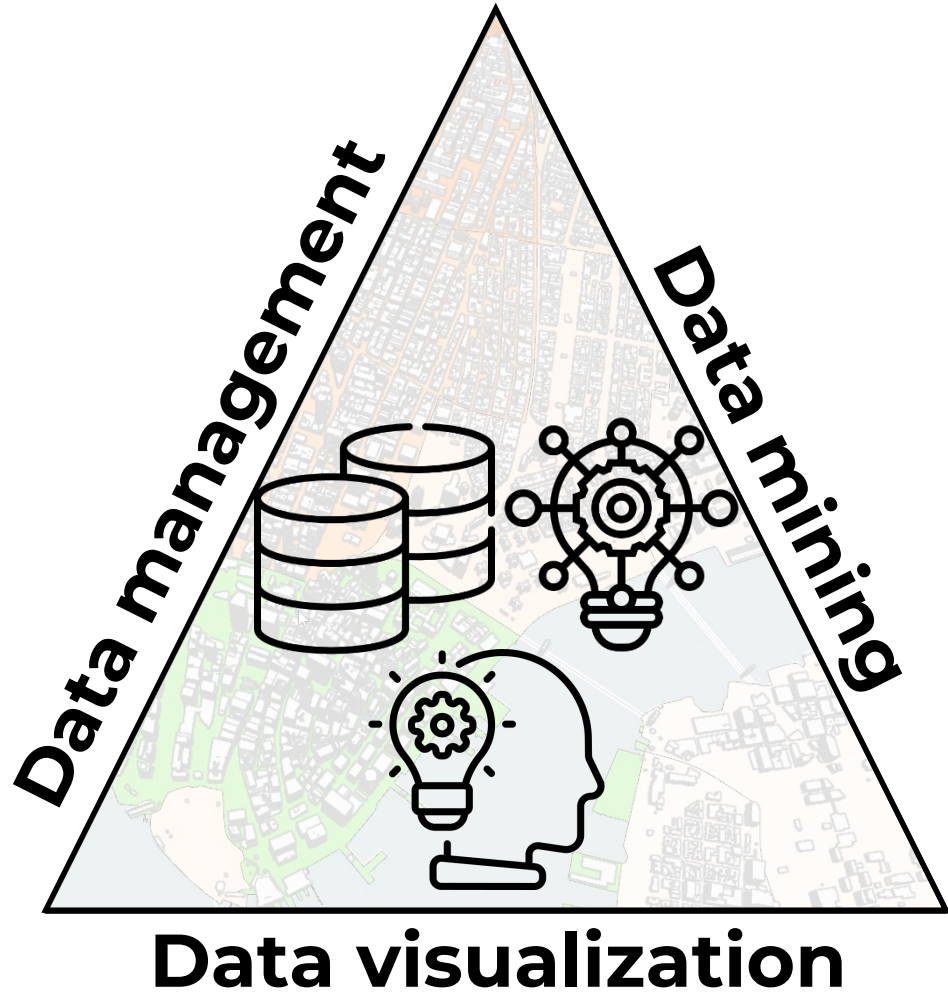
Visual Analytics as Dataflows



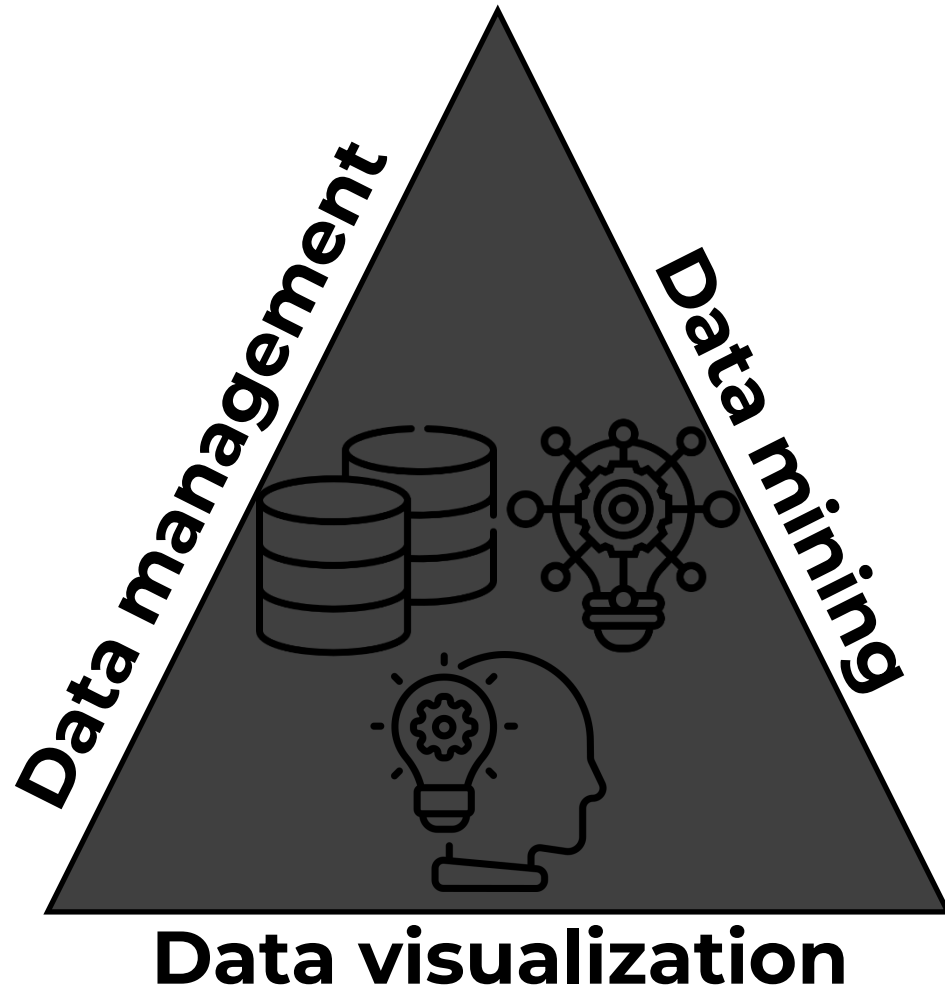
Curio
(Moreira et al., 2024)



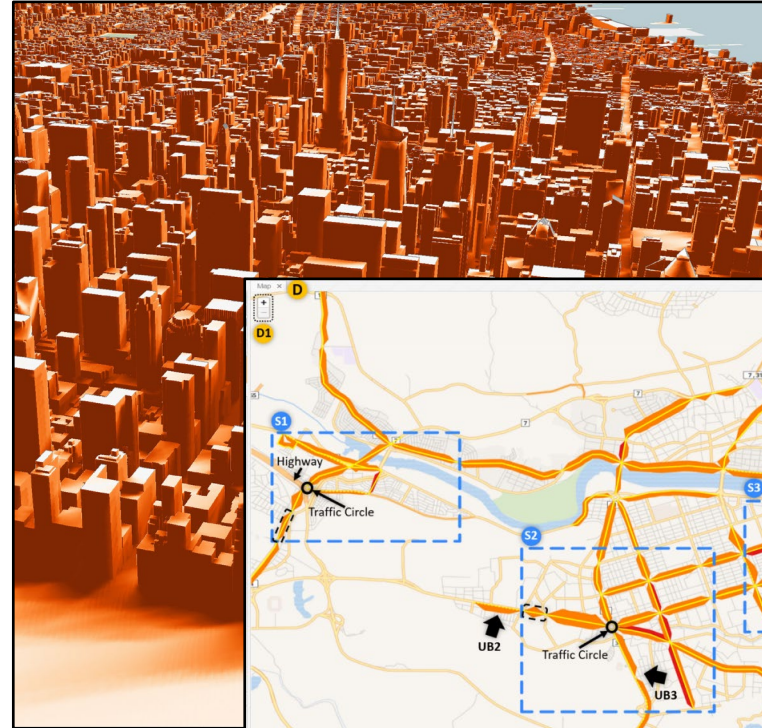
Urban Visual Analytics



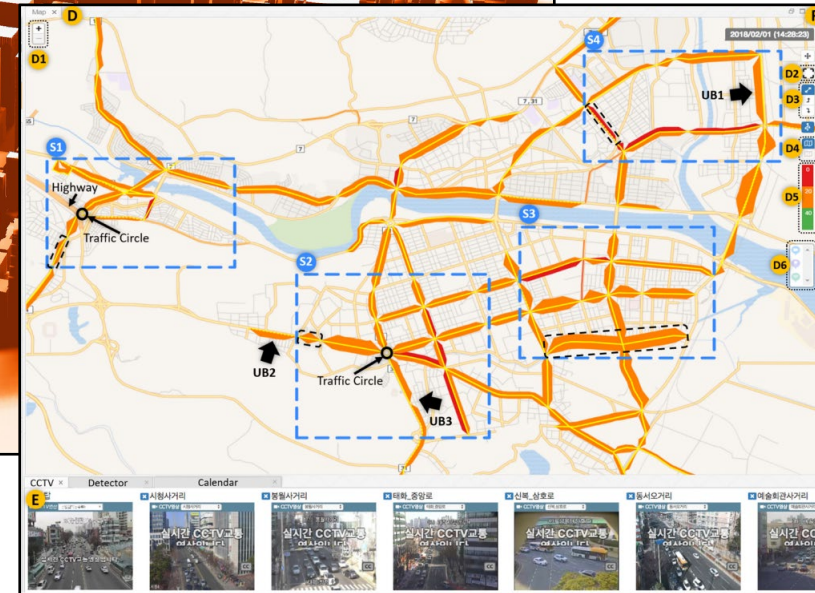
Urban Visual Analytics



Shadow Accrual Maps [Miranda et al., 2018]

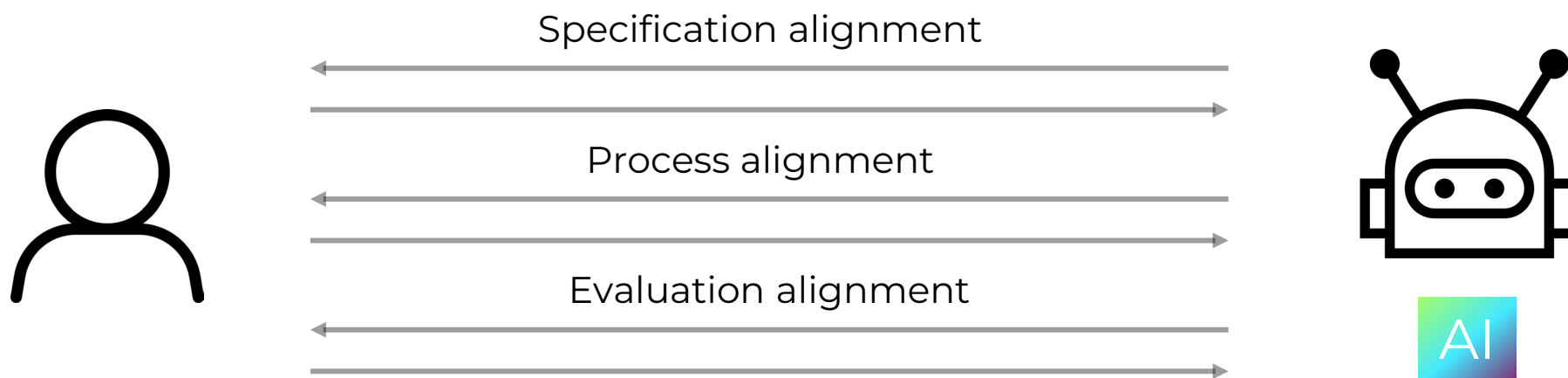


Insights
Patterns
Interventions



Visual Analysis of Uncertainty in Trajectories [Lu et al., 2014]

Intent alignment



AI alignment: Specification, process, and evaluation alignment [Terry et al., 2023]

An aerial, top-down view of a city street grid at night. The streets are illuminated by warm yellow streetlights, creating a grid pattern of light and shadow. The surrounding areas are filled with dark, rectangular shapes representing houses and buildings. Some windows are lit up, and a few cars are visible on the streets. The overall scene is a high-contrast, dark environment with the primary light source being the streetlights.

How to lower the barriers that prevent urban experts from directly engaging in the creation and adaptation of VA systems?

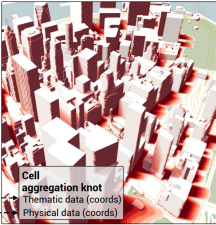
Introducing Urbanite

Indirect manipulation

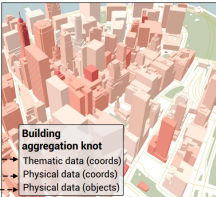
```

knots: [
  {
    "name": "shadowAverage",
    "integration_scheme": [
      {"spatial_relation": "nearest",
       "layer_in": {"name": "shadows",
                    "level": "coord"},
       "layer_out": {"name": "shadows", "level": "coord"}}
    ],
    "name": "buildings",
    "integration_scheme": [
      {"layer_in": {"data": {"url": "buildings.utk"}}}
    ]
  },
  {
    "name": "shadowAverage",
    "integration_scheme": [
      {"spatial_relation": "nearest",
       "layer_in": {"data": {"url": "shadows.utk"},
                    "level": "coord"},
       "layer_out": {"name": "shadows", "level": "coord"}},
      {"spatial_relation": "contains",
       "layer_in": {"name": "shadows", "level": "coord"},
       "layer_out": {"name": "buildings", "level": "objects"},
       "operation": "mean"}
    ]
  }
]

```



Cell aggregation knot
Thematic data (coords)
Physical data (coords)

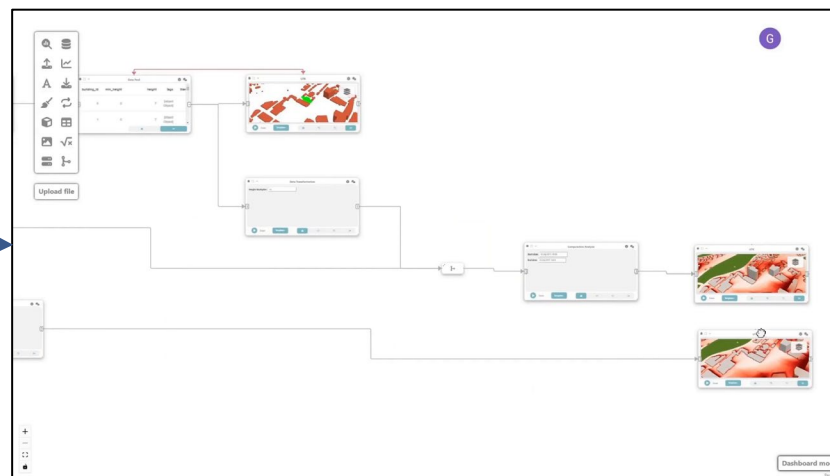


Building aggregation knot
Thematic data (coords)
Physical data (objects)

UTK
(Moreira et al., 2023)



Indirect + Direct



Curio
(Moreira et al., 2024)



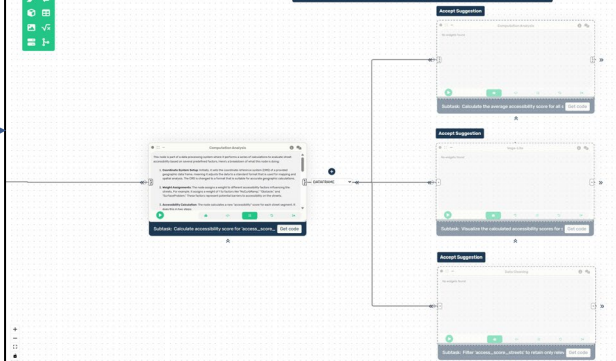
Indirect + Direct + Declarative

Analyzing Accessibility

```

Load the 'access_utk' dataset from a local CSV file, and clean the data by filtering relevant fields. Fill missing values, and creating a GeoDataFrame for point generation based on latitude and longitude. Next, load the 'access_score_neighborhood' GeoJSON dataset, retaining key fields for further analysis, and then clean this data by removing normalized attribute count fields and removing rows with null values. Finally, calculate the average accessibility score for each neighborhood.

```



LLM Assistant

Need to load and visualize the 'access_utk' dataset. How can I do this?

From a local file

How can I define how you will load the 'access_score_neighborhood' GeoJSON dataset. Will it be from a local file, or if you prefer to download it from the internet?

Also local file

Finally, let's clarify how you will load the 'access_score_neighborhood' GeoJSON dataset. Will it be from a local file or from the internet?

No

How can I define the type of analysis you would like to perform on these datasets. What specific analysis do you plan to conduct with the 'access_score_neighborhood' and 'access_utk' datasets?

I need to calculate some type of accessibility score based on latitude and longitude for each dataset that has information about a point in the network. I also want to calculate an accessibility score for each neighborhood.

Got your message.

Urbanite
(Moreira et al., 2025)



Back to Projects

File

Upload Dataset

Dashboard Mode

Provenance

Sidewalk analysis

Click here or interact with the LLM to define your task

LLM >>

LLM Assistant

I want to load and visualize two crowdsourced sidewalk datasets 'access_attributes' as CSV and 'access_score_neighborhood' as GeoJSON.

Now let's define how you're going to load the data. How are you planning to load the CSV file for 'access_attributes'? Is it located on your local file system, or will you be downloading it from an online source?

They are both locally stored

Next, let's discuss the type of analysis you want to perform with these datasets. What specific analysis or computations do you want to run on the 'access_attributes' and 'access_score_neighborhood' data?

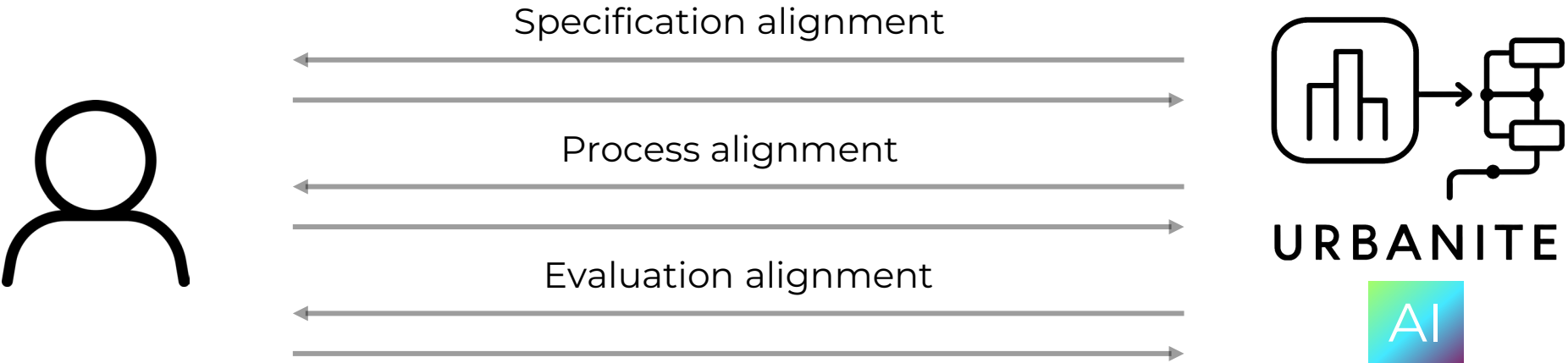
I want to calculate some type of accessibility score based on attribute counts in each dataset that has information about problems on the sidewalk. I also want to calculate an uncertainty score based on how users agree on data points.

Now let's define the visualizations you would like to create. How do you want

Type your message...

102

Intent alignment



AI alignment: Specification, process, and evaluation alignment [Terry et al., 2023]

Design goals

Specification (S)

- Enable natural language specifications of modular dataflows.

Process (P)

- Support user control during dataflow authoring.

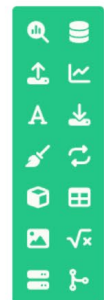
Evaluation (E)

- Enable inspection of AI-generated outputs.
- Trace and visualize the provenance of data transformations and human-AI interactions.

Crosscut (CC)

- Support iterative refinement across alignment stages.
- Support multi-level interaction within alignment stages.

Analyzing Accessibility



Load the 'access_attributes' dataset from a local CSV file, and clean the data by filtering relevant fields, filling missing values, and creating a GeoDataFrame for point geometries based on latitude and longitude. Next, load the 'access_score_neighborhood' GeoJSON dataset, retaining key fields for further analysis, and then clean this data by join ing normalized attribute count fields and removing rows with missing geometry and neighborhood names. Similarly load and clean the 'access_score_streets' GeoJSON dataset. Generate suggestions

Accept Suggestion

Computation Analysis

No widgets found

Subtask: Calculate the average accessibility score for all s

Get code

Accept Suggestion

Vega-Lite

No widgets found

Subtask: Visualize the calculated accessibility scores for s

Get code

Accept Suggestion

Data Cleaning

No widgets found

Subtask: Filter 'access_score_streets' to retain only relev

Get code

Computation Analysis

This node is part of a data processing system where it performs a series of calculations to evaluate street accessibility based on several predefined factors. Here's a breakdown of what this node is doing:

- Coordinate System Setup:** Initially, it sets the coordinate reference system (CRS) of a provided geographic data frame, meaning it adjusts the data to a standard format that is used for mapping and spatial analysis. The CRS is changed to a format that is suitable for accurate geographic calculations.
- Weight Assignments:** The node assigns a weight to different accessibility factors influencing the streets. For example, it assigns a weight of 1 to factors like "NoCurbRamp," "Obstacle," and "SurfaceProblem." These factors represent potential barriers to accessibility on the streets.
- Accessibility Calculation:** The node calculates a new "accessibility" score for each street segment. It does this in two steps:

DATAFRAME

Get code

Computation Analysis

```
1 import numpy as np
2
3 gdf_streets = arg.set_crs(4326)
4 gdf_streets = gdf_streets.to_crs(3395)
5
6 w_noCurb = 1
7 w_obstacle = 1
8 w_surfaceProblem = 1
9 w_curb = 1
10
11 numerator = (w_noCurb * gdf_streets["NoCurbRamp"] +
12             w_obstacle * gdf_streets["Obstacle"] +
13             w_surfaceProblem * gdf_streets["SurfaceProblem"])
14
15 denominator = w_curb * gdf_streets["CurbRamp"]
16
17 gdf_streets["accessibility"] = np.where(
18     denominator == 0,
```

Subtask: Calculate accessibility score for 'access_score_

Get code

LLM Assistant

I want to load and visualize three crowdsourced Sidewalk datasets 'access_attributes' as CSV, 'access_score_neighborhood' as GeoJSON, and 'access_score_streets' as GeoJSON.

Now let's define how you will load the three datasets. Can you specify how you intend to load the 'access_attributes' CSV file? Will it be from a local file or downloaded from a URL?

From a local file

Now let's define how you will load the 'access_score_neighborhood' GeoJSON dataset. Will it also be from a local file, or do you prefer to download it from the internet?

Also local file

Finally, let's clarify how you will load the 'access_score_streets' GeoJSON dataset. Will it be from a local file as well?

Yes

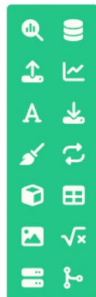
Now let's define the type of analysis you would like to perform on these datasets. What specific analysis do you plan to conduct with the 'access_attributes', 'access_score_neighborhood', and 'access_score_streets' datasets?

I want to calculate some type of accessibility score based on attribute counts in each dataset that has information about problems on the sidewalk. I also want to calculate an uncertainty score based on how many

Type your message...



Analyzing Accessibility



Task

Load the 'access_attributes' dataset from a local CSV file, and clean the data by filtering relevant fields, filling missing values, and creating a GeoDataFrame for point geometries based on latitude and longitude. Next, load the 'access_score_neighborhood' GeoJSON dataset, retaining key fields for further analysis, and then clean this data by joining normalized attribute count fields and removing rows with missing geometries and neighborhood names. Similarly, load and clean the 'access_score_streets' GeoJSON dataset. Generate suggestions

LLM chat

Suggestions

Explanation

Node logic

Subtask

LLM Assistant

I want to load and visualize three crowdsourced Sidewalk datasets 'access_attributes' as CSV, 'access_score_neighborhood' as GeoJSON, and 'access_score_streets' as GeoJSON.

Now let's define how you will load the three datasets. Can you specify how you intend to load the 'access_attributes' CSV file? Will it be from a local file or downloaded from a URL?

From a local file

Now let's define how you will load the 'access_score_neighborhood' GeoJSON dataset. Will it also be from a local file, or do you prefer to download it from the internet?

Also local file

Finally, let's clarify how you will load the 'access_score_streets' GeoJSON dataset. Will it be from a local file as well?

Yes

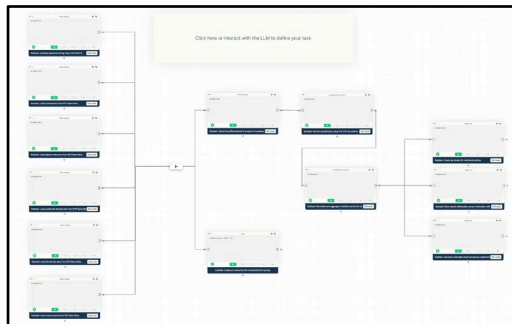
Now let's define the type of analysis you would like to perform on these datasets. What specific analysis do you plan to conduct with the 'access_attributes', 'access_score_neighborhood', and 'access_score_streets' datasets?

I want to calculate some type of accessibility score based on attribute counts in each dataset that has information about problems on the sidewalk. I also want to calculate an uncertainty score based on how many

Type your message...



Urbanite



```
report pandas to get
import geopandas as gpd
from shapely.geometry import Point

df_point = gpd

columns_to_keep = ['access_type', 'access_score_neighborhood', 'access_score_streets', 'agree_count', 'disagree_count', 'unsure_count']
df_point_filtered = df_point[columns_to_keep]

df_point_filtered['access_type'] = df_point_filtered['access_type'].fillna('none')
df_point_filtered['access_score_neighborhood'] = df_point_filtered['access_score_neighborhood'].fillna(0)
df_point_filtered['access_score_streets'] = df_point_filtered['access_score_streets'].fillna(0)
df_point_filtered['agree_count'] = df_point_filtered['agree_count'].fillna(0)
df_point_filtered['disagree_count'] = df_point_filtered['disagree_count'].fillna(0)
df_point_filtered['unsure_count'] = df_point_filtered['unsure_count'].fillna(0)

df_point_filtered['geometry'] = df_point_filtered.apply(lambda row: Point(row['longitude'], row['latitude']), axis=1)

gdf_point = gpd.GeoDataFrame(df_point_filtered, geometry='geometry', crs='EPSG:4326')

return gdf_point
```

Subtask: Clean and transform the 'access_attributes' dataset [Get code](#)

Data Loading

Load CSV:

Choose File No file chosen

Max Building Height (m): 10

Neighborhood: Manhattan

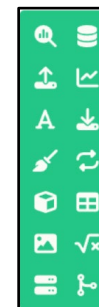
Done Templates </> </>

Scopes of specification (what)	NL	UI	Grammar	Code
Dataflow	✓	✓	✓	✓
Module	✓	✓	✓	✓
Parameter		✓	✓	✓

Load the 'access_attributes' dataset from a local CSV file, clean the data by filtering relevant fields and filling missing values for latitude, longitude, agree, disagree, and unsure counts while creating a GeoDataFrame for point geometries. Load the 'access_score_neighborhood' dataset from the 'access_score_neighborhood' .geojson' file, retaining key fields like coverage, average attribute count, neighborhood name, and geometry for further analysis. Load

[Generate suggestions](#)

Subtask: Clean and transform the 'access_attributes' dataset [Get code](#)



Load CSV:

Choose File No file chosen

Max Building Height (m): 10

Neighborhood: Manhattan

Done Templates </> </>

UTK powered by ace

```
1 {
2   "components": [
3     {
4       "map": {
5         "camera": {
6           "position": [
7             -8239611,
8             4941390.5,
9             1000000000000000.0
10          ]
11        }
12      }
13    ]
14  }
```

Templates </> </>

Data Cleaning

```
report pandas to get
import geopandas as gpd
from shapely.geometry import Point

df_point = gpd

columns_to_keep = ['access_type', 'access_score_neighborhood', 'access_score_streets', 'agree_count', 'disagree_count', 'unsure_count']
df_point_filtered = df_point[columns_to_keep]

df_point_filtered['access_type'] = df_point_filtered['access_type'].fillna('none')
df_point_filtered['access_score_neighborhood'] = df_point_filtered['access_score_neighborhood'].fillna(0)
df_point_filtered['access_score_streets'] = df_point_filtered['access_score_streets'].fillna(0)
df_point_filtered['agree_count'] = df_point_filtered['agree_count'].fillna(0)
df_point_filtered['disagree_count'] = df_point_filtered['disagree_count'].fillna(0)
df_point_filtered['unsure_count'] = df_point_filtered['unsure_count'].fillna(0)

df_point_filtered['geometry'] = df_point_filtered.apply(lambda row: Point(row['longitude'], row['latitude']), axis=1)

gdf_point = gpd.GeoDataFrame(df_point_filtered, geometry='geometry', crs='EPSG:4326')

return gdf_point
```

Done Templates </> </>

Task: You will load three datasets: 'access_attributes' from a local CSV file, and both 'access_score_neighborhood' and 'access_score_streets' from local GeoJSON files. You will compute an accessibility score based on attribute counts and an uncertainty score based on user agreement. For visualizations, you will generate four map-based visualizations: two for accessibility (neighborhood and street) and two for uncertainty (neighborhood and street). Additionally, you will create a plot showing accessibility by neighborhood that is linked with interactions to the map visualizations, enabling dynamic exploration of the data. If you would like to change it let me know!

[Apply task](#)

Type your message...

Blended Interfaces

Sehi et al., 2025 (P1)

IEEE TRANSACTIONS ON VISUALIZATION AND COMPUTER GRAPHICS, VOL. 31, NO. 1, JANUARY 2025

459

Learnable and Expressive Visualization Authoring Through Blended Interfaces

Sehi L'Yi , Astrid van den Brandt , Etowah Adams , Huyen N. Nguyen , and Nils Gehlenborg 

Abstract—A wide range of visualization authoring interfaces enable the creation of highly customized visualizations. However, prioritizing expressiveness often impedes the learnability of the authoring interface. The diversity of users, such as varying computational skills and prior experiences in user interfaces, makes it even more challenging for a single authoring interface to satisfy the needs of a broad audience. In this paper, we introduce a framework to balance learnability and expressivity in a visualization authoring system. Adopting insights from learnability studies, such as multimodal interaction and visualization literacy, we explore the design space of blending multiple visualization authoring interfaces for supporting authoring tasks in a complementary and flexible manner. To evaluate the effectiveness of blending interfaces, we implemented a proof-of-concept system, *Blace*, that combines four common visualization authoring interfaces—template-based, shelf configuration, natural language, and code editor—that are tightly linked to one another to help users easily relate unfamiliar interfaces to more familiar ones. Using the system, we conducted a user study with 12 domain experts who regularly visualize genomics data as part of their analysis workflow. Participants with varied visualization and programming backgrounds were able to successfully reproduce unfamiliar visualization examples without a guided tutorial in the study. Feedback from a post-study qualitative questionnaire further suggests that blending interfaces enabled participants to learn the system easily and assisted them in confidently editing unfamiliar visualization grammar in the code editor, enabling expressive customization. Reflecting on our study results and the design of our system, we discuss the different interaction patterns that we identified and design implications for blending visualization authoring interfaces.

Index Terms—Visualization authoring, blended interfaces, genomics data visualization

1 INTRODUCTION

A wide range of visualization authoring interfaces has been introduced in the visualization field to enable the creation of highly customized visualizations [43, 49, 62, 63, 68, 79, 83]. However, prioritizing expressiveness often impedes the learnability of the authoring interface. For example, visual builder interfaces [12, 35, 49, 68, 83], which typically offer visual encoding palettes along with a visualization canvas, enable fine-grained customization but at the cost of a steep learning curve [48]. The trade-off between expressivity and learnability has been discussed as an essential design consideration for visualization authoring systems in prior studies [18, 64, 69] (Fig. 1).

Supporting learnability often involves not only making the system easy to learn but also supporting a broad range of users [15]. Users with varied computational skills and prior experiences in specific user interfaces find it challenging to satisfy their diverse needs with a single authoring interface. For example, genomics is a typical domain that shares this challenge. In this domain, experts with different roles and backgrounds, such as computational biologists, clinicians, and experimentalists, frequently visualize data as an essential part of their analysis workflows. However, according to our user interviews [42], domain experts have strong preferences for specific visualization authoring techniques (e.g., graphical user interface and programming-based), and they vastly diverge. This challenges the creation of widely learnable and highly expressive visualization authoring systems.

To balance the learnability and expressivity of a visualization authoring system, we adopt ideas from learnability studies [31, 45, 65, 72]. Using the design strategies of multimodal interactions [45], we explore different ways to combine multiple authoring interfaces in a complementary manner. Moreover, following the learning-by-analogy ap-

proach [2], which showed its effectiveness in visualization literacy [65], visualization onboarding [75], and visualization education [28], we propose to tightly connect combined authoring interfaces to offer easy ways for users to learn unfamiliar—but potentially more expressive—interfaces (e.g., code editors) by relating them with familiar interfaces (e.g., template-based interfaces).

In this paper, we introduce a general framework to blend multiple authoring interfaces, called *blended interfaces*. Using the framework, we explore the design space of blending multiple visualization authoring interfaces to support authoring tasks in a complementary and flexible manner. The essential building blocks of blended interfaces are (1) composing two or more authoring interfaces, (2) supporting one or more types of cooperative interactions [45], and (3) visually linking interfaces. To evaluate the effectiveness of blending interfaces, we further implement a genomics visualization authoring system—*Blace*—that combines four common visualization authoring interfaces—template-based, shelf configuration, natural language, and code editing interfaces. The system consistently highlights state differences across all four authoring interfaces upon mouse hovering (i.e., visual linking) to enable users to learn unfamiliar interfaces (e.g., the code editor) after interacting with the system. Using *Blace*, we conducted a user study with 12 domain experts. All participants with varied skill sets in genomics data analysis, programming, and visualization were able to successfully reproduce complex visualization examples without a guided tutorial in the study. Feedback from a post-study qualitative questionnaire further suggests that blending interfaces enabled participants to learn the system easily and assisted them in confidently editing unfamiliar visualization grammar in the code editor, enabling expressive customization. Reflecting on our study results and the design of our system, we discuss the different interaction patterns that we identified and design implications for blending visualization authoring interfaces.

The key contribution of our work is three-fold:

- A conceptual framework of *blended interfaces* to explore its design space;
- An implementation of *Blace* for genomics data visualization; and
- A user evaluation with domain experts ($n=12$) to demonstrate the effectiveness of blended interfaces.

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Date of publication 10 September 2024; date of current version 29 November 2024.

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Digital Object Identifier no. 10.1109/TVCG.2024.3456598

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LEARNABILITY

EXPRESSIVITY



urbantk.org



electronic
visualization
laboratory



UNIVERSITY OF
ILLINOIS CHICAGO

Blended Interfaces (Formalization)

Interface := (Template-Based, Shelf Configuration, Code Editor)

Cooperation := (
 (Template-Based, Specialization, (Data, Encoding, Style)_{high-level changes}),
 (Shelf Configuration, Specialization, Encoding),
 (Code Editor, Specialization, Encoding),
)

Linking := (Interface, State, Complete Bidirectional Linking)

Ivy [McNutt and Chugh, 2021]



A

Natural Language

</> Code Editor

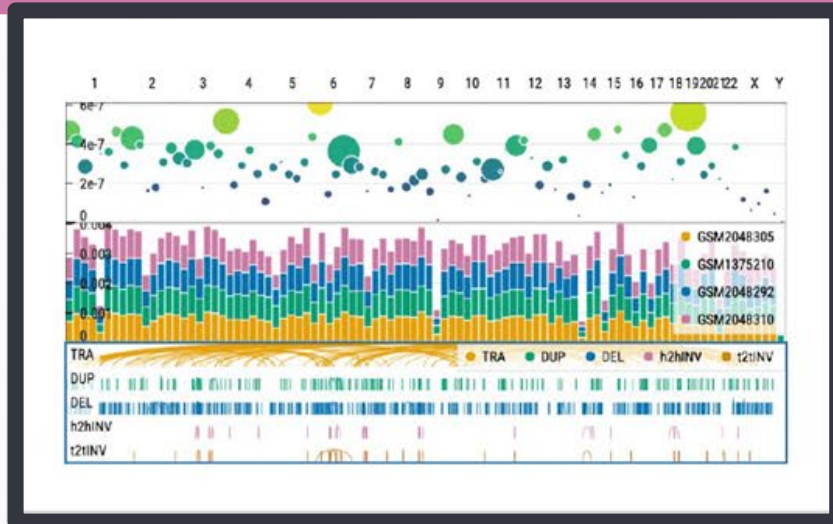
NATURAL LANGUAGE INTERFACE

AI Assistant

The visualization type has been changed to Multi-Row Arc Diagram .

You

Ask AI Assistant to customize the selected track... (e.g., "change mark color to red" or "change to heatmap")



B

</> CODE EDITOR

Apply Changes

```
1 {
2   "id": "track-2",
3   "width": 600,
4   "height": 100,
5   "data": {
6     "type": "csv",
7     "url": "https://somatic-browser-test.s3.amazonaws.com/SRR7898985/SRR7898985.gripps.filtered.bedpe",
8     "genomicFieldsToConvert": [
9       {
10        "chromosomeField": "chrom1",
11        "genomicFields": [
12          "start1",
13          "end1"
14        ]
15      },
16      {
17        "chromosomeField": "chrom2",
18        "genomicFields": [
19          "start2",
20          "end2"
21        ]
22      }
23    ]
24  }
25 }
```

C

TEMPLATES

TRACKS

Switch the visualization type of a selected track

Arc Diagram
Use the circular arcs to represent connections between two genomic regions.

Colored Arc Diagram
Use the circular arcs with different colors to represent connections between two genomic regions and a nominal field.

Elliptical Arc Diagram
Use the elliptical arcs with different colors to represent connections between two genomic regions and a nominal field.

Multi-Row Arc Diagram
Use the elliptical arcs with different colors in multiple rows to represent connections between two genomic regions and two nominal fields.

Multi-Row Block Diagram
Use the rectangles in different colors to represent the genomic range and two nominal fields.

Coverage Line Track
Use lines to represent the frequency of data points along the genomic axis. The frequency is calculated and then mapped to the vertical positions of lines.

Multi-Row Bubble Chart
Use points with varying sizes in multiple rows to represent the distance of genomic positions and a nominal field. The distance is calculated and then mapped to the size of points.

D

SHELF CONFIGURATION

TRACK ENCODING

Mark

Fields (Found from the CSV file)

N chrom1

G start1

G end1

N chrom2

G start2

G end2

N sv_id

pe_support

N strand1

N strand2

N svclass

N svmethod

Channels

x

G start1

xe

G end2

y

row

N svclass

color

N svclass

size

stroke

N svclass

strokeWidth

opacity

HISTORY

USER INTERACTIONS

Hover to see the visualization in the corresponding step. Click to revert back to the step.

Latest

Changed to Multi-Row Arc Diagram

Edited Code

Edited Code

Changed to Multi-Row Arc Diagram

Edited Code

Changed to Stacked Bar Chart

Encoded "value_min" to "size"

Encoded "value" to "color"

Selected Point mark

Selected Area mark

Selected Line mark

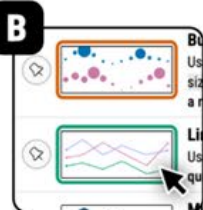
Changed to Scatterplot

Oldest

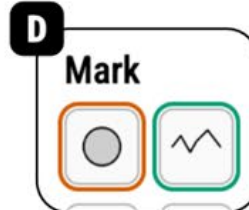
Initial Visualization

110

A Change the visualization type to Line Chart ?



C -1 +1



E -2 +1

F size # value

Natural Language </> Code Editor

NATURAL LANGUAGE INTERFACE

AI Assistant Change the visualization type to Line Chart ?

You Ask AI Assistant to customize the selected track... (e.g., "change mark color to red" or "change to heatmap")

Preview

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 X Y

0.0005 0.001

GSM2048305 GSM1375210 GSM2048292 GSM2048310

TRA DUP DEL h2hINV i2hINV

P_99, Training Session

TEMPLATES -1 +1

TRACKS

Switch the visualization type of a selected track

Scatterplot Use points with different colors to represent a quantitative field and a nominal field.

Multi-Row Scatterplot Use points with different colors in multiple rows to represent a quantitative field and two nominal fields.

Bubble Chart Use points with different colors and varying sizes to represent two quantitative fields and a nominal field.

Line Chart Use lines with different colors to represent a quantitative field and a nominal field.

Multi-Row Area Chart Use filled areas in multiple rows to represent a quantitative field and a nominal field.

Stacked Area Chart Use filled areas that are stacked to represent a quantitative field and a nominal field.

Stacked Bar Chart Use stacked bars to represent a quantitative field and a nominal field.

Multi-Row Bar Chart Use bars in multiple rows to represent a quantitative field and a nominal field.

Heatmap Use the color of rectangles to represent quantitative values along the genomic axis.

SHELF CONFIGURATION -2 +1

TRACK ENCODING

Mark

Fields (Found from the MULTIVIEW file)

N category # value

G position G start

G end # value_min

value_max

Channels

x G position

xe

y # value

row

color N category

size # value

stroke

strokeWidth

opacity

HISTORY

USER INTERACTIONS

Hover to see the visualization in the corresponding step. Click to revert back to the step.

Latest

Changed to Bubble Chart

Changed to Multi-Row Scatterplot

Changed to Bubble Chart

Changed to Multi-Row Arc Diagram

Changed to Multi-Row Block Diagram

Changed to Multi-Row Arc Diagram

Edited Code

Edited Code

Changed to Multi-Row Arc Diagram

Edited Code

Changed to Stacked Bar Chart

Encoded "value_min" to "size"

Encoded "value" to "color"

Selected Point mark

Selected Area mark

Selected Line mark

Changed to Scatterplot

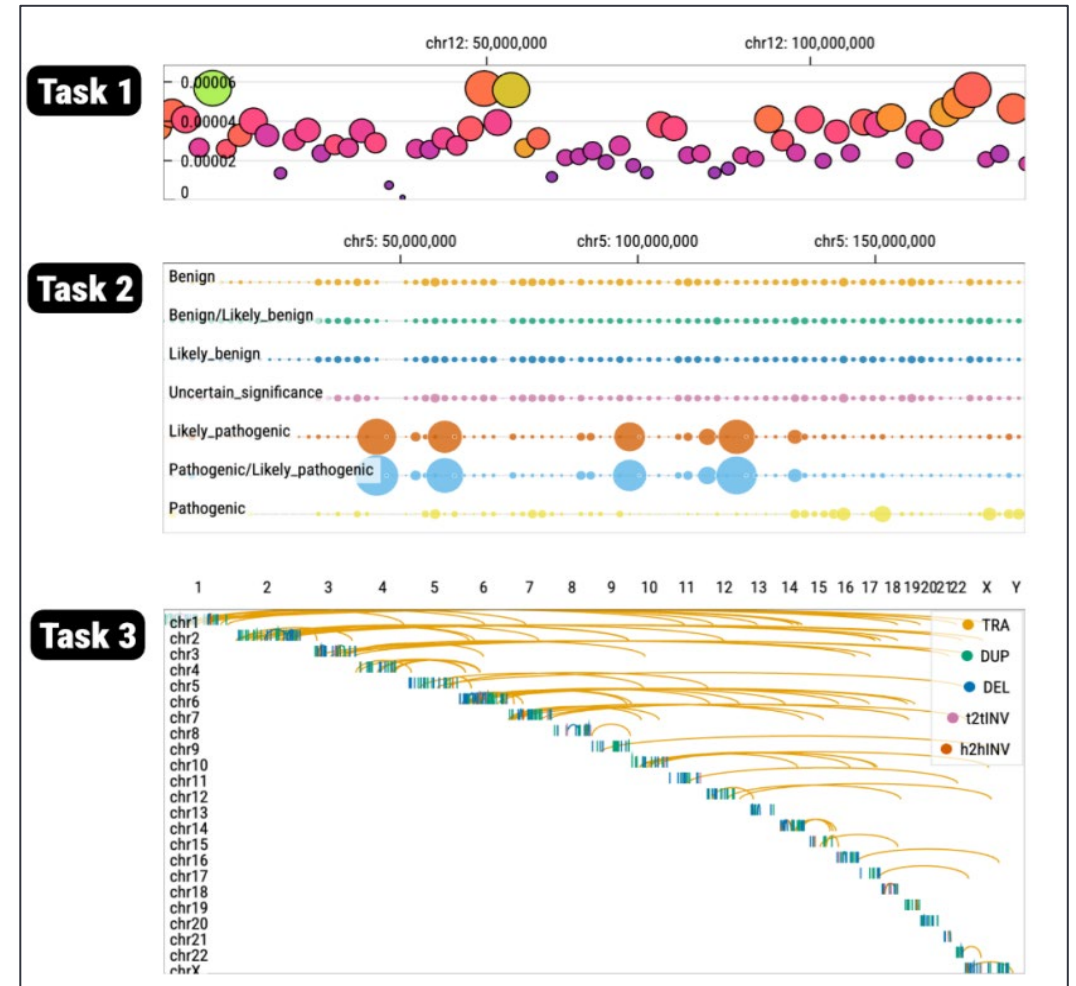
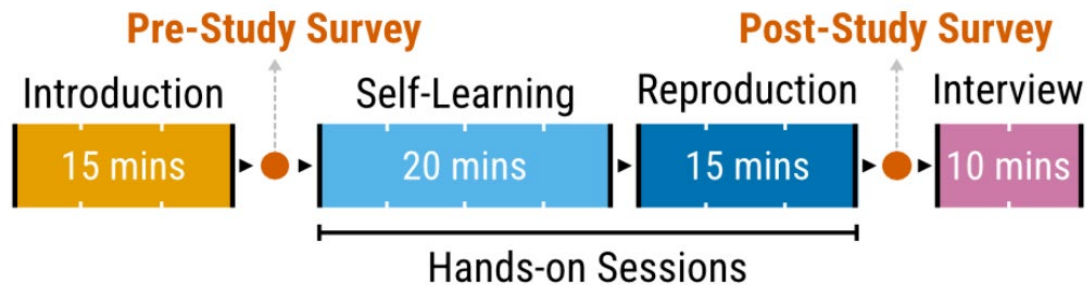
Oldest

Initial Visualization

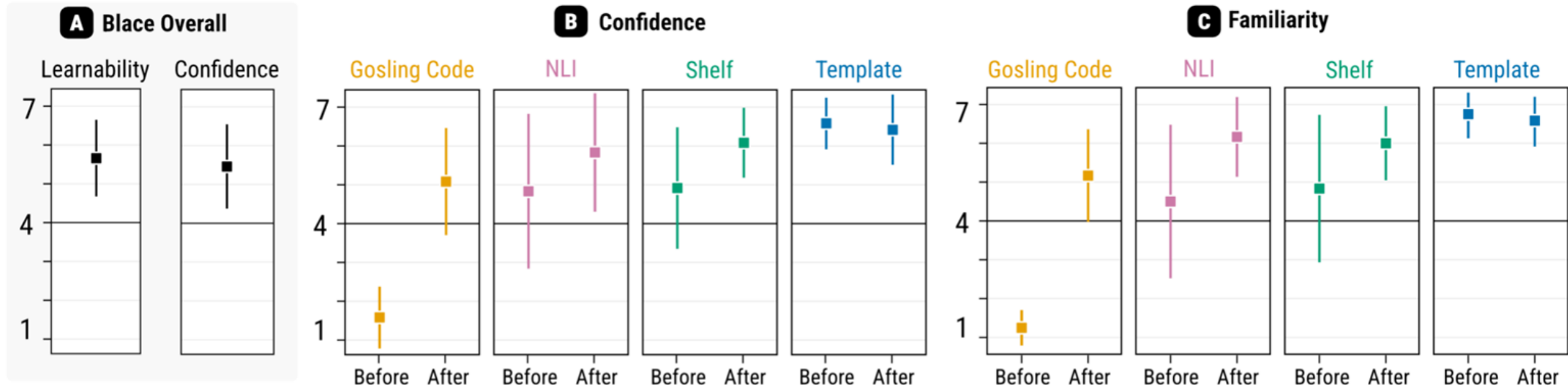
</> CODE EDITOR Apply Changes -9 +1

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66 66 "mark": "point",
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68 68 "x": {
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70 70 "type": "genomic"
71 71 },
72 72 "type": "quantitative",
73 73 "grid": true
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75 75 },
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78 78 "type": "quantitative",
79 79 "range": [
80 80 1,
81 81 15
82 82 ],
83 83 },
84 84 "color": {
```

Blended Interfaces (Evaluation)



Blended Interfaces (Results)



No assistance needed

Gosling fam. (1.3 to 5.2) and conf (1.1 to 5.1)

High confidence

Linking helped a lot in learning Gosling

Zhao et al., 2025 (P2)

1830IEEE TRANSACTIONS ON VISUALIZATION AND COMPUTER GRAPHICS, VOL. 31, NO. 3, MARCH 2025

LEVA: Using Large Language Models to Enhance Visual Analytics

Yuheng Zhao, Yixing Zhang, Yu Zhang, Xinyi Zhao, Junjie Wang, Zekai Shao, Cagatay Turkey, and Siming Chen

Abstract—Visual analytics supports data analysis tasks within complex domain problems. However, due to the richness of data types, visual designs, and interaction designs, users need to recall and process a significant amount of information when they visually analyze data. These challenges emphasize the need for more intelligent visual analytics methods. Large language models have demonstrated the ability to interpret various forms of textual data, offering the potential to facilitate intelligent support for visual analytics. We propose LEVA, a framework that uses large language models to enhance users’ VA workflows at multiple stages: onboarding, exploration, and summarization. To support onboarding, we use large language models to interpret visualization designs and view relationships based on system specifications. For exploration, we use large language models to recommend insights based on the analysis of system status and data to facilitate mixed-initiative exploration. For summarization, we present a selective reporting strategy to retrace analysis history through a stream visualization and generate insight reports with the help of large language models. We demonstrate how LEVA can be integrated into existing visual analytics systems. Two usage scenarios and a user study suggest that LEVA effectively aids users in conducting visual analytics.

Index Terms—Insight recommendation, mixed-initiative, interface agent, large language models, visual analytics.

I. INTRODUCTION

VISUAL analytics (VA) combines data analysis techniques with visualizations for effective understanding, reasoning and decision-making on the basis of large and complex datasets [21], [22], [55]. However, the VA processes often require significant effort on the side of the user, resulting in a less efficient data interpretation and analysis process [58]. Several challenges underpin this inefficiency: a steep learning curve

with VA systems’ onboarding [50], a tendency to lose direction in data exploration [29], and the difficulty of summarizing final insights [8]. Such issues emphasize the need for a more intelligent approach to VA.

To specify the challenges further: first, the challenge of onboarding stems from unfamiliarity with a VA system, leading to reduced efficiency in grasping visual mappings and interactions [5], [41]. Second, during data exploration, the numerous avenues to analyze data and the lack of structured guidance and direction make it difficult to uncover insights [6]. Third, keeping track of and aggregating key findings is often difficult and summarizing insights is a time-consuming task that demands significant effort and attention to detail [3], [33]. These challenges highlight the need for an intelligent framework to support VA processes and foster efficacy across these pivotal stages.

To address these problems, researchers focus on enhancing VA through visualization onboarding [49], interaction recommendation and guidance [4], [29], [47], and result summarization [8], [44]. However, these methods often do not take advantage of advancements in intelligent algorithms and are not easily adaptable to various VA systems. There is a lack of generalizable intelligent approaches that can work across different VA systems.

The Large Language Models (LLMs) exhibit broader knowledge and problem-solving abilities, making it possible to address the above challenges in the three stages of VA. First, as LLMs can interpret visualizations’ declarative grammar [13], [35], we can try to propose the grammar for VA systems and use it as input for LLMs to generate onboarding tutorials. Additionally, LLMs can process various data types [39], making it possible to recommend insights by analyzing both the system status and underlying data, assisting users’ exploration. Finally, LLMs exhibit powerful summarization capabilities [17], which may help to summarize the exploration process and generate a report with rich forms.

In this paper, we propose a framework named LEVA (LLM-Enhanced Visual Analytics) that uses LLMs to enhance visual analytics in three stages of the workflow. In the onboarding phase, we provide a solution that enables LLMs to interpret visualizations in each of the views and these views’ relationships based on a specification of the VA system. This enables the flexible creation of tutorials for various VA systems. In the exploration phase, we design an insight recommendation strategy that guides LLMs in recommending insights based on the understanding of the system, analytical task, user’s interaction,

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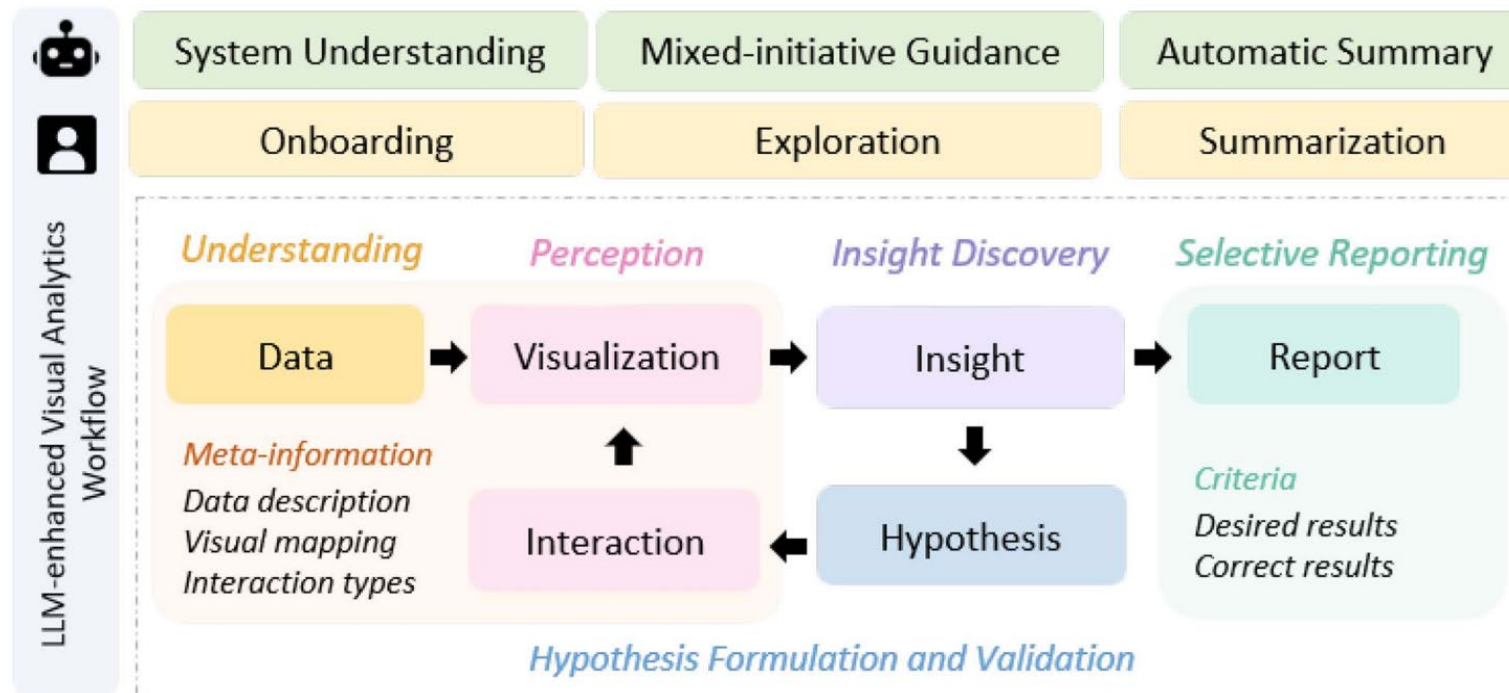
Onboarding

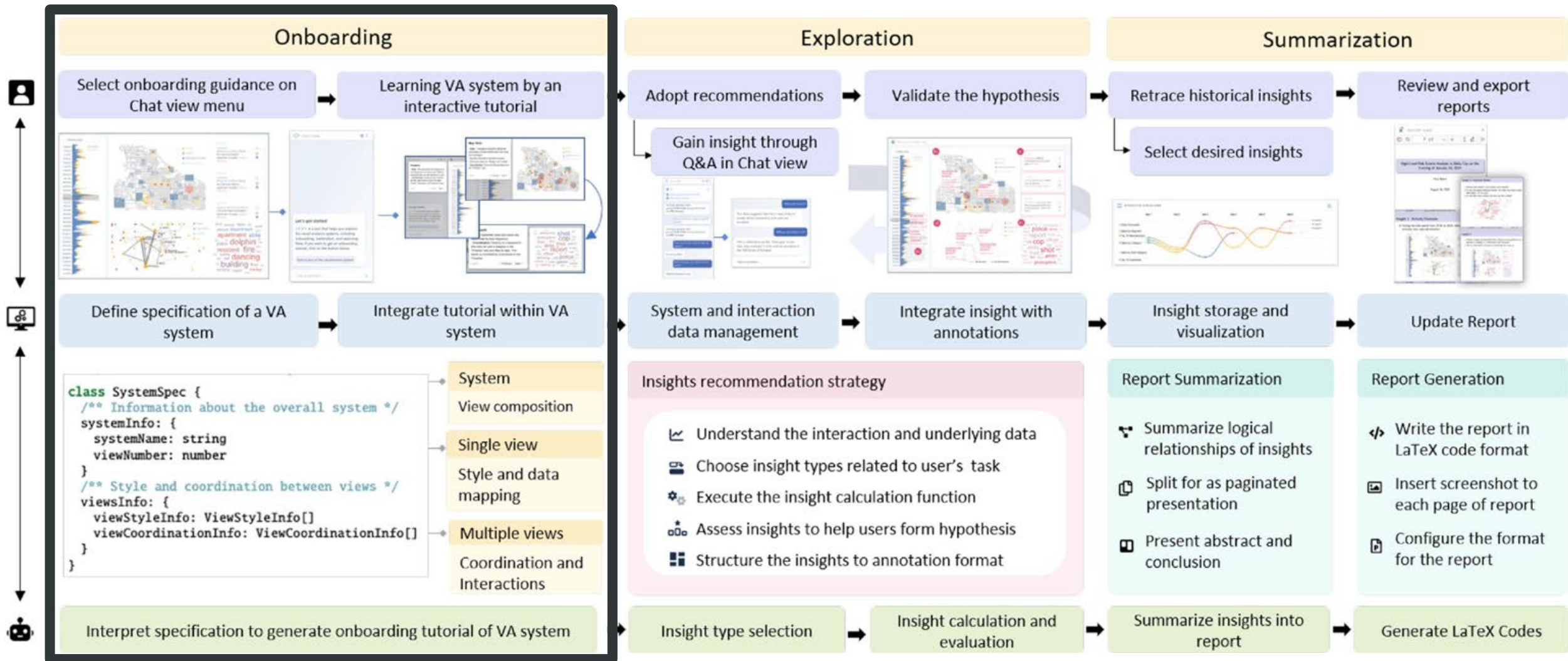
Exploration

Summarization



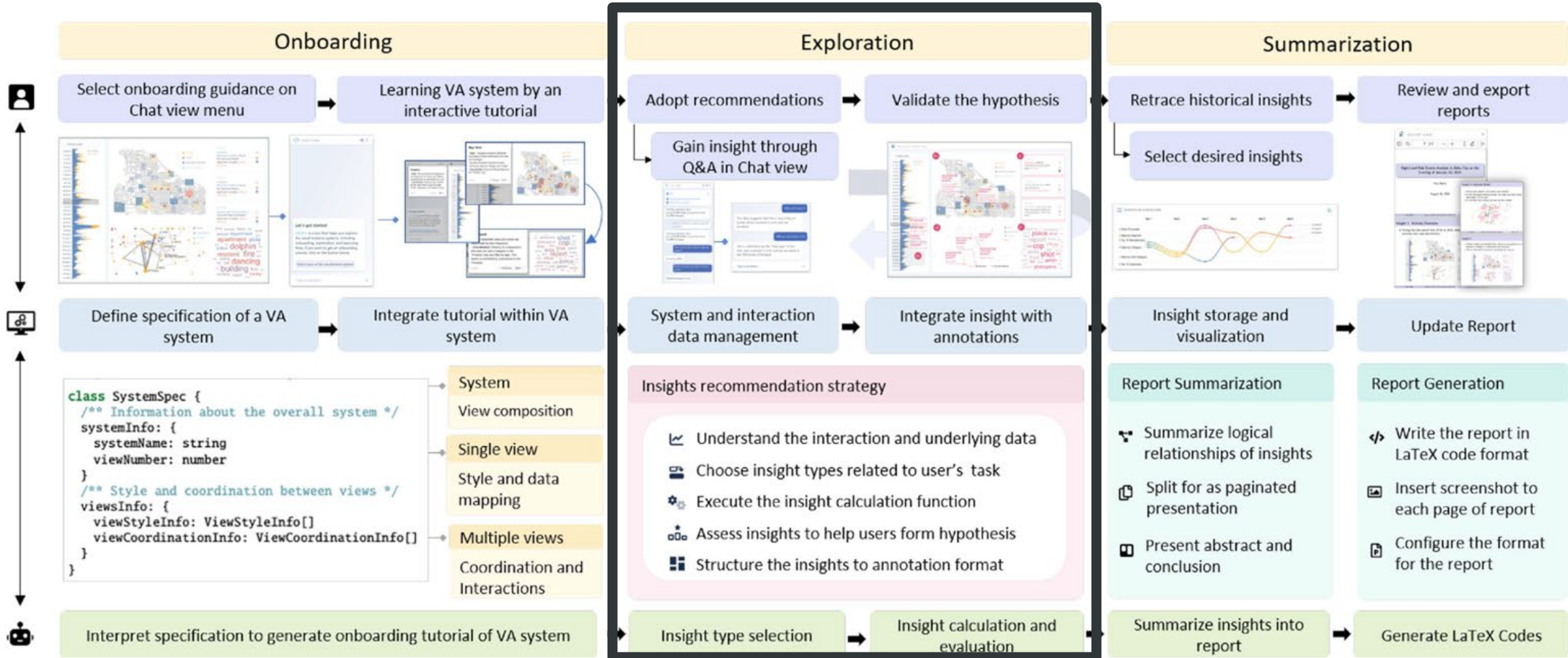
LEVA (Proposed Solution)





LEVA (Formalization)

System-level	View-level	Interaction-level
<pre> 1 class SystemSpec { 2 /** Information about the overall system */ 3 systemInfo: { 4 systemName: string 5 viewNumber: number 6 } 7 /** Style and coordination between views */ 8 viewsInfo: { 9 viewStyleInfo: ViewStyleInfo[] 10 viewCoordinationInfo: ViewCoordinationInfo[] 11 } 12 } </pre>	<pre> 1 class EncodingInfo { 2 field: string; 3 type: string; 4 description: string; 5 } 6 class ViewStyleInfo { 7 viewName: string 8 /** Information about styles of multiple encodings */ 9 layers: { 10 markType: string 11 encoding: { 12 x?: EncodingInfo[]; 13 y?: EncodingInfo[]; 14 color?: EncodingInfo[]; 15 size?: EncodingInfo[]; 16 /** Detail level data field */ 17 lod?: EncodingInfo[]; 18 } 19 /** Display prompt information for mouse hover */ 20 tooltip?: EncodingInfo[]; 21 }[] 22 } </pre>	<pre> 1 class ViewCoordinationInfo { 2 sourceViewName: string 3 targetViewName: string string[] 4 /** Coordination type */ 5 coordinationType: string 6 interaction: { 7 /** Type of interaction */ 8 type: string 9 /** Interaction's effect on target */ 10 effect?: { 11 /** Action type */ 12 action: string 13 targetViewName: string 14 /** Data category for action */ 15 category: string 16 /** Control the result */ 17 changeby: string 18 } 19 }[] 20 } </pre>



CHAT VIEW

☒ All
☒ Gelatogalore Shooting Incident
☒ Biker Henri Incident

You have selected a time period:19:09-19:18, during which there are 108 messages.

What are the main events in high-risk regions?

You have selected a time period:19:35-19:43, during which there are 270 messages.

What are the main events in high-risk regions?

Summary added.

Which messages are related to these events?

Related messages found.

Which nodes are related to these events?

Related nodes found.

You added a keyword van into events list.

Which events related to this keyword?

Story added to timeline.

Type a question...



ORIGINAL SYSTEM VIEW

mbdata/ccdata

17:05:00
17:13:20
17:21:40
17:30:00
17:38:20
17:46:40
17:55:00
18:03:20
18:11:40
18:20:00
18:28:20
18:36:40
18:45:00
18:53:20
19:01:40
19:10:00
19:18:20
19:26:40
19:35:00
19:43:20
20:00:00
20:08:20
20:16:40
20:25:00
20:33:20
20:41:40
20:50:00
20:58:20
21:06:40
21:15:00
21:23:20

mbdata
ccdata

HIT AND RUN
19:20:00 - 19:23:54
PULLED OVER
19:40:00 - 19:40:29
HOSTAGE SITUATION
19:58:13 - 20:10:38
STANDOFF
20:16:14 - 21:14:47

GELATOGALORE SHOOTING INCIDENT
Shooting incident involving a black van and police at Gelatogalore leading to a cop's death.

BIKER HENRI INCIDENT
Biker Henri was hit and consoled with coffee at a local shop.

GELATOGALORE SHOOTING INCIDENT
van guys

GELATOGALORE SHOOTING INCIDENT
lights gelato

BIKER HENRI INCIDENT
Henri

GELATOGALORE SHOOTING INCIDENT
Gelato Galore

van guys

#troubleatgelato

Person
Organization
Location
Tag

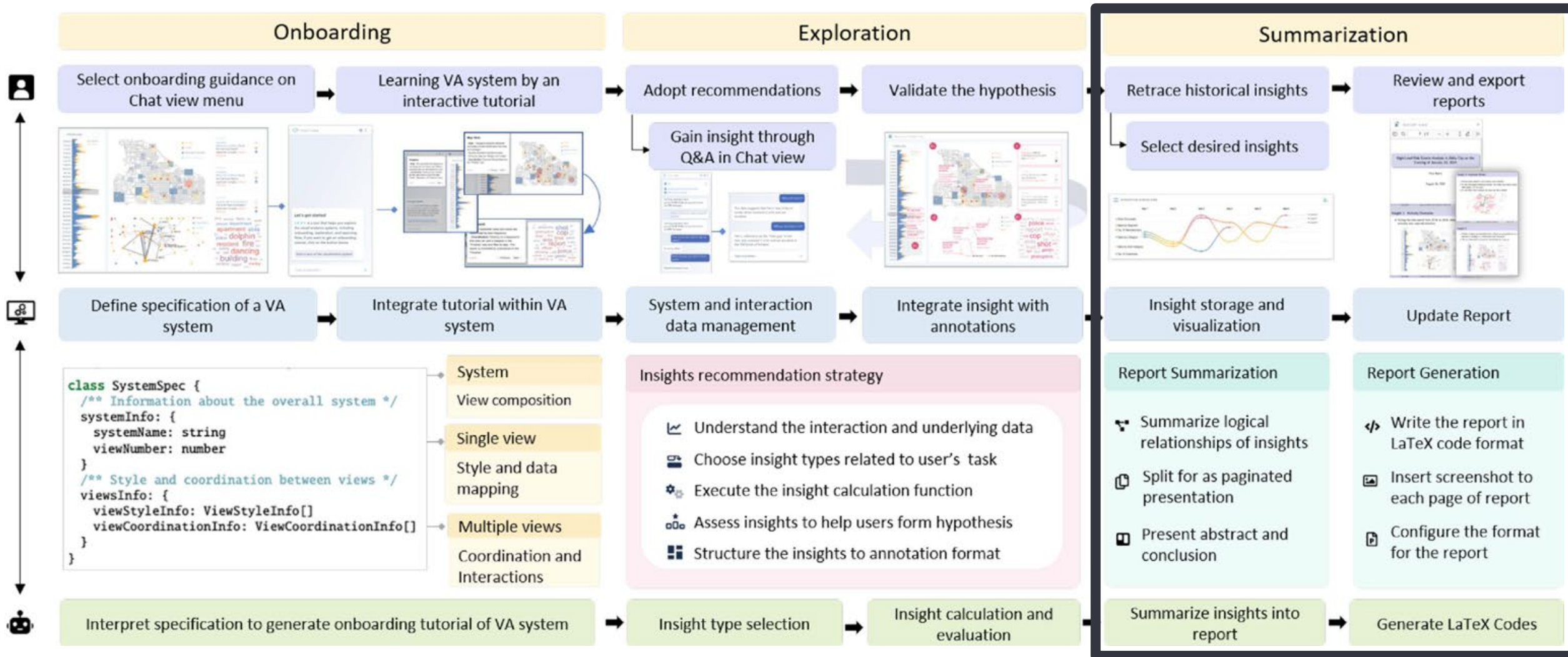
Recommended Insights

19:39:07
Call Center: PURSUIT-CONTINUES-POLICE UNITS AND #253 #357

19:39:40
sofitees: @gelatogalore Wow ! crazy driving into #somebodywantgelatobad

19:40:00
truccotrucco: wow some crazy black van just got pulled over in the parking lot - dinner & a show !

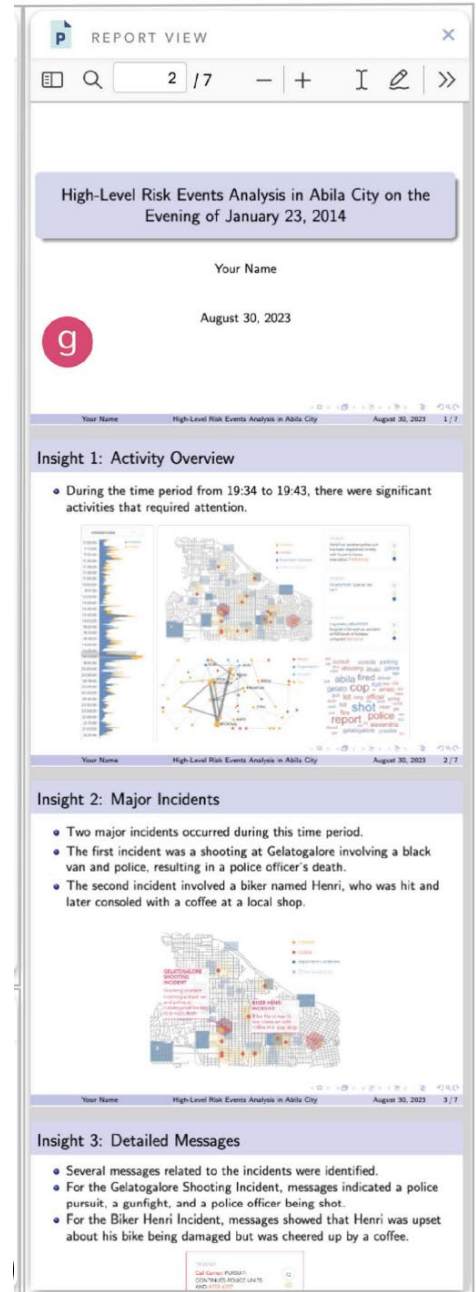
parking sit shooting pls
result near police driver null
red report ithaki got wtf
ran hit cop officer like
give abila lot one going
exit ago fired car shot van
gun run fire gelato bike
bullet erratic gelatogalore 500 city say



C Analysis History



D Generated Report



A Conversation Control

CHAT VIEW

☒ All

☒ Gelatogalore Shooting Incident

☒ Biker Henri Incident

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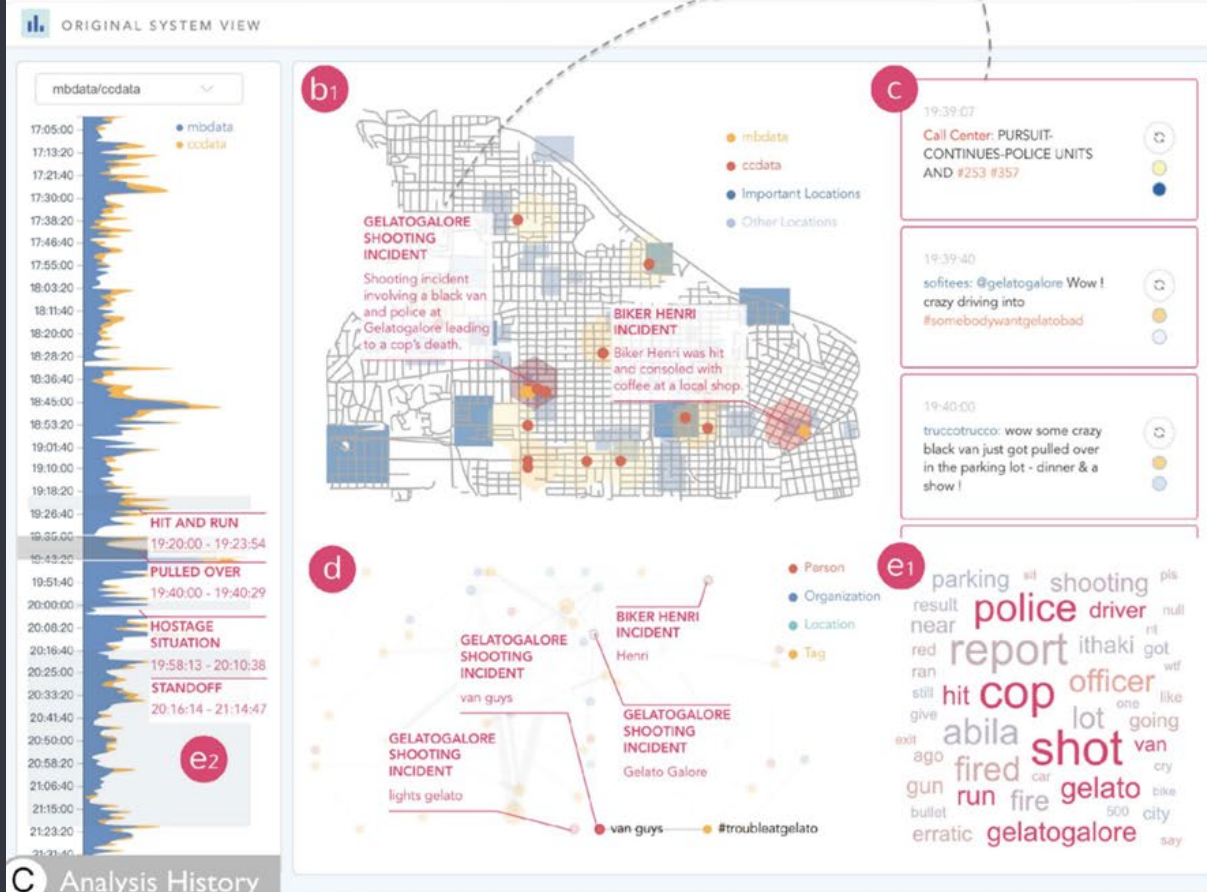
You added a keyword van into events list.

Which events related to this keyword?

Story added to timeline.

Type a question...

B Recommended Insights



D Generated Report

REPORT VIEW

2 / 17

High-Level Risk Events Analysis in Abila City on the Evening of January 23, 2014

Your Name

August 30, 2023

Insight 1: Activity Overview

- During the time period from 19:34 to 19:43, there were significant activities that required attention.

Insight 2: Major Incidents

- Two major incidents occurred during this time period.
- The first incident was a shooting at Gelatogalore involving a black van and police, resulting in a police officer's death.
- The second incident involved a biker named Henri, who was hit and later consoled with a coffee at a local shop.

Insight 3: Detailed Messages

- Several messages related to the incidents were identified.
- For the Gelatogalore Shooting Incident, messages indicated a police pursuit, a gunfight, and a police officer being shot.
- For the Biker Henri Incident, messages showed that Henri was upset about his bike being damaged but was cheered up by a coffee.

A Conversation Control

CHAT VIEW

☒ All

☒ Gelatogalore Shooting Incident

☒ Biker Henri Incident

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Which events related to this keyword?

Story added to timeline.

Type a question...

B Recommended Insights

ORIGINAL SYSTEM VIEW

mbdata/ccdata

17:05:00
17:13:20
17:21:40
17:30:00
17:38:20
17:46:40
17:55:00
18:03:20
18:11:40
18:20:00
18:28:20
18:36:40
18:45:00
18:53:20
19:01:40
19:10:00
19:18:20
19:26:40
19:35:00
19:43:20
19:51:40
20:00:00
20:08:20
20:16:40
20:25:00
20:33:20
20:41:40
20:50:00
20:58:20
21:06:40
21:15:00
21:23:20
21:31:40

HIT AND RUN
19:20:00 - 19:23:54

PULLED OVER
19:40:00 - 19:40:29

HOSTAGE SITUATION
19:58:13 - 20:10:38

STANDOFF
20:16:14 - 21:14:47

b1

GELATO GALORE SHOOTING INCIDENT

Shooting incident involving a black van and police at Gelatogalore leading to a cop's death.

BIKER HENRI INCIDENT

Biker Henri was hit and consoled with coffee at a local shop.

c

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19:40:00
truccotruc: wow some crazy black van just got pulled over in the parking lot - dinner & a show!

d

GELATO GALORE SHOOTING INCIDENT

van guys

GELATO GALORE SHOOTING INCIDENT

Henri

GELATO GALORE SHOOTING INCIDENT

Gelatogalore

lights gelato

van guys

#troubleatgelato

e1

parking wil shooting pls
result near police driver null
report ithaki got wtf
ran hit cop officer like
still give abila lot going
ago fired car shot van
gun run fire gelato bike
bullet erratic gelatogalore say

e2

Person
Organization
Location
Tag

C Analysis History

INTERACTION STREAM VIEW

step 1 step 2 step 3 step 4 step 5 step 6 step 7 step 8

Timeline

Risk Map

Graph

Message

Keyword

round 1

round 2

D Generated Report

REPORT VIEW

2 / 7

High-Level Risk Events Analysis in Abila City on the Evening of January 23, 2014

Your Name

August 30, 2023

Insight 1: Activity Overview

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Insight 2: Major Incidents

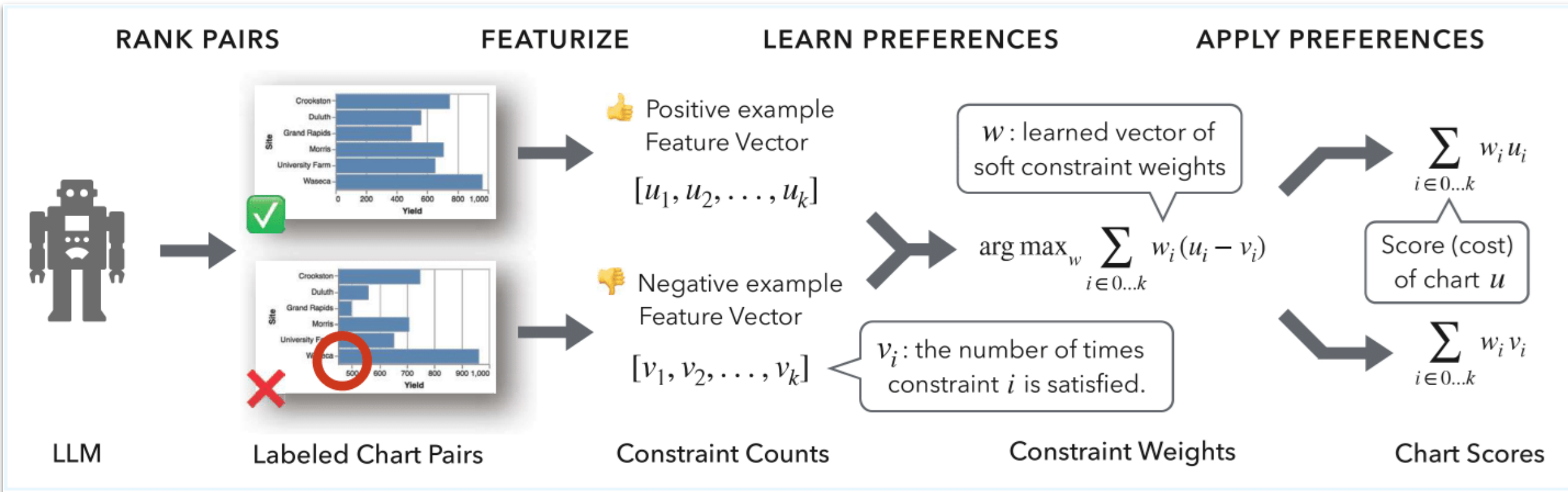
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DracoGPT (DracoGPT-Rank)



DracoGPT (DracoGPT-Recommend)

